

v1 교재(Klein 유기화학 1~23장) 인용 논문·반응 추출본

대응 질문 ①·② · 스캔 PDF를 판독한 원자료. 각 장 「참고문헌」과 각주 문제를 대응시켰다. SMILES는 RDKit으로 검증. [H]/[M]/[L]·[high]/[med]/[low]는 신뢰도.

Klein 유기화학 — Chapters 1-6: summaries + literature-based problems

Source: /root/.claude/uploads/e196aa5e-9821-56b2-b877-2f07b790959e/2cf3abfb-1-6.pdf (52 scanned pages; PDF page N = p-NN.jpg). "PDF p" = PDF page index; "book p" = printed page number. SMILES: stereo is given only where the drawing shows wedges/dashes; otherwise omitted. Confidence marks: [high]/[med]/[low].

Chapter 1 — 일반화학 복습하기 (A Review of General Chemistry) (PDF p1-8, book p58-65)

Note: the PDF begins mid-summary (book p58, from §1.6 onward); the §1.1-1.5 summary text is not in these pages.

Key concepts (summary, §1.6-1.13 + skill builders 1.1-1.10)

- §1.6 Atomic orbitals, nodes ($\psi = 0$); electron density; filling orbitals: Aufbau, Pauli exclusion, Hund's rule.
- §1.7 Valence bond theory: bonding = constructive overlap of atomic orbitals; σ bonds have density on the bond axis.
- §1.8 MO theory / LCAO: bonding MO (constructive), antibonding MO (destructive); MOs belong to the whole molecule; HOMO and LUMO.
- §1.9 Hybridization: sp^3 (methane, tetrahedral), sp^2 (ethylene, trigonal planar, $\sigma + \pi$), sp (acetylene, linear, $\sigma + 2\pi$). Triple bond shorter and stronger than double, which is shorter and stronger than single.
- §1.10 VSEPR geometry; steric number = σ bonds + lone pairs $\rightarrow$ tetrahedral, trigonal pyramidal, bent, trigonal planar, linear.
- §1.11 Dipole moment (μ , debye); % ionic character; molecular dipole = vector sum of bond dipoles.
- §1.12 Intermolecular forces: dipole-dipole, hydrogen bonding (higher bp), London dispersion (larger surface area $\rightarrow$ stronger; branching lowers bp).
- §1.13 Solubility: "like dissolves like"; soaps are amphiphilic (hydrophilic head, hydrophobic tail), forming micelles.
- Skill builders: constitutional isomers; Lewis dot structures; formal charge; partial charges (δ^+/δ^-); electron configurations; hybridization from bond types (4 single = sp^3 , 1 double = sp^2 , triple = sp); geometry via steric number; net molecular dipole; predicting physical properties (dipole-dipole, H-bonding, carbon count/branching).

Literature footnotes, chapter 1 (참고문헌, book p65)

1. Tetrahedron Lett. 2012, 53, 4536-4537.
2. Org. Lett. 1999, 1, 1123-1125.
3. Tetrahedron Lett. 2003, 44, 9165-9167.
4. Tetrahedron 2006, 62, 10968-10979.
5. Org. Lett. 1999, 1, 1713-1715.
6. J. Org. Chem. 2009, 74, 1099-1113.
7. Tetrahedron Lett. 2001, 42, 8579-8582.
8. Org. Lett. 2001, 3, 3843-3846.
9. J. Am. Chem. Soc. 2009, 131, 8833-8838.

1.68 (종합 문제; book p63, PDF p6) — footnote 1 → Tetrahedron Lett. 2012, 53, 4536-4537

- Compounds: tert-butyl isocyanide **A**, $(\text{CH}_3)_3\text{C}-\text{N}^+\equiv\text{C}^- \rightarrow$ tert-butyl isocyanide dichloride (isonitrile dihalide / carbonimidic dichloride) **B**, $(\text{CH}_3)_3\text{C}-\text{N}=\text{CCl}_2$. Highlighted: the N and terminal C of A; the N and C of B. Context: converting an isonitrile to its dihalide temporarily "masks" isonitrile reactivity.
- SMILES: A CC(C)(C)[N+]#[C-] [high]; B CC(C)(C)N=C(Cl)Cl [high]. No reagents shown (the conversion uses Cl_2 , which is not drawn).
- Questions: (a) hybridization of the highlighted atoms in A; (b) which atomic orbital holds the carbon lone pair in A; (c) the C-N-C angle in A; (d) hybridization of the highlighted atoms in B; (e) which orbital holds the N lone pair in B; (f) the C-N-C angle in B.
- Expected answers: (a) N sp, C sp; (b) an sp orbital; (c) $\sim 180^\circ$; (d) N sp², C sp²; (e) an sp² orbital; (f) $\sim 120^\circ$ (somewhat less because of the lone pair). [high]
- Concept: hybridization and geometry; lone-pair orbital type.

1.69 (book p63, PDF p6) — footnote 2 → Org. Lett. 1999, 1, 1123-1125

- Context: branching / steric hindrance ("입체 장애"); more hindered C=C π bonds react more slowly.
- Molecule drawn: $(\text{CH}_3)_2\text{C}=\text{CH}-\text{CH}_2-\text{CH}_2-\text{CH}(\text{CH}_3)-\text{CH}=\text{CH}_2$, i.e. 3,7-dimethylocta-1,6-diene (dihydromyrcene). SMILES CC(C)=CCCC(C)C=C [med-high, read from the expanded drawing].
- Question: identify each π bond and say which is more sterically hindered. Note: hindrance is not the same as steric number.
- Answer: the trisubstituted C₆=C₇ (with the two methyls) is more hindered; the terminal monosubstituted CH=CH₂ is less hindered (so selective reagents attack the terminal alkene). [high]

1.70 (book p63) — no footnote

Bond-length table, C-X in Å:

	X = F	X = Cl	X = Br	X = I
sp ³ C	1.40	1.79	1.97	2.16
sp ² C	1.34	—	1.88	2.10
sp C	1.27	1.63	1.79	—

It asks whether C(sp²)-Cl or C(sp)-I is shorter. It is listed here only for completeness because it has no literature footnote.

1.71 (book p63, PDF p6) — footnote 3 → Tetrahedron Lett. 2003, 44, 9165-9167

- PET tracer [18F]fluorocholine synthesis: 18F-CH₂CH₂-Br + Me₂N-CH₂CH₂-OH (2-(dimethylamino)ethanol) → [18F]F-CH₂CH₂-N⁺(CH₃)₂-CH₂CH₂-OH Br⁻ (N-(2-[18F]fluoroethyl)-N,N-dimethylethanolammonium bromide, i.e. [18F]fluoroethylcholine). The drawing shows an ethyl linker on the 18F side.
- SMILES: [18F]CCBr, CN(C)CCO, C[N+](C)(CC[18F])CCO.[Br-] [high].
- Questions: (a) hybridization of every atom other than C, H and O in the three compounds (lone pairs are not drawn); (b) the C-N-C angles in the product.
- Answers: F sp³; Br sp³ (in the alkyl bromide); amine N sp³; ammonium N sp³ (Br⁻ treated as having no hybridization, or sp³ by convention); all C-N-C angles ~109.5°. [high]

1.72 (도전 문제; book p64, PDF p7) — footnote 4 → Tetrahedron 2006, 62, 10968-10979

- Epichlorohydrin (1, 2-(chloromethyl)oxirane, C1CC1Cl) + phenol (2, Oc1ccccc1), NaOH, heat → 3 1-chloro-3-phenoxypropan-2-ol Oc1ccccc1CC1 + 4 phenyl glycidyl ether C1CC1Cc2ccccc2 [high].
- Observation: at room temperature the two liquid products are hard to separate, but on heating they separate easily. Explain.
- Expected answer: 3 has an O-H and forms intermolecular hydrogen bonds; 4 has none. 3 therefore has a much higher boiling point, so heating (distillation) separates 4 (more volatile) from 3. [med-high]
Concept: hydrogen bonding and boiling point.

1.73 (book p64, PDF p7) — footnote 5 → Org. Lett. 1999, 1, 1713-1715

- Phenalamide A2 (antibiotic, antifungal, antiviral natural product): a boronate ester used in the first total synthesis. Structure: an allylic pinacol boronate, CH₂=CH-CH(Bpin)-(4H-1,3-dioxin-4-yl). The ring is a six-membered O-CH₂-O-CH=CH-CH ring.
- SMILES C=CC(B1OC(C)(C)C(C)C1)C1C=CCOC1 [med; ring connectivity read from the drawing].
- Questions: (a) boron hybridization; (b) predict the O-B-O angle and suggest why the real angle deviates; (c) draw all lone pairs (hint: no formal charges).
- Answers: (a) sp²; (b) predicted 120°, actual smaller (~110°) because B is in a five-membered dioxaborolane ring (pentagon interior ~108°), so ring strain/geometry constrains it; (c) two lone pairs on each O (4 O = 8 lone pairs); B has none (empty p orbital). [high]

1.74 (book p64, PDF p7) — footnote 6 → J. Org. Chem. 2009, 74, 1099-1113

- Oxazolidinone synthesis: hydroximoyl chloride 1, benzohydroximoyl chloride PhC(Cl)=N-OH O=N=C(Cl)c1ccccc1, → nitrile oxide 2, benzonitrile oxide PhC≡N⁺-O⁻ [O-][N+]#Cc1ccccc1 (drawn without charges as C≡N-Ö:) [high].
- Questions (book p64 right column): (a) the missing formal charges in 1 and 2; (b) which is more soluble in a polar solvent, and why; (c) how much the C-C-N angle increases from 1 to 2.
- Answers: (a) 1 has none; in 2 N is +1 and O is -1; (b) 2 is more soluble (charge-separated, large dipole; 1 can also H-bond, so this is debatable; textbook answer likely 2) [med]; (c) 120° → 180°, an increase of ~60° [high].

1.75 (book p64, PDF p7) — footnote 7 → Tetrahedron Lett. 2001, 42, 8579-8582

- Estradiol derivative (breast-cancer treatment candidate). Estradiol skeleton (aromatic A ring with 3-OH, 18-CH₃), with C17 bearing OH and a C≡C-CH=CH-CH=C=CH₂ chain (17 α -alkynyl enyne-allene; Ca = terminal CH₂, Cb = central allene C, Cc = allene CH).
- SMILES (no stereo shown) CC12CCC3C4CCC(O)CC4CCC3C1CCC2(O)C#CC=CC=C [med-high].
- Questions: (a) hybridization of Ca, Cb, Cc; (b) the H-Ca-Cb angle; (c) the Ca-Cb-Cc angle; (d) Cb has two π bonds; draw the relative orientation of its two p orbitals and the p orbitals of Ca and Cc.
- Answers: (a) Ca sp², Cb sp, Cc sp²; (b) ~120°; (c) 180°; (d) Cb's two p orbitals are perpendicular, so the Ca and Cc p orbitals are perpendicular to each other (the allene's terminal substituent planes are orthogonal). [high]

1.76 (book p64, PDF p7) — footnote 8 → Org. Lett. 2001, 3, 3843-3846

- Enzyme-mimic / foldamer study compound: Boc-Leu-NH-iPr, (CH₃)₃C-O-C(=O)-NH-CH(CH₂CH(CH₃)₂)-C(=O)-NH-CH(CH₃)₂. Highlighted: the carbamate C=O oxygen (Boc) and the N(-H) of the isopropylamide.
- SMILES CC(C)CC(NC(=O)OC(C)(C)C)C(=O)NC(C)C [high; no stereo drawn].
- Questions: (a) "the compound has two N-C-N units with different bond angles"; predict the difference and explain (as printed; the wording looks garbled, likely meaning the two amide/carbamate N environments or the C-N-C angles, so the intended comparison is uncertain); (b) the highlighted regions prefer to sit close in 3D; explain the interaction and draw it.
- Answer (b): an intramolecular hydrogen bond, N-H (isopropylamide) $\cdots$ O=C (Boc carbamate), closing a 7-membered ring (γ -turn/C7-type). [high]

1.77 (book p65, PDF p8) — footnote 9 → J. Am. Chem. Soc. 2009, 131, 8833-8838

- Molecularly imprinted polymers. Guests: bisphenol A **1** CC(C)(c1ccc(O)cc1)c1ccc(O)cc1 and 4,4'-methylenedianiline **2** Nc1ccc(Cc2ccc(N)cc2)cc1 (drawn with a CH₂ linker; [med] that the linker is CH₂). Polymer A binding site: two benzoic-acid (-COOH) groups facing each other. Polymer B binding site: two pyridine N's.
- Facts: A binds **1** and **2** about equally (limited selectivity); B binds **1** strongly and **2** weakly.
- Task: draw how **1** and **2** fit each cavity and explain the selectivity.
- Answer: COOH is both an H-bond donor and acceptor, so it H-bonds with Ar-OH and Ar-NH₂ alike (no discrimination). Pyridine N is only an acceptor and pairs best with the stronger, more acidic O-H donor of the phenol; aniline N-H is a weaker donor, so **2** binds weakly. [med-high]

Chapter 2 — 분자 표현법 (Molecular Representations) (PDF p9-16, book p101-108)

Key concepts (개념과 용어의 복습)

- §2.1 Lewis, partially condensed and condensed structures; a molecular formula gives no structural information.
- §2.2 Bond-line structures (C and most H not drawn).

- §2.3 Functional groups determine chemistry.
- §2.4 Formal charge; carbon with a charge has 3 bonds.
- §2.5 Lone pairs are usually omitted in bond-line structures and must be inferred.
- §2.6 Wedges/dashes for 3D; Fischer and Haworth projections.
- §2.7-2.13 Resonance: structures linked by double-headed arrows in brackets; resonance stabilization = delocalization of charge. Curved-arrow rules: don't break single bonds; never exceed an octet for second-row elements. Show all formal charges. Five patterns: (1) allylic lone pair, (2) allylic carbocation, (3) lone pair adjacent to C⁺, (4) π bond between atoms of different electronegativity, (5) conjugated π bonds in a ring. Significance rules: max full octets > fewest formal charges > negative charge on the more electronegative atom; equivalent structures contribute equally. Resonance hybrid shown with partial bonds/charges. Delocalized lone pairs occupy p orbitals (atom becomes sp²); localized ones don't; an atom with both a π bond and a lone pair doesn't delocalize both.
- Skill builders 2.1-2.10 cover each of these.

Literature footnotes, chapter 2 (참고문헌, book p108)

1. Org. Biomol. Chem. 2015, 13, 7813-7821.
2. Org. Lett. 2015, 17, 1529-1532.
3. Cryst. Growth Des. 2012, 12, 4311-4314.
4. J. Am. Chem. Soc. 1971, 93, 4332-4334.
5. Angew. Chem. Int. Ed. 2005, 44, 1012-1044.
6. Chem. Rev. 1993, 93, 381-433.
7. J. Chem. Ed. 2006, 83, 1534-1536.
8. J. Chem. Ed. 2006, 83, 287-289.
9. Soft Matter 2012, 8, 5486-5492.
10. Langmuir 2012, 28, 14567-14572.
11. J. Am. Chem. Soc. 2009, 131, 11478-11484.
12. J. Org. Chem. 1997, 62, 3200-3207.
13. Chem. Rev. 1993, 93, 381-433.
14. J. Invest. Dermatol. 2014, 134, 2988-2990.

Footnote 1 marker not found in any problem on these pages. I checked 2.63-2.67 at high resolution: 2.63 cycloserine, 2.64 ramelteon, 2.65, 2.66, 2.67 show no superscript. The marker may sit in chapter body text outside this excerpt; it is not assigned to any problem here.

2.68 (종합 문제; book p106, PDF p14) — footnote 2 → Org. Lett. 2015, 17, 1529-1532

- (+)-Dragmacidin D (marine sponge alkaloid; antiviral, antifungal, antibacterial). Structure: 2-aminoimidazolium, an imidazole ring with C2=N⁺H₂, two ring N-H, and C4-R. R is a CH(CH₃) (wavy/hashed stereocenter) attached at C4 of a 7-hydroxyindol-3-yl unit, linked at indole C3 to a pyrazin-2(1H)-one, whose other carbon bears a 6-bromoindol-3-yl.
- SMILES (stereo omitted) CC(c1c[nH]c(=[NH2+])[nH]1)c1ccc(O)c2[nH]cc(-c3cnc(-c4c[nH]c5cc(Br)ccc45)c(=O)[nH]3)c12 [med; pyrazinone regiochemistry read from the drawing].
- Question: the cation is delocalized; draw all resonance structures delocalizing the charge, then the hybrid.

- Answer: three equivalent-type guanidinium-like structures with the + on each of the three N's (exocyclic NH₂ and the two ring NH); the hybrid has a partial positive charge (δ^+) on all three N's and partial C=N character. [high]

2.69 (book p106, PDF p14) — footnote 3 → Cryst. Growth Des. 2012, 12, 4311-4314

- CL-20 (hexanitrohexaazaisowurtzitane) and HMX (octahydro-1,3,5,7-tetranitro-1,3,5,7-tetrazocine). A 2:1 CL-20:HMX cocrystal is more powerful than HMX with similar sensitivity.
- SMILES: HMX C1N(CN(CN(CN1[N+](=O)[O-])[N+](=O)[O-])[N+](=O)[O-])[N+](=O)[O-]; CL-20 cage O=[N+](=[O-])N1C2C3N(C4C(N3[N+](=O)[O-])N(C1C(N2[N+](=O)[O-])N4[N+](=O)[O-])[N+](=O)[O-])[N+](=O)[O-] [med for the CL-20 SMILES; well-known structure].
- Questions: (a) molecular formulas; (b) is the lone pair on a ring N localized or delocalized?
- Answers: (a) CL-20 = C₆H₆N₁₂O₁₂; HMX = C₄H₈N₈O₈; (b) delocalized into the N=O of the nitro group (nitramine N-NO₂ resonance), so the ring N is sp²/trigonal planar. [high]

2.70 (book p106, PDF p14) — footnote 4 → J. Am. Chem. Soc. 1971, 93, 4332-4334

- W. S. Johnson biomimetic progesterone synthesis (polyene cyclization). Intermediate shown: a tetracyclic D-homo/A-ring cyclopentene steroid skeleton (cyclopentene A ring with a methyl on the double bond, angular methyls, trans ring fusions with H's dashed/wedged), where C17 bears an exocyclic C=C(CH₃)-O-C⁺ ring: a 1,3-dioxolan-2-ylum cation (an enol ether attached to a dioxolanyl cation).
- Question: use resonance to explain why this intermediate is especially stable.
- Answer: the carbocation C2 of the dioxolane is flanked by two O atoms, each with a lone pair that delocalizes the + charge (oxocarbenium resonance). Including the third, vinyl-ether O, the + is shared over three O's (orthoester-derived trialkoxycarbenium-type cation); all atoms keep octets. [high]
- SMILES: uncertain (full tetracyclic skeleton stereochemistry); cation fragment C(C)(=C<steroid>)O[C+]1OCCO1.

2.71 (book p107, PDF p15) — footnote 5 → Angew. Chem. Int. Ed. 2005, 44, 1012-1044

- Misassigned natural-product structures (structure revision). (a) Cholesterol: the originally proposed (wrong) structure, a strange polycyclic with an OH, a cyclopentene and the same C₈ side chain, vs. the correct one, C[C@H](CCCC(C)C)[C@H]1CC[C@@H]2[C@@H]3CC=C4[C@@H](O)CC[C@]4(C)[C@H]3CC[C@]12C. (b) Porritoxin: proposed = benzoxazocine lactam (8-membered ring with C(=O)NH-CH₂CH₂-O-CH₂); correct = isoindolinone N-(2-hydroxyethyl), with a prenyloxy group, aromatic CH₃ and OCH₃.
- Correct porritoxin SMILES CC(C)=CCO1CC2C(=O)N(CCOC2C(OC)C1C [med].
- Task: identify all functional groups in each pair; compare similarities and differences.
- Answer: (a) both have a secondary alcohol, alkene and alkyl side chain; the proposed structure has a different ring system and no C₅=C₆ trisubstituted alkene in the same place. (b) Both have an ether (prenyl aryl ether, an alkene), an aryl methyl ether and an amide; the proposed one is a cyclic ether + secondary lactam (8-membered), the correct one a tertiary γ -lactam + primary alcohol (same formula C₁₇H₂₃NO₄? [low]).

2.72 (book p107, PDF p15) — footnote 6 → Chem. Rev. 1993, 93, 381-433

- Triarylmethane dyes: generic Ar₂C=C₆H₄=N⁺R₂ Cl⁻ with one ring bearing NR₂, one para-X. Basic green 4 (malachite green: X = H, R = CH₃); Basic red 9 (pararosaniline: X = NH₂, R = H); Basic violet 4 (ethyl violet: X = N(C₂H₅)₂, R = C₂H₅).

- Malachite green SMILES CN(C)c1ccc(cc1)C(=C1C=CC(C=C1)=[N+](C)C)c1ccccc1.[Cl-] [high].
- Questions: (a) resonance structures showing delocalization of the + charge for basic green 4; (b) which has greater resonance stabilization, basic green 4 or basic violet 4?
- Answers: (a) + on each of the two N's and on the central C, plus ring ortho/para carbons; (b) basic violet 4: its third ring carries an NMe₂ lone pair that can delocalize the charge over 3 N atoms. [high]

2.73 (book p107, PDF p15; IR problem) — footnote 7 → J. Chem. Ed. 2006, 83, 1534-1536

- Vinyl acetate **1** (C=CC(=O)O) → poly(vinyl acetate) **2**; partial hydrolysis → **3** (some OAc remaining); NaOH/H₂O full hydrolysis → poly(vinyl alcohol) **4**.
- Question: how to use IR to confirm **3** → **4** (complete hydrolysis of acetate).
- Answer: the ester C=O stretch (~1740 cm⁻¹) and C-O ester stretches (~1240 cm⁻¹) disappear completely; the broad O-H (~3200-3500 cm⁻¹) remains/grows. [high]

2.74 (book p107, PDF p15; IR problem) — footnote 8 → J. Chem. Ed. 2006, 83, 287-289

- Knoevenagel/lactonization: 4-(diethylamino)salicylaldehyde **1** CCN(CC)c1ccc(C=O)c(O)c1 + ethyl acetoacetate CCOC(=O)CC(C)=O, piperidine → 3-acetyl-7-(diethylamino)coumarin **2** CCN(CC)c1ccc2cc(C(C)=O)c(=O)oc2c1 [high].
- Question: give at least three IR observations that confirm **1** → **2**.
- Answer: phenol O-H broad band disappears; aldehyde C-H stretches (~2720/2820 cm⁻¹) disappear; aldehyde C=O (~1660-1700, conjugated/H-bonded) replaced by lactone C=O (~1720-1740) plus ketone C=O (~1680); the ester ethyl C-H/C-O pattern of acetoacetate is absent. [high]

2.75 (도전 문제; book p107, PDF p15) — footnote 9 → Soft Matter 2012, 8, 5486-5492

- Gelator intermediate: N-(3,4,5-tris(dodecyloxy)benzyl)-4-formylbenzamide. SMILES CCCCCCCCCCCCOc1cc(CNC(=O)c2ccc(C=O)cc2)cc(OC12CCCCCCCCCCC1)OCCCCCCCCCCC [high].
- Questions: (a) identify the functional groups; (b) of the two C-N bonds, one rotates freely and one is restricted; identify the restricted one and explain with resonance.
- Answers: (a) three ethers (aryl alkyl), a secondary amide, an aldehyde, two aromatic rings; (b) the amide C(=O)-N bond has partial double-bond character (N lone pair delocalized onto O); the benzylic CH₂-N bond rotates freely. [high]

2.76 (book p108, PDF p16) — footnote 10 → Langmuir 2012, 28, 14567-14572

- Imine formation RCHO + R'NH₂ → RCH=NR'. Reactants: **A** = 3,5-bis(10-(pyridinium-1-yl)decyloxy)benzaldehyde (dication); **B** = 1,5-diaminopentane (cadaverine) NCCCCN; **C** = p-xylylenediamine NCc1ccc(CN)cc1; **D** = 3,4-bis(10-pyridiniumdecyloxy)benzaldehyde.
- A SMILES O=Cc1cc(OC12CCCCCCCCC[n+]2CCCCC2)cc(OC12CCCCCCCCC[n+]2CCCCC2)c1 [high]; D the analogous 3,4-isomer.
- Questions: (a) draw the products: i. A + C (2:1); ii. C + D (1:2); iii. A + B (2:1); iv. B + D (1:2); (b) the relationship between the products of iii and iv.
- Answers: (a) bis-imines ArCH=N-linker-N=CHAr. (b) iii and iv are constitutional isomers (3,5- vs 3,4-substitution; same formula). [high]

2.77 (book p108, PDF p16) — footnote 11 → J. Am. Chem. Soc. 2009, 131, 11478-11484

- Amino-acid-derived bolaamphiphile gelator: Pro-Val-NH-(CH₂)₈-NH-Val-Pro (L-prolyl-L-valyl amide on both ends of a 1,8-diaminooctane), drawn with stereo wedges.

- SMILES CC(C)[C@H](NC(=O)[C@H]1CCCN1)C(=O)NCCCCCCCNC(=O)[C@H](NC(=O)[C@H]1CCCN1)C(C)C [med; spacer counted as 8 CH2 from the zigzag; L-configurations assumed from the wedges].
- Questions: (a) all six N-H's can H-bond, but amide N-H's are stronger donors than amine N-H's; explain with resonance; (b) draw the intermolecular H-bonds in a stack of 3 molecules: 8 H-bonds from the central molecule's amide H's/O's to the two neighbors.
- Answer (a): resonance puts $\delta+$ on the amide N ($N^+=C-O^-$), making N-H more polarized/acidic. [high]

2.78 (book p108, PDF p16) — footnote 12 → J. Org. Chem., 1997, 62, 3200-3207

- Tropolone **1** (2-hydroxycyclohepta-2,4,6-trien-1-one, O=C1C=CC=CC=C1O, Kekulé O=C1C=CC=CC=C1O) is acidic (→ tropolonate anion **2**) and basic (→ cation **3**, 1,2-dihydroxytropylium).
- Questions: (a) resonance structures of **2** and **3**, and why they are stable; (b) C-C bond lengths alternate in **1** but are similar in **2** and **3**; explain; (c) the intramolecular H-bond in **3** is much weaker than in **1**; explain.
- Answers: (a) **2**: the - charge is delocalized over both O's and the ring; **3**: the + is delocalized onto ring carbons, giving an aromatic tropylium (6π) character; (b) delocalization equalizes the bonds (aromatic tropylium/tropolonate character); (c) in **3** both O's are OH donors (no good acceptor lone pair/C=O). The O's bear δ+, making them poor acceptors. [med-high]

2.79 (book p108, PDF p16) — footnote 13 → Chem. Rev. 1993, 93, 381-433

- Basic red 1 (rhodamine 6G): xanthene with 3-NHEt/6= N^+HEt , 2,7-dimethyl, 9-(2-ethoxycarbonylphenyl); Cl^- . The text refers to "문제 2.74" dyes, actually the triarylmethane dyes of 2.72.
- SMILES CCNc1cc2oc3cc(=[NH+](CC)c(C)cc3c(-c3ccccc3C(=O)OCC)c2cc1C.[Cl-] [high].
- Question: one of the four rings does not participate effectively in resonance (little positive charge); identify it and explain.
- Answer: the pendant 2-(ethoxycarbonyl)phenyl ring. It is twisted roughly perpendicular to the xanthene plane by steric clash (ortho ester + xanthene H's), so its p orbitals cannot overlap; also, putting + on the ring next to the electron-withdrawing ester is unfavorable. [high]

2.80 (book p108, PDF p16) — footnote 14 → J. Invest. Dermatol. 2014, 134, 2988-2990

- Tofacitinib (JAK inhibitor; approved for psoriasis/RA; promotes hair regrowth). SMILES CC1CCN(C(=O)CC#N)CC1N(C)c1ncnc2[nH]ccc12 (drug is (3R,4R)) [high].
- Task: identify every carbon and fill in the missing H count.
- Answer: C16H20N6O; H per carbon: CH3 (ring Me) 3, C4-H 1, C5 H2, C6 H2, C2 H2, C3-H 1, N-CH3 3, C=O 0, CH2CN 2, CN 0, pyrimidine C2-H 1, C4 0, C4a 0, C7a 0, pyrrole C5-H 1, C6-H 1; NH 1. [high]

Chapter 3 — 산과 염기 (Acids and Bases) (PDF p17-24, book p139, 141-147; book p140 is NOT in the PDF)

Key concepts (개념과 용어의 복습 §3.1-3.9 + skill builders)

- §3.1 Brønsted-Lowry acid = proton donor, base = proton acceptor; gives conjugate acid + conjugate base.

- §3.2 Curved arrows show mechanism; proton transfer always uses at least two curved arrows.
- §3.3 Equilibria are described by K_a (pK_a) rather than K_{eq} ; typical pK_a -10 to 50; strong acid = low pK_a . The equilibrium favors the weaker acid (higher pK_a).
- §3.4 Qualitative acidity from conjugate-base stability: ARIO, i.e. (1) Atom (electronegativity within a row, size within a group), (2) Resonance, (3) Induction (e.g. halogens), (4) Orbital ($sp > sp^2 > sp^3$ carbanion stability). ARIO is the usual priority, with exceptions.
- §3.5 The equilibrium favors the more stable negative charge.
- §3.6 Leveling effect: in basic solution no base stronger than the solvent's conjugate base; in acidic solution no acid stronger than the solvent's conjugate acid.
- §3.7 Solvent effects (e.g. large/bulky bases are less effectively solvated) explain some pK_a differences.
- §3.8 Counter-ions (cations) accompany negatively charged bases; their choice rarely matters in this book.
- §3.9 Lewis acid = electron-pair acceptor; Lewis base = electron-pair donor.
- Skill builders 3.5-3.12: ranking conjugate-base stability by atom/resonance/induction/orbital; combining all four (ARIO exceptions); predicting equilibrium without pK_a ; choosing reagents for proton transfer (water is not a suitable proton source for carboxylate); identifying Lewis acids/bases ($BH_3 \cdot THF$).
- In-chapter problem 3.33 (book p139; 학습능력 응용): Nazarov cyclization, the dienone 1 (2-(3,4-dihydro-2H-pyran-6-yl) enone with α -Me, β -Ph) + $AlCl_3 \rightarrow$ cyclopentenone 2; draw resonance of the carbonyl- $AlCl_3$ Lewis complex. **No footnote marker** visible.

Literature footnotes, chapter 3 (참고문헌, book p147)

1. Org. Lett. 2003, 5, 4931-4934.
2. Tetrahedron Lett. 2012, 53, 5749-5752.
3. J. Am. Chem. Soc. 2009, 131, 8720-8721.
4. J. Am. Chem. Soc. 1984, 106, 4186-4188.
5. J. Chem. Ed. 1981, 58, 79-80.
6. Tetrahedron Lett. 2007, 48, 8787-8789.
7. J. Org. Chem. 2009, 74, 2433-2437.
8. Org. Lett. 2002, 4, 2645-2648.
9. J. Org. Chem. 1981, 46, 4327-4331.
10. Tetrahedron Lett. 2012, 53, 4051-4053.
11. J. Am. Chem. Soc. 2009, 131, 8720-8721. (identical to ref 3)

Footnote 1 marker not found on the available pages. I checked 3.33, 3.55-3.61 and 3.60 (amphotericin B); it probably sits on the missing book p140 or in the chapter body.

Possible book misassignment (my inference, not printed): refs 3 and 4 seem swapped relative to their topics. JACS 1984, 106, 4186 is Schreiber's asteltoxin synthesis (fits 3.63), while JACS 2009, 131, 8720 is the genetic-encoding paper on alkyne-lysine labeling (fits 3.64/3.72, which also cites 11 = the same paper). As printed: 3.63 $\rightarrow$ 3, 3.64 $\rightarrow$ 4.

3.62 (book p145, PDF p22) — footnote 2 $\rightarrow$ Tetrahedron Lett. 2012, 53, 5749-5752

- Total synthesis of (-)-seimatopolide A (antidiabetic). Acid: an oxonium ion, i.e. a secondary alkyl O bearing both an H^+ and a MOM group ($R-O^+(H)-CH_2OCH_3$), on a chain bearing OTBS (alkyl chain drawn

as "CH₃CH₂(CH₂)₇"), a second OMOM, and a terminal vinyl. Base: Hünig's base (N,N-diisopropylethylamine, CCN(C(C)C)C(C)C).

- Substrate SMILES (stereo omitted; chain length per drawing) CCCCCCCCC(O[Si](C)(C)C(C)(C)C)CC(OCOC)C(C=C)[O+][COC] [med].
- Questions: (a) identify the acid and base, draw the products with mechanism; (b) predict the equilibrium position using the pK_a table.
- Answers: the base N removes the O-H proton → neutral MOM ether + iPr₂EtNH⁺. Oxonium pK_a ≈ -2 to -3.5 vs trialkylammonium ≈ 10-11, so the equilibrium lies far to the products. [high] (This is the last step of MOMCl/DIPEA protection.)

3.63 (book p145, PDF p22) — footnote 3 → J. Am. Chem. Soc. 2009, 131, 8720-8721 (as printed; see swap note)

- Asteltoxin (from *Aspergillus stellatus*; inhibitor of E. coli BF1-ATPase); S. L. Schreiber synthesis. **1** = 4-methoxy-5,6-dimethyl-2H-pyran-2-one COc1cc(=O)oc(C)c1C [high]; strong base → anion **2** (deprotonated 6-CH₃ → CH₂⁻). Asteltoxin is shown: a bis-THF (furofuran) core linked by a triene to the 4-methoxy-5-methyl-2-pyrone.
- Question: draw all resonance structures of anion **2** and identify the major contributor.
- Answer: four contributors, with - on: exocyclic CH₂; ring C5 (methyl-bearing carbon); C3 (α to C=O); carbonyl O (enolate). Major contributor: - on the carbonyl oxygen (most electronegative atom). [high]

3.64 (book p146, PDF p23) — footnote 4 → J. Am. Chem. Soc. 1984, 106, 4186-4188 (as printed; see swap note)

- Compound designed for site-specific protein labeling: Nε-(propargyloxycarbonyl)-L-lysine. SMILES N[C@@H](CCCCNC(=O)OCC#C)C(=O)O [high; L from the hashed CO₂H].
- Question: identify the four most acidic protons and rank them by increasing acidity.
- Answer: amine N-H (pK_a ~38) < terminal alkyne C-H (~25) < carbamate N-H (~17-20, amide-like) < CO₂H (~4-5). [med: the carbamate NH vs alkyne CH order depends on the book's table; a carbamate is amide-like (~17), so it is more acidic than the alkyne]

3.65 (IR problem; book p146, PDF p23) — footnote 5 (on part b) → J. Chem. Ed. 1981, 58, 79-80

- Deuterium labeling: isobutane **1** → (Br₂, hv) tert-butyl bromide **2** → (Mg, Et₂O) t-BuMgBr **3** → (D₂O) 2-deuterio-2-methylpropane **4**. SMILES: CC(C)C, CC(C)(C)Br, CC(C)(C)[Mg]Br, [2H]C(C)(C)C [high].
- Questions: (a) redraw **3** as the ion pair t-Bu⁻ BrMg⁺ and show the mechanism **3** → **4** (carbanion deprotonates D₂O → C-D + DO⁻); (b) IR of **4**: signals at 1250-1500, 2180, and 2800-3000 cm⁻¹; locate the C-D stretch and justify.
- Answer: C-D at 2180 cm⁻¹. $\nu \propto \sqrt{k/\mu}$ and D roughly doubles the reduced mass, so $\nu(\text{C-D}) \approx \nu(\text{C-H})/\sqrt{2} \approx 2100\text{-}2200\text{ cm}^{-1}$. [high]

3.66 (IR problem; book p146, PDF p23) — footnote 6 → Tetrahedron Lett. 2007, 48, 8787-8789

- Bengamide synthesis (methionine aminopeptidase inhibitors). **1** = a 1,2-di-O-benzoyl polyol: BzO-CH₂-CH(OBz)-CH(OH)-CH(OTMS)-CH(OMe)-CH=CH-Ph (E-styryl). **2** = TMSCl; intermediate **3** = O-silyl oxonium (R-O⁺(H)SiMe₃); base **4** = 2,6-lutidine; product **5** = TMS ether.
- SMILES **1** (stereo omitted) O=C(OC(OC(=O)C1CCCC1)C(O)C(O[Si](C)(C)C)C(OC)/C=C/C1CCCC1)C1CCCC1 [med].

- Questions: (a) draw 5 and the mechanism from 3 (lutidine removes O-H⁺); (b) use pKa to judge whether 4 is a suitable base (oxonium ~ -2 to -3 vs 2,6-lutidinium ~6.7, so yes, the equilibrium strongly favors 5; lutidine is hindered and non-nucleophilic); (c) IR check 1 → 5: the broad O-H stretch (3200-3600 cm⁻¹) disappears. [high]

3.67 (도전 문제; book p146, PDF p23) — footnote 7 → J. Org. Chem. 2009, 74, 2433-2437

- (+)-Coronafacic acid (core of the phytotoxin coronatine) synthesis: a bicyclo[3.1.0]hexan-2-one-1-carboxylic acid (-)-menthyl ester + 2,2-dimethylpropane-1,3-diol, p-TsOH (cat.), -H₂O → 5,5-dimethyl-1,3-dioxane ketal (spiro), 89%.
- SMILES SM O=C1CCC2CC12C(=O)OC1CC(C)CCC1C(C)C; product CC1(C)COC2(CCC3CC23C(=O)OC2CC(C)CCC2C(C)C)OC1 [med; stereo omitted].
- Questions: (a) curved-arrow mechanism of the first step: protonation of the ketone C=O by TsOH; (b) does the equilibrium favor the protonated ketone? Support with pKa.
- Answer: no. Protonated ketone pKa ≈ -7 vs TsOH ≈ -2.8 (or -1), so the equilibrium favors the neutral ketone + TsOH; only a small fraction is protonated (enough for catalysis). [high]

3.68 (book p146, PDF p23) — footnote 8 → Org. Lett. 2002, 4, 2645-2648

- Phakellin (3, marine antibiotic natural product) synthesis. 1 = cyclo(L-Pro-L-Pro) diketopiperazine, O=C1[C@@H]2CCCN2C(=O)[C@@H]2CCCN12 [high; cis H's wedged]; LDA → 2 (enolate) → ... → 3 (pyrrole-fused tetracyclic 2-aminoimidazoline; SMILES uncertain).
- Questions: (a) the most acidic proton in 1 and its conjugate base 2 (α-C-H at the Pro α-carbon, next to the amide C=O; the enolate is resonance-delocalized onto O); (b) why LDA is suitable: quantitatively, diisopropylamine pKa ~36 vs α-H of an amide ~25-30, so it is favorable; qualitatively, it is bulky and non-nucleophilic and does not attack C=O; (c) mechanism 1 → 2. [high]

3.69 (book p146-147) — no footnote (E. J. Corey ginkgolide B synthesis). Compound 1 is a spiro THF/cyclopentanone with tBu; LDA → enolate 3a↔3b; PhNTf₂ (4) → enol triflate 5 + PhN(Tf)⁻Li⁺ (6). Listed for completeness.

3.70 (book p147, PDF p24) — footnote 9 → J. Org. Chem. 1981, 46, 4327-4331 (Bordwell DMSO acidities)

- Data (most acidic CH₂, DMSO pKa): CH₃COCH₂CO₂CH₂CH₃ (1) 14.1; PhCH₂CN (2) 21.9; PhCH₂CO₂CH₂CH₃ (3) 22.7; PhCH₂CON(CH₃)₂ (4) 26.6.
- Question: compare conjugate bases of 5 diphenylmethane (c1ccc(cc2ccccc2)cc1) and 6 phenylacetone (CC(=O)Cc1ccccc1); decide which is more acidic with all resonance contributors, and which contributor is more effective.
- Answer: 6 is much more acidic (DMSO pKa ≈ 19.9 vs 32.2). The enolate places - on oxygen (electronegative), which stabilizes it more than delocalization into a phenyl ring (- only on ring carbons). From the data, carbonyl/nitrile stabilization > phenyl. [high]

3.71 (book p147, PDF p24) — footnote 10 → Tetrahedron Lett. 2012, 53, 4051-4053

- 7-Desmethoxyfusarentin synthesis (cytotoxic to MCF-7). Imidazole (c1c[nH]cn1) + oxonium of a 2,6-disubstituted tetrahydropyran-4-ol carrying O⁺(H)-TBS at C4 (2-propyl, 6-CH₂OR).
- SMILES (stereo omitted) CCCC1CC(CC(COR)O1)[OH+][Si](C)(C)C(C)(C)C [med].

- Question: predict products and mechanism. Answer: imidazole's sp² N deprotonates the O-H⁺ → TBS ether + imidazolium. Imidazolium pK_a ~7 vs oxonium ~ -2, so the products are strongly favored. [high]

3.72 (book p147, PDF p24) — footnote 11 → J. Am. Chem. Soc. 2009, 131, 8720-8721

- Same lysine derivative as 3.64 (N[C@@H](CCCCNC(=O)OCC#C)C(=O)O). Predict the products of 1 mol of compound with: (a-d) 1, 2, 3, 4 mol EtNa; (e-h) 1, 2, 3, 4 mol EtONa.
- Answer (EtNa: conjugate acid ethane pK_a ~50, so it deprotonates all four sites in order of acidity): 1 eq → carboxylate; 2 eq → + carbamate N⁻; 3 eq → + acetylide; 4 eq → + amide (NH⁻) on the α-amine. EtONa (EtOH pK_a ~16): only the CO₂H is deprotonated irreversibly; more EtONa gives at most partial carbamate N-H deprotonation (pK_a ~17, an equilibrium); the alkyne and amine are not deprotonated. [med; order of steps 2/3 as in 3.64]

3.73 (book p147) — no footnote (piperidine alkaloid from Streptomyces; protonation at N, pK_a data -2.2 vs 10.6 given). Listed for completeness.

Chapter 4 — 알케인과 사이클로알케인 (Alkanes and Cycloalkanes) (PDF p25-33, book p188-196)

Key concepts (개념과 용어의 복습 §4.1-4.15 + skill builders 4.1-4.13)

- §4.1 Alkanes = saturated hydrocarbons; nomenclature.
- §4.2 IUPAC naming in four steps (parent incl. cycloalkanes; substituents, alkyl and complex/branched, with allowed common names; numbering/locants; alphabetical assembly with di/tri... ignored). Bicyclic naming: "bicyclo[x.y.z]" with bridge lengths in brackets; numbering starts at a bridgehead and follows the longest bridge.
- §4.3 Number of constitutional isomers grows with size; use IUPAC names to avoid duplicates.
- §4.4 Heat of combustion (-ΔH°) compares the stability of isomeric alkanes (branched = more stable).
- §4.5 Petroleum: fractional distillation, cracking, reforming (octane rating).
- §4.6-4.8 Conformations and Newman projections; dihedral/torsional angle; staggered (low) vs eclipsed (high); torsional strain (ethane ~12 kJ/mol; propane larger); butane anti < gauche (gauche steric interaction).
- §4.9 Cycloalkane angle strain + torsional strain; heat of combustion per CH₂.
- §4.10 Cyclohexane chair (no torsional or angle strain); boat (flagpole + eclipsing), twist boat.
- §4.11-4.13 Axial/equatorial; ring flip; 1,3-diaxial interactions favor equatorial substituents; for polysubstituted rings, sum the axial energy costs (e.g. Et 8.0, Me 7.6, Cl 2.0 kJ/mol in skill builder 4.13).
- §4.14 cis/trans (clearest in Haworth projections).
- §4.15 cis- vs trans-decalin are stereoisomers (no ring-flip interconversion); norbornane = bicyclo[2.2.1]heptane; camphor, camphene; diamond lattice of chairs.

Literature footnotes, chapter 4 (참고문헌, book p196)

1. J. Am. Chem. Soc. 2009, 131, 8766-8768.
2. J. Am. Chem. Soc. 2012, 134, 8078-8081.

3. J. Am. Chem. Soc. 2009, 131, 8766-8768.
4. J. Phys. Chem. A 2002, 106, 1642-1646.
5. J. Org. Chem. 2012, 77, 7479-7486.
6. J. Am. Chem. Soc. 1990, 112, 893-894.
7. J. Am. Chem. Soc. 2009, 131, 8766-8768.
8. J. Am. Chem. Soc. 1961, 83, 606-614.
9. Bioorg. Med. Chem. Lett. 2012, 22, 1756-1760. (Refs 1, 3 and 7 are the same paper, as printed.)

4.67 (종합 문제; book p194-195, PDF p32) — footnote 1 (on "titin") → J. Am. Chem. Soc. 2009, 131, 8766-8768

- A structural unit used in synthetic polymers that mimic the backbone of the muscle protein titin: a 2-ureido-4(1H)-pyrimidinone (UPy) unit, shown as an N-C(=O)-NH-C2 urea attached to a pyrimidinone (ring N1-H, C4=O, ring N3), with ring substituents (wavy bonds) at C5 and C6. Core SMILES O=C(N*)Nc1nc(=O)c(*)c(*)[nH]1 [med].
- Task: in its most stable conformation, identical units form one intramolecular H-bond and four intermolecular H-bonds; draw two such units showing all interactions.
- Answer: intramolecular NH(urea, terminal N-H)⋯N3 (ring) H-bond (6-membered pseudo-ring) makes the unit planar and presents a DDAA array (ring N1-H, inner urea N-H, urea C=O⋯ / C4=O, N3). Two units dimerize through a self-complementary quadruple H-bond (Meijer UPy dimer). [med-high]

4.68 (book p195, PDF p32) — footnote 2 (on "Aspergillus nomius") → J. Am. Chem. Soc. 2012, 134, 8078-8081

- Aspernomine (polycyclic cytotoxic indole diterpenoid from *Aspergillus nomius*); a proposed biosynthetic route. Drawn: an indoline/aniline N-H fused carbocyclic skeleton with a cyclohexanone, two angular methyls (Me, Me) on adjacent ring-fusion carbons, a secondary OH, and a homoprenyl side chain CH₂CH₂CH=C(CH₃)₂ at a ring-fusion carbon. SMILES: uncertain (complex polycycle; not determined).
- Questions: (a) two six-membered rings are drawn as chairs; is their junction like cis- or trans-decalin? (b) Is the aromatic ring axial or equatorial? (c) For the 6-carbon acyclic substituent on a bridgehead carbon, is it axial or equatorial with respect to each chair? (d) The approximate dihedral angle between the two methyls on the chairs.
- Tentative answers [low; depend on 3D reading]: (a) trans-decalin-like (the angular groups anti); (d) ~60° (gauche, adjacent ring-fusion methyls in a trans-diaxial-avoiding arrangement), or 180° if both are axial on a trans junction. **I could not verify from the 2D drawing; consult the original.**

4.69 (도전 문제; book p195, PDF p32) — footnote 3 (on "수소 결합 상호작용") → J. Am. Chem. Soc. 2009, 131, 8766-8768

- Butane-2,3-diol, drawn as a sawhorse/zigzag along C2-C3. My CIP assignment from the wedges is (2R,3R): C2 has H hashed, CH₃ and OH in plane; C3 has HO wedged, H hashed, CH₃ in plane. SMILES C[C@H](O)[C@H](O)O = (R,R) [med for the stereo reading].
- Task: draw the three staggered conformers about C2-C3 (Newman); considering gauche interactions and hydrogen bonding, decide the most stable.
- Answer: the conformer with CH₃/CH₃ anti and OH/OH gauche (allows an intramolecular O-H⋯O hydrogen bond) is most stable. [med-high]

4.70 (book p195, PDF p32) — footnote 4 → J. Phys. Chem. A 2002, 106, 1642-1646

- Methanol rotation about C-O: staggered vs eclipsed; barrier ≈ 4.2 kJ/mol (1 kcal/mol).
- Questions: (a) Newman projections of both; (b) the energy cost of a lone-pair/C-H eclipsing interaction.
- Answer: the eclipsed form has one H/H eclipse (~ 4.0 kJ/mol, from ethane 12/3) + two lone-pair/C-H eclipses. $4.2 - 4.0 = 0.2$ kJ/mol, so ~ 0.1 kJ/mol each (essentially negligible). [high]

4.71 (book p195-196, PDF p32-33) — footnote 5 → J. Org. Chem. 2012, 77, 7479-7486

- Fluorophore: 1,4-bis(1-R-1H-1,2,3-triazol-4-yl)phenazine (R = polyether). Planar conformers A and B differ by 180° rotation of each triazole about its aryl-triazole C-C bond. SMILES (R = Me placeholder) Cn1cc(-c2ccc(-c3cn(C)nn3)c3nc4ccccc4nc23)nn1 [high].
- Questions: (a) which σ bonds must rotate for A $\rightarrow$ B, and what happens to the geometry during rotation (the triazoles twist out of the phenazine plane, temporarily breaking π -conjugation); (b) A is 8.96 kcal/mol more stable than B; explain.
- Answer (b): in A, the triazole C5-H points at the phenazine N (stabilizing C-H $\cdots$ N interaction) and the triazole N lone pairs point away. In B, the triazole N2/N3 lone pairs face the phenazine N lone pairs (electrostatic lone-pair repulsion), and the R groups/H interact sterically. [med]

4.72 (book p196, PDF p33) — footnote 6 (on "모두 수평방향 형태") → J. Am. Chem. Soc. 1990, 112, 893-894

- all-trans-1,2,3,4,5,6-hexaethylcyclohexane **1** adopts the all-equatorial chair; all-trans-1,2,3,4,5,6-hexaisopropylcyclohexane **2** has a very unstable all-equatorial form and prefers all-axial.
- SMILES **1** (stereo omitted) CCCC1(CC)C(CC)C(CC)C(CC)C1CC; **2** CC(C)C1C(C(C)C)C(C(C)C)C(C(C)C)C(C(C)C)C1C(C)C [high constitution].
- Questions: (a) why adjacent equatorial iPr groups clash severely while Et groups don't; draw the all-equatorial chair of **2** and a Newman projection down the ring-CH(CH₃)₂ bond; (b) why the all-axial chair of **2** is lower in energy (Newman down the C(ring)-iPr bond).
- Answers: (a) Et can rotate its CH₃ away so that only H's face the neighbor. iPr has two CH₃'s, so in any rotamer one methyl is gauche to/eclipses the adjacent equatorial iPr (gear clash; each substituent is flanked by two neighbors). (b) In the all-axial form, alternating up/down axial iPr groups are 1,3-related only to other iPr groups. Each iPr can orient its methine H toward the ring interior/axial neighbors and its methyls outward, avoiding the vicinal gauche clashes, so it is less strained overall. [med-high]

4.73 (book p196, PDF p33) — footnote 7 (on "titin") → J. Am. Chem. Soc. 2009, 131, 8766-8768

- Same ureidopyrimidinone (UPy) concept as 4.67, drawn as R/R'-substituted 2-(3-R''-ureido)pyrimidin-4(1H)-one (ring N1-H, C4=O, N3; urea NH-C(=O)-NHR''). SMILES O=C(NR'')Nc1nc(=O)c(R)c(R')[nH]1 [med].
- Task and answer as in 4.67: one intramolecular H-bond (terminal urea N-H $\cdots$ N3) plus four intermolecular H-bonds in the dimer (quadruple DDAA $\cdot$ AADD). [med-high]

4.74 (book p196, PDF p33) — footnote 8 → J. Am. Chem. Soc. 1961, 83, 606-614

- Two diastereomeric γ -lactones fused to a trans-decalin (linear tricyclic: decalin + γ -butyrolactone on the 2,3-positions of ring B). Fusion H's: top wedge, bottom hash (trans-decalin) in both. Compound **1**: ring-B C-CH₂ bond hashed, C-O wedge. Compound **2**: C-CH₂ wedge, C-O hashed (each lactone itself trans-fused). The heats of combustion differ by 17.2 kJ/mol.

- SMILES (constitution) O=C1CC2CC3CCCC3CC2O1 [high]; stereo omitted.
- Questions: (a) redraw 1 and 2 with chairs and their lowest-energy conformers; (b) which has the larger heat of combustion, and why?
- Answer (my configurational analysis): in the rigid trans-decalin, 1 has both lactone ring bonds equatorial (a trans-diequatorial 1,2-fusion, strain-free chair). In 2 the lactone bonds would be trans-diaxial (180° apart), which a 5-membered ring cannot span. Ring B cannot flip (trans-decalin), so it must adopt a twist-boat. **2 has the larger heat of combustion** (17.2 kJ/mol less stable). [med-high]

4.75 (book p196, PDF p33) — footnote 9 (on "비타민 D의 활성") → Bioorg. Med. Chem. Lett. 2012, 22, 1756-1760

- Vitamin D mimics: R-O-CH₂-(bicyclo[2.2.2]octan-1-yl)-(CH₂)₄-C(OH)(CH₂CH₃)₂. The highlighted hydrocarbon core = bicyclo[2.2.2]octane + CH₂ (methyl) + the 5-ethylheptyl chain (OH and OR excluded).
- SMILES of the whole (R = H) OCC12CCC(CCCC(O)(CC)CC)(CC1)CC2 [med; chain counted as 4 CH₂ then C(OH)].
- Task: give the systematic name of the hydrocarbon core.
- Answer: **1-(5-ethylheptyl)-4-methylbicyclo[2.2.2]octane** [med; depends on the chain count].

(4.36-4.66 have no literature footnotes.)

Chapter 5 — 입체이성질현상 (Stereoisomerism) (PDF p34-41, book p230-237)

Key concepts (개념과 용어의 복습 §5.1-5.10 + skill builders 5.1-5.9)

- §5.1 Constitutional isomers vs stereoisomers; cis/trans for disubstituted cycloalkanes and alkenes.
- §5.2 Chirality = non-superimposable mirror image; chirality center (four different groups); one center → enantiomers.
- §5.3 CIP R/S: prioritize by atomic number, lowest priority on a dash, 1→2→3 clockwise = R.
- §5.4 Polarimetry; optically active; specific rotation [α] = $\alpha/(c \cdot l)$; dextrorotatory (+)/levorotatory (-); racemic mixture; enantiomeric excess.
- §5.5 Diastereomers; at most 2ⁿ stereoisomers.
- §5.6 Rotational vs reflectional symmetry; a plane of symmetry → achiral; meso compounds (fewer than 2ⁿ stereoisomers).
- §5.7 Fischer projections (horizontal = wedges, vertical = dashes).
- §5.8 Conformationally mobile compounds (butane, cis-1,2-dimethylcyclohexane) are optically inactive (rapid interconversion of enantiomeric conformers).
- §5.9 A chirality center is not required for chirality: atropisomers (hindered rotation), allenes.
- §5.10 Resolution with chiral resolving agents and chiral column chromatography.
- Skill builders: cis/trans; locating chirality centers; drawing enantiomers; CIP assignment; specific rotation; % ee; enantiomer vs diastereomer; meso; Fischer → R/S.

Literature footnotes, chapter 5 (참고문헌, book p237)

1. J. Am. Chem. Soc. 2012, 134, 13554-13557.

2. Org. Lett. 2012, 14, 3878-3881.
3. J. Am. Chem. Soc. 2012, 134, 8054-8057.
4. J. Am. Chem. Soc. 2011, 133, 14892-14895.
5. Langmuir 2012, 28, 14039-14044.
6. J. Org. Chem. 2001, 66, 2072-2077.
7. Org. Lett. 2012, 14, 934-937.
8. J. Org. Chem. 2012, 77, 7777-7792.
9. Chem. Eng. News 2015, 93, 7.

5.63 (종합 문제; book p236, PDF p40) — footnote 1 → J. Am. Chem. Soc. 2012, 134, 13554-13557

- A 1,2,4-trioxane with anti-osteosarcoma activity (the text says "골육종", osteosarcoma). Bicyclic: cyclohex-2-en-1-one fused (C4a/C8a) to a 1,2,4-trioxane; C4a bears CH₃ (wedge) and the O-O; C8a bears H (wedge) and the ether O; trioxane C3 (acetal CH) bears 3-methylisoxazol-5-yl (hashed) and H.
- SMILES (constitution) O=C1CC2OC(C3CC(C)OC2(C)C=C1 [high]; stereo [low-med]: my reading gives cis ring fusion with C8a = S, C4a = R, C3 = S, but **verify**.
- Task: assign R/S to each chirality center; draw the possible stereoisomers that have specific relationships (enantiomer, diastereomers) to the drawn structure.
- Answer: 3 chirality centers → up to $2^3 = 8$ stereoisomers (4 enantiomeric pairs); the enantiomer inverts all three; diastereomers invert one or two centers. [high for the counting]

5.64 (book p236, PDF p40) — footnote 2 → Org. Lett. 2012, 14, 3878-3881

- Coibacin B (anti-inflammatory, potent anti-leishmanial natural product): 6-substituted 5,6-dihydro-2H-pyran-2-one (α,β -unsaturated δ -lactone). C6 bears H (hashed) and the chain (wedge): (E)-CH=CH-(E)-CH=CH-CH₂-CH₂-(E)-CH=CH-(trans-2-methylcyclopropyl).
- SMILES (stereo omitted) CC1CC1/C=C/CC/C=C/C=C/C1CC=CC(=O)O1 [med].
- Questions: (a) assign the configuration of each chirality center; (b) with the alkene geometry fixed, how many stereoisomers?
- Answers: (a) lactone C6 = S by my reading [med]; the two cyclopropane carbons are trans-disubstituted (H's drawn trans), with R/S not worked out [low]; (b) 3 chirality centers → $2^3 = 8$. [high]

5.65 (book p236, PDF p40) — footnote 3 → J. Am. Chem. Soc. 2012, 134, 8054-8057

- Redox-switchable chiral copper catalyst: Cu(I) form gives the S enantiomer, Cu(II) form the R enantiomer. Reaction: Michael addition of diethyl malonate (CCOC(=O)CC(=O)OCC) to trans- β -nitrostyrene ([O-][N+](=O)/C=C/c1ccccc1), with catalyst (Cu(I) or Cu(II) form), base and solvent → diethyl 2-(2-nitro-1-phenylethyl)malonate.
- Product (S) SMILES CCOC(=O)C(C(=O)OCC)[C@H](C([N+](=O)[O-])c1ccccc1 [med-high; CIP: CH₂NO₂ > Ph > CH(CO₂Et)₂ > H].
- Data (% ee of (S)-product / % yield): toluene 24/55; THF 48/33; CH₃CN 72/55; CHCl₃ 30/40; CH₂Cl₂ 46/44; hexane 51/30.
- Questions: (a) draw the major products with the Cu(I) and Cu(II) catalysts and state the catalyst/stereochemistry; (b) compute %S and %R for each solvent; which is best?
- Answers: %S/%R = toluene 62/38; THF 74/26; CH₃CN 86/14; CHCl₃ 65/35; CH₂Cl₂ 73/27; hexane 75.5/24.5. Best: **CH₃CN** (highest ee and the joint-highest yield). [high]

5.66 (book p236, PDF p40) — footnote 4 → J. Am. Chem. Soc. 2011, 133, 14892-14895

- Tandem catalytic dehydrogenation + Diels-Alder: hex-5-en-1-yl acetate (CC(=O)OCCCC=C) is dehydrogenated by a special catalyst to a 1,3-diene (hexa-3,5-dienyl acetate), which reacts with N-phenylmaleimide (O=C1C=CC(=O)N1c1ccccc1). The achiral SM gives 4 stereoisomeric products (2 major, 2 minor). Major shown: cis-fused 3a,4,7,7a-tetrahydro-2-phenylisoindole-1,3-dione with 4-(2-acetoxyethyl), drawn with ring-fusion H's hashed and the CH₂CH₂OAc wedge.
- SMILES (constitution) CC(=O)OCCC1C=CCC2C(=O)N(c3ccccc3)C(=O)C12 [high].
- Questions: (a) which is the enantiomer of the shown product; draw the other major product; (b) the two minor products keep cis fusion but differ in the relative configuration of the third chirality center; draw them; (c) the relationship between the two minor products; (d) the relationship between major and minor products.
- Answers: (a) the other major product is the enantiomer of the drawn (endo) adduct (racemic, since no chiral catalyst/SM); (b) the exo adducts (the CH₂CH₂OAc on the opposite face relative to the fusion H's); (c) enantiomers; (d) diastereomers. [high]

5.67 (도전 문제; book p237, PDF p41) — footnote 5 → Langmuir 2012, 28, 14039-14044

- Sugar-derived gelator (self-assembles into fibers in cooling toluene → gel): methyl 4,6-O-(4-aminobenzylidene)- α -D-glucopyranoside. From the drawing: C1-OMe hashed, C2-OH hashed, C3-OH wedge, H4 wedge, H5 hashed; trans-fused 1,3-dioxane; the relative configuration is consistent with α -gluco [med; D assumed].
- SMILES (no stereo) COc1oc2c(Oc3ccc(N)cc3)OC2C(O)c1O [high constitution].
- Task: redraw with bridgehead H's axial and both non-aromatic rings as chairs.
- Answer: a trans-decalin-like fused bis-chair: H4 and H5 axial (anti); the aryl on the acetal C equatorial; C2-OH and C3-OH equatorial; C1-OMe axial (α anomer). [med-high]

5.68 (book p237, PDF p41) — footnote 6 → J. Org. Chem. 2001, 66, 2072-2077

- A rigid polycyclic cage ketone (C=O on a one-carbon bridge over a fused polycyclic cage). Probably D3-trishomocubaneone (pentacyclo[6.3.0.0^{2,6}.0^{3,9}.0^{5,8}]undecan-4-one) [low; identity not confirmed from the drawing]. SMILES: uncertain.
- Questions: (a) rotational symmetry? (b) reflectional symmetry? (c) chiral? If so, draw the enantiomer.
- Expected answers: (a) yes, a C₂ axis through the C=O; (b) no mirror plane; (c) chiral, so draw the mirror image. [med; this matches the problem's intent, which is a chiral molecule with rotational but no reflectional symmetry]

5.69 (book p237, PDF p41) — footnote 7 → Org. Lett. 2012, 14, 934-937

- Meloscine (natural product) 19-step synthesis via a key allene intermediate. Allene C=C=C: one terminus bears 2-bromophenyl and CH₂CH₂N₃ (2-azidoethyl, N=N⁺=N⁻); the other bears CH₂OSi(iPr)₃ (hashed) and an alkenyl/alkyl chain (wedge) ending in a bicyclic orthoester (OBO-type, three O's; exact linker uncertain).
- SMILES: uncertain (side chain/orthoester not resolved); allene core [N-]=[N+]=NCCC(c1ccccc1Br)=C=C(CO[Si](C)(C)C)(C(C)C)C(C)C(C)C...
- Questions: (a) Newman projection looking down the C=C=C from the left (front and back carbons separated by the sp carbon; substituent pairs perpendicular); (b) bond-line drawing (wedge/dash) of the

enantiomer, plus its Newman; (c) the two diastereomers in bond-line and Newman form.

- Concept: axial chirality of 1,3-disubstituted allenes (the planes of the terminal substituents are orthogonal). (c) presupposes a second stereo element in the side chain (not resolvable from my reading). [med]

5.70 (book p237, PDF p41) — footnote 8 → J. Org. Chem. 2012, 77, 7777-7792

- Ten stereoisomeric sugar-derivative (iminosugar) targets, 2,5-dideoxy-2,5-imino-hexitols (2,5-bis(hydroxymethyl)pyrrolidine-3,4-diols, e.g. DMDP-type), are made by multistep synthesis from D-glucuronolactone (as its 1,2-O-isopropylidene-furanurono-6,3-lactone) **1D** or its enantiomer **1L**. Configurations at C2, C3 and C5 can be retained or inverted, but C4 cannot be inverted.
- SMILES: 1D (constitution) CC1(C)OC2OC3C(O)C(=O)OC3C2O1; products OCC1NC(CO)C(O)C1O [high constitution].
- Questions: (a) draw 1L (the mirror image of 1D: invert every center); (b) how many stereoisomers are possible for 1D? (c) draw the 8 chiral products (4 enantiomeric pairs) and 2 achiral (meso) products; (d) only 4 products (incl. the 2 achiral ones) can be made from either 1D or 1L; identify them and explain.
- Answers: (b) 5 chirality centers (C1-C5) → $2^5 = 32$ in principle [med]; (c) 4 centers in a C2-symmetric skeleton → $16 - 6 \text{ duplicates} = 10$ (8 chiral + 2 meso) [high]; (d) [low] the products accessible from both enantiomeric starting materials are those whose fixed C4 configuration can appear as either "C3" or "C4" after renumbering, i.e. the meso pair plus one C2-symmetric pair. **Verify against the source.**

5.71 (book p237, PDF p41) — footnote 9 → Chem. Eng. News 2015, 93, 7

- Levetiracetam, (S)-2-(2-oxopyrrolidin-1-yl)butanamide, anti-epileptic (FDA approval of a 3D-printed fast-dissolving porous tablet). $[\alpha]^{20} = -93.4$. SMILES CC[C@@H](C(N)=O)N1CCCC1=O [high].
- Questions: (a) observed rotation for 500 mg in 10 mL acetone, 10.0 cm cell; (b) levetiracetam is the (S)-enantiomer of racemic etiracetam (memory enhancer); where does the name come from?
- Answers: (a) $\alpha = [\alpha] \cdot c \cdot l = -93.4 \times 0.0500 \text{ g/mL} \times 1.00 \text{ dm} = -4.67^\circ$; (b) "lev-" = levorotatory (–) enantiomer of etiracetam. [high]

(5.29-5.62 have no literature footnotes; 5.29 atorvastatin, 5.30 ixabepilone and 5.31 paclitaxel are drug examples without citations.)

Chapter 6 — 화학반응성과 메커니즘 (Chemical Reactivity and Mechanisms) (PDF p42-52, book p274-284)

Key concepts (개념과 용어의 복습 §6.1-6.12 + skill builders 6.1-6.6)

- §6.1 Enthalpy ΔH (heat of reaction); bond dissociation energy (homolytic cleavage); exothermic vs endothermic.
- §6.2 Entropy; spontaneity requires $\Delta S_{\text{total}} > 0$; $\Delta S_{\text{sys}} > 0$ when the number of molecules or conformational freedom increases.
- §6.3 $\Delta G < 0$ = exergonic (spontaneous); $\Delta G > 0$ = endergonic.
- §6.4 Equilibrium: K_{eq} relates to ΔG ($\Delta G < 0 \rightarrow K_{\text{eq}} > 1$); thermodynamics.
- §6.5 Kinetics: rate equations (1st/2nd/3rd order); activation energy E_a ; temperature and catalysts.
- §6.6 Energy diagrams: transition states (maxima), intermediates (minima); Hammond postulate.

- §6.7 Polar (ionic) reactions: nucleophiles (lone pairs, π bonds, inductive effects), electrophiles (inductive $\delta+$, empty p orbital, carbocations); polarizability.
- §6.8 Curved arrows: four arrow-pushing patterns: (1) nucleophilic attack, (2) loss of a leaving group, (3) proton transfer, (4) rearrangement (hydride/methyl shift). Carbocation stability $3^\circ > 2^\circ > 1^\circ$ (hyperconjugation).
- §6.9 Every ionic mechanism combines these four patterns; concerted steps use two patterns at once.
- §6.10 Arrow rules: tail on a bond or lone pair; head to form a bond or lone pair; never give a 2nd-row element (C, N, O, F) a fifth orbital/expanded octet.
- §6.11 Carbocation rearrangements occur only when they give a more stable cation (none from 3° or resonance-stabilized allylic cations).
- §6.12 Irreversible arrows: poor leaving group + strong nucleophile; proton transfers with a large pKa difference; carbocation rearrangements. Otherwise use equilibrium arrows.
- Skill builders: ΔH° from BDEs (example: isobutane + $\text{Cl}_2 \rightarrow$ tert-butyl chloride + HCl , $\Delta H^\circ = -138 \text{ kJ/mol}$); identifying nucleophilic/electrophilic centers; arrow-pushing patterns and their sequences; drawing curved arrows; predicting carbocation rearrangements.

Literature footnotes, chapter 6 (참고문헌, book p284)

1. J. Am. Chem. Soc. 1990, 112, 2003-2005.
2. J. Am. Chem. Soc. 2012, 134, 17877-17880.
3. J. Org. Chem. 2007, 72, 793-798.
4. Org. Lett. 1999, 1, 3-5.
5. Org. Lett. 2012, 14, 4710-4713.
6. J. Am. Chem. Soc. 1954, 76, 4749-4751.
7. J. Am. Chem. Soc. 1979, 101, 7130-7131.
8. Org. Lett. 1999, 1, 537-539.
9. Org. Lett. 2005, 7, 515-517.
10. Org. Biomol. Chem. 2013, 11, 2498-2513.

Footnote 1 marker not found. 6.52 (Eschenmoser-type fragmentation of an epoxy N-chlorohydrazone, below) shows no superscript at high resolution. Nor do 6.47-6.51. The marker may be on a page outside this excerpt or was omitted in print.

6.52 (book p282) — NO footnote visible. Listed for context only.

α,β -Epoxy ketone N-chlorohydrazone (2,3-epoxy-3-methylcyclohexanone-derived, N-N(H)Cl) + $\text{HO}^- \rightarrow$ vinyl azo (N=N-Cl) alkoxide $\rightarrow$ hept-6-yn-2-one ($\text{CC(=O)CCCC\#C}$) + N_2 + Cl^- . Draw the curved arrows (Eschenmoser-Tanabe-type fragmentation).

6.53 (book p282, PDF p50) — footnote 2 $\rightarrow$ J. Am. Chem. Soc. 2012, 134, 17877-17880

- Base-catalyzed (cat. MeO^- in MeOH) rearrangement of **1** into the propellane isomer **5**. **1** = a tricyclic butenolide: a cyclopentene-fused α,β -unsaturated γ -lactone (butenolide) whose spiro/quaternary γ -carbon carries a $\text{CH}_2\text{CH}_2\text{-O}$ bridge forming a tetrahydrofuran to a bridgehead CH (oxa-bridged).
- Steps shown: **1** + $\text{MeO}^- \rightleftharpoons$ **2** (carbanion at the allylic bridgehead CH γ to the butenolide; extended dienolate; major contributor shown separately) $\rightleftharpoons$ **3** (β -elimination: ring C-O bond breaks $\rightarrow$

cyclopentadiene-fused butenolide + pendant $\text{CH}_2\text{CH}_2\text{O}^- \rightleftharpoons \mathbf{4}$ (intramolecular oxa-Michael addition of the alkoxide to the diene/butenolide, giving a lactone enolate; new THF ring) $\rightleftharpoons$ (MeOH) $\mathbf{5}$ (protonation, regenerating MeO^-).

- Task: curved arrows for each step and the major resonance contributor of $\mathbf{2}$.
- Answers: $1 \rightarrow 2$ proton transfer (MeO^- removes the γ /allylic H, which is acidic because the anion is a vinylogous enolate). The major contributor of $\mathbf{2}$ has the $-$ charge on the lactone carbonyl oxygen (dienolate O^-). $2 \rightarrow 3$: loss of a leaving group (E1cB -type ring opening expelling alkoxide). $3 \rightarrow 4$: nucleophilic attack (conjugate addition). $4 \rightarrow 5$: proton transfer from MeOH, regenerating the catalyst. [med-high]
- SMILES: not determined with confidence (cage connectivity) — **uncertain**.

6.54 (book p282, PDF p50) — footnote 3 $\rightarrow$ J. Org. Chem. 2007, 72, 793-798

- Probe of an intramolecular elimination (stereochemistry with D labels). $\mathbf{1}$ = tert-butyl 3-acetoxy-2,3-dideuteriobutanoate (C3 bears CH_3 , D, OAc; C2 bears H, D, CO_2tBu ; specific stereo drawn with wedges/dashes).
- SMILES (stereo omitted) [2H]C(C)(OC(=O)C([2H])C(=O)OC(C)(C)C [high constitution].
- Proposed path: HO^- adds to the acetate $\text{C}=\text{O} \rightarrow$ tetrahedral alkoxide $\rightleftharpoons$ concerted syn-elimination: the alkoxide O^- removes the α -H through a cyclic TS, $\text{C}=\text{C}$ forms and the C3-O bond breaks, releasing acetic acid (as the hydrate/orthoacid, losing HO^-) and tert-butyl (E)-2,3-dideuteriobut-2-enoate (C/C([2H])=C([2H])/C(=O)OC(C)(C)C, with the D's trans to each other as drawn). Then $\text{HO}^- + \text{AcOH} \rightarrow \text{AcO}^- + \text{H}_2\text{O}$.
- Task: complete the mechanism with curved arrows matching each step's bond changes.
- Answer: as above. The patterns are nucleophilic attack; then concerted proton transfer + loss of the leaving group (intramolecular, syn); then proton transfer. The D-labeling reveals the syn geometry. [med-high]

6.55 (book p282-283, PDF p50-51) — footnote 4 (on "균류", coprophilous fungi) $\rightarrow$ Org. Lett. 1999, 1, 3-5

- Biomimetic synthesis inspired by preussomerins (antifungals from dung-loving fungi). SM: a spiroketal linking a tetrahydronaphthoquinone (enedione ring) to naphthalene-1,8-diol, with the naphthalene bearing a para-OH (4-hydroxy). $\text{LiOH} \rightarrow$ naphthoxide $\rightarrow$ (one step) rearranged bis-spiroketal anion: the former quinone ring becomes an aromatic hydroquinone mono-anion, and the naphthalene ring becomes a 4-oxo-cyclohexadienone whose C1 is a new ketal carbon bonded to the former quinone oxygen. Then H_2O protonates the phenoxide.
- Questions: (a) curved arrows for each step; (b) explain the observed stereochemistry.
- Answers: (a) deprotonation (proton transfer); an intramolecular redox-type step in which the electron-rich naphthoxide pushes electrons ($\text{O}^- \rightarrow \text{C4}=\text{O}$) and the quinone oxygen becomes nucleophilic, forming the new C1-O ketal bond and aromatizing the quinone ring to a hydroquinone anion; then protonation. [low: the single polar arrow set is not obvious; it may be stepwise via SET; consult the paper]. (b) The new O-C1 bond forms intramolecularly on the only face the rigid spiro framework allows, so a single diastereomer (the cis-fused bis-ketal shown with wedges) results. [med]

6.56 (book p283, PDF p51) — footnote 5 $\rightarrow$ Org. Lett. 2012, 14, 4710-4713

- (+)-Aureol biosynthesis (selective anticancer activity against certain lung and colon cancers): carbocation $\mathbf{A} \rightarrow$ carbocation $\mathbf{B}$. $\mathbf{A}$ is a trans-decalin (gem-dimethyl at C4, angular CH_3 at C10) with the cation at C8

(bearing CH₃), adjacent to C₉, which bears CH₂-(2,5-dihydroxyphenyl). B has the cation at C₁₀ (ring fusion); C₉ now carries CH₃ + CH₂Ar; C₈ carries H + CH₃ (stereo per wedges).

- Answer (mechanism): (1) a 1,2-hydride shift from C₉ to C₈ (cation now at C₉, tertiary) then (2) a 1,2-methyl shift of the angular CH₃ from C₁₀ to C₉ → tertiary cation at C₁₀ = B (a "fries"-type backbone rearrangement). Each migrating group moves suprafacially on the face it leaves from: the H and CH₃ are trans-diaxial/antiperiplanar to the empty p orbital. This sets the observed relative configurations (the C₈ methyl and the C₉ methyl/benzyl relationships). [med-high]
- SMILES: uncertain for full stereo; A constitution CC1(C)CCCC2(C)C1CC[C+](C)C2Cc1cc(O)ccc1O [med].

6.57 (book p283, PDF p51) — footnote 6 → J. Am. Chem. Soc. 1954, 76, 4749-4751 (Woodward, total synthesis of strychnine)

- Last stage of Woodward's synthesis: isostrychnine-type **1** (α,β-unsaturated lactam + allylic CH₂OH) with HO⁻ → **2** (deprotonation at the γ-C-H giving an extended lactam dienolate) → H₂O → **3** (reprotonation at the α-carbon, moving the C=C into conjugation in a new position) → HO⁻ → **4** (alkoxide by O-H deprotonation) → **5** (intramolecular oxa-Michael addition forms the ether ring; lactam enolate) → H₂O → **6** (–)-strychnine.
- Strychnine SMILES O=C1C[C@H]2OCC=C3CN4CC[C@]56[C@@H]4C[C@H]3[C@H]2[C@H]5N1c1cccc16 [med; standard literature SMILES].
- Questions: (a) all curved arrows and the arrow-pushing patterns 1→6 (proton transfer, proton transfer, proton transfer, nucleophilic attack (conjugate addition), proton transfer); (b) for 4→5, identify the nucleophilic center (alkoxide O) and the electrophilic center (β-carbon of the α,β-unsaturated lactam), and justify the electrophile's electron deficiency via resonance (C=C-C=O ↔ ⁺C-C=C-O⁻); (c) 4→5 creates a new chirality center; give its configuration and why the other is not observed.
- Answer (c): the configuration is the one in natural strychnine (the ether O on the face set by the cage). The alkoxide can reach only one face of the alkene intramolecularly; the other configuration would require an impossibly strained ring. [med; R/S not assigned]

6.58 (도전 문제; book p283, PDF p51) — footnote 7 (on "Isocoma wrightii") → J. Am. Chem. Soc. 1979, 101, 7130-7131 (Pirrung, isocomene)

- Isocomene (sesquiterpene, triquinane): a carbocation rearrangement that is neither a methyl nor a hydride shift. SM cation: a tertiary cation on a 6-membered ring fused to a cyclobutane (bicyclo[4.2.0]) that is spiro/fused to a cyclopentane, with methyl groups → product: a triquinane tertiary cation (angular triquinane skeleton).
- Questions: (a) identify the migrating carbon and draw the mechanism; (b) a new chirality center forms with only one configuration (diastereoselective because two diastereomeric cations are possible); draw the unformed diastereomer and explain why it is unfavorable (hint: models); (c) the cation goes from tertiary to tertiary; suggest the driving force.
- Answers: (a) a 1,2-alkyl shift: a cyclobutane ring C-C bond (the carbon bonded to the cation-adjacent carbon) migrates to the cation, i.e. a ring expansion of a cyclobutylcarbiny cation to a cyclopentyl cation; (b) the migrating bond must be aligned (antiperiplanar) with the empty p orbital; the alternative diastereomer would need the bond on the other face, giving a strained trans-fused bicyclo[3.3.0] (diquinane) system; (c) relief of cyclobutane ring strain (4- → 5-membered ring). [med-high]

6.59 (book p283-284, PDF p51-52) — footnote 8 → Org. Lett. 1999, 1, 537-539

- Thermal N₂ loss (250 °C) from a 4-amino-pyrrolo-fused 1,2,3-triazine **1** (R₁/R₂-substituted pyrrole) gives 2-amino-3-cyanopyrrole **4**. Proposed (later refuted) mechanism: via **2** (ring-opened diazonium/amidine anion: C(=NH)-N⁻H ... pyrrole-N≡N⁺) and **3** (a fused azetine: 2-imino-azetidine/amidine four-membered ring), then ring opening to **4**. ¹⁵N-labeling: **1** with ¹⁵NH₂ → **4a** (¹⁵N in the nitrile, ¹⁵N≡C-) observed; **4b** (¹⁵N in the NH₂) not observed.
- Questions: (a) curved arrows for **2** → **3** (the amide-type N⁻ attacks the C bearing N₂⁺, with loss of N₂; four-membered ring formation); (b) explain with resonance why the proposed mechanism predicts formation of **4b**.
- Answer (b): in **3** the exocyclic =NH and the ring NH form an amidine (H-N=C-NH ↔ ⁻/⁺ resonance, plus tautomerization), making the two N atoms equivalent. Ring opening could then put either N in the nitrile, so a ~1:1 mixture of **4a** and **4b** would be expected. The absence of **4b** rules this path out. [high]
- Core SMILES of **4** (R₁ = R₂ = H placeholder) Nc1[nH]ccc1C#N [high].

6.60 (book p284, PDF p52) — footnote 9 → Org. Lett. 2005, 7, 515-517

- Lewis-acid-promoted rearrangement: **1** = 6-methyl-2-phenyl-1,3-dioxane bearing a spiro-cyclopropane at C4 (cyclopropanone ketal-type), CC1CC2(CC2)OC(C2CCCC2)O1 [med-high] → **2** = ring-opened oxocarbenium (PhCH=O⁺-CH(Me)-) with a cyclopropanolate (cyclopropyl-O⁻ on the spiro carbon) → **3** = cis-2-methyl-7-phenyloxepan-4-one, CC1CC(=O)CCC(C2CCCC2)O1 (drawn 2,7-cis with wedges) [high constitution].
- Questions: (a) arrows for **1** → **2** (the Lewis acid coordinates O; the acetal C-O bond to the cyclopropyl-bearing O breaks, giving an oxocarbenium + cyclopropoxide); (b) arrows for **2** → **3** (hint: redraw **2**).
- Answer (b): the cyclopropanolate acts as a homoenolate. The O⁻ lone pair re-forms C=O, the strained cyclopropane C-C bond breaks, and the resulting CH₂ nucleophile (σ bond) attacks the oxocarbenium carbon, closing the 7-membered oxepanone (relief of cyclopropane strain). Redraw **2** so that the CH₂ is near the C=O⁺ carbon. [high]

6.61 (book p284, PDF p52) — footnote 10 → Org. Biomol. Chem. 2013, 11, 2498-2513

- Natural products from *Sarcostemma acidum* (folk antitussive); a strong acid (TsOH) converts one natural product into another by cyclization. Model: **1** = 2,3-dimethyl-4,4-diphenylbut-3-en-1-ol, CC(CO)C(C)=C(C1CCCC1)C1CCCC1 [high] → TsOH → **2** = 2,3-dimethyl-4-phenyl-1,2-dihydronaphthalene, CC1Cc2ccccc2C(C2CCCC2)=C1C [high].
- Task: identify the sequence of arrow-pushing patterns.
- Answer: (1) proton transfer (TsOH protonates OH → -OH₂⁺); (2) concerted nucleophilic attack by the phenyl π bond + loss of the leaving group (H₂O); intramolecular Friedel-Crafts-type S_N2 at the primary carbon gives an arenium (σ-complex) cation; (3) proton transfer (H₂O removes the ring H, restoring aromaticity). [high]

(6.19-6.51 have no literature footnotes.)

Global notes / caveats

- Pages missing from this PDF excerpt: chapter 1 summary §1.1-1.5 (book p57), chapter 3 book p140 (rest of the summary + skill builders 3.1-3.4), and early summary text of other chapters is partial. Chapter 2's

practice problems before 2.37 and the chapter bodies are not included.

- Footnote numbers whose markers I could **not** find on any page here: ch2 ref 1 (Org. Biomol. Chem. 2015, 13, 7813), ch3 ref 1 (Org. Lett. 2003, 5, 4931), ch6 ref 1 (J. Am. Chem. Soc. 1990, 112, 2003). They are probably cited in chapter body text (not in this excerpt).
- Duplicated references as printed: ch2 refs 6 = 13 (Chem. Rev. 1993, 93, 381); ch3 refs 3 = 11 (J. Am. Chem. Soc. 2009, 131, 8720); ch4 refs 1 = 3 = 7 (J. Am. Chem. Soc. 2009, 131, 8766).
- Likely book mis-assignment: ch3 refs 3/4 (asteltoxin vs. alkyne-lysine) appear swapped relative to the problem topics (my inference, flagged above).
- Every footnote-to-problem mapping above was read from high-resolution (220-500 dpi) crops.

Klein 유기화학 — Chapters 7-11 (first half: PDF p-01 … p-26)

Source: /root/.claude/uploads/e196aa5e-9821-56b2-b877-2f07b790959e/991a3849-7-11.pdf (PDF page N = p-NN.jpg). Confidence tags: [H] high, [M] medium, [L] low.

Chapter 7 — 치환반응 (Substitution Reactions) (book pp. 333-341; PDF p-01 … p-09)

Key reactions / concepts from summary (반응의 복습, p-01-p-04)

- **SN2**: concerted, backside attack, rate = $k[\text{substrate}][\text{Nu}]$; methyl $> 1^\circ > 2^\circ > 3^\circ$; inversion of configuration (stereospecific). Favoured by strong Nu, good LG, polar aprotic solvent (DMSO, DMF). E.g. (S)-2-bromobutane + $\text{HO}^- \rightarrow$ (R)-2-butanol; $\text{EtCl} + \text{NaSH} \rightarrow \text{EtSH}$.
- **SN1**: stepwise (LG loss = rate-determining $\rightarrow$ carbocation $\rightarrow$ Nu attack), rate = $k[\text{substrate}]$; $3^\circ > 2^\circ > 1^\circ >$ methyl; racemization (slight excess inversion due to ion pair). Favoured by weak Nu, excellent LG, protic solvent. Possible extra steps: proton transfer before LG loss ($\text{OH} \rightarrow \text{OH}_2^+$), carbocation rearrangement, deprotonation after neutral Nu attack (solvolysis). E.g. 3° bromide + $\text{NaCN} \rightarrow$ pair of enantiomers.
- **Activating OH**: $\text{ROH} + \text{TsCl/pyridine} \rightarrow \text{ROTs}$ (retention), then NaCN/DMSO (SN2, inversion); $\text{ROH} + \text{HX}$ (SN1 for 3°).
- Leaving groups: halides, sulfonates (tosylate most common).

Literature-based problems (참고문헌 on p-09, book p. 341)

#	Problem (book page)	fn	Citation
1	7.77 (p.339)	1	Tetrahedron 2004, 60, 9599-9614.
2	7.78 (p.339)	2	J. Chem. Ed. 2009, 86, 850-852.
3	7.79 (p.339)	3	J. Chem. Ed. 2009, 86, 853-855.
4	7.80 (p.339)	4	Org. Lett. 2000, 2, 2717-2719.
5	7.81 (p.339)	5	J. Am. Chem. Soc. 1980, 102, 6161-6163.
6	7.82 (p.339-340)	6	J. Am. Chem. Soc. 1965, 87, 287-291.
7	7.83 (p.340)	7	Tetrahedron Lett. 1967, 8, 4941-4944.
8	7.84 (p.340)	8	Org. Lett. 2010, 12, 136-139.

#	Problem (book page)	fn	Citation
9	7.85 (p.340)	9	J. Am. Chem. Soc. 1951, 73, 22-23.
10	7.86 (p.340)	10	J. Am. Chem. Soc. 2004, 126, 6868-6869.
11	7.87 (p.340)	11	J. Org. Chem. 2012, 77, 3181-3190.
12	7.88 (p.341)	12	J. Am. Chem. Soc. 1982, 104, 6460-6462.

7.77 — cyoctol (anti-androgen / male-pattern baldness agent), fn 1, Tetrahedron 2004, 60, 9599-9614 (종합 문제, p.339)

- SM: (R)-3-methoxypentan-1-ol (OMe on wedge), CC[C@H](OC)CCO [H for constitution; R assigned from drawing, M].
- Conditions: 1) TsCl, pyridine; 2) NaI; 3) PPh₃, heat → "?".
- Asks: (a) draw the products of the sequence and explain why step 3 is favourable (a neutral nucleophile gives an ion pair as product); (b) rationale for step 2.
- Answer [H]: 1) primary tosylate (C-O retained); 2) Finkelstein SN₂ → primary iodide (I⁻ better LG / better substrate for weakly basic neutral PPh₃, NaOTs precipitates/acetone-type driving force); 3) SN₂ by PPh₃ → (R)-(3-methoxypentyl)triphenylphosphonium iodide CC[C@H](OC)CC[P+](c1ccccc1)(c1ccccc1)c1ccccc1.[I-] (Wittig salt). Stereocentre untouched (reaction at C1). Step 3 favoured: P is a large, polarizable, strong neutral nucleophile; primary unhindered carbon; iodide excellent LG.
- Concept: activating OH as sulfonate; Finkelstein; SN₂ with neutral Nu → phosphonium salt.

7.78 — 2-naphthol Williamson ether synthesis (IR-monitored teaching lab), fn 2, J. Chem. Ed. 2009, 86, 850-852

- SM 1: 2-naphthol Oc1ccc2ccccc2c1; NaOH → 2 (sodium 2-naphthoxide); then n-butyl iodide → 3.
- Asks: (a) structure/mechanism for 2, and justify why hydroxide is basic enough (pK_a naphthol ≈ 9.5 vs water 15.7 → equilibrium lies far to phenoxide); (b) structure/mechanism of 3.
- Answer [H]: 3 = 2-butoxynaphthalene CCCCOc1ccc2ccccc2c1 (C₁₄H₁₆O) via SN₂ of naphthoxide on 1° BuI. (c) on p-07: product purified by recrystallization from water; must be dried before IR because water O-H band would mask disappearance of the phenol O-H; complete conversion shown by absence of broad O-H stretch (~3200-3600 cm⁻¹).
- Concept: acidity of phenols, Williamson ether synthesis (SN₂), IR monitoring.

7.79 — Bromotriphenylmethane solvolysis, fn 3, J. Chem. Ed. 2009, 86, 853-855

- SM 1: Ph₃C-Br BrC(c1ccccc1)(c1ccccc1)c1ccccc1; ROH → Ph₃C-OR: 2a R = H (triphenylmethanol), 2b R = CH₃, 2c R = CH₂CH₃.
- Asks: (a) mechanism 1 → 2b; (b) all three conversions are almost instantaneous — draw structure that justifies this; (c) why IR is ideal to follow 1 → 2a but other spectroscopy is better for 1 → 2b / 2c.
- Answer [H]: SN₁ (solvolysis): ionization to trityl cation (highly resonance-stabilized over three rings — the structure to draw), attack by MeOH, deprotonation. IR: 2a shows new O-H stretch; 2b/2c ethers have no diagnostic new IR band (C-O only) → NMR better (OCH₃ singlet / OCH₂CH₃ quartet-triplet).

7.80 — (±)-camerodnanol (from *Echinops giganteus* flower essential oil), fn 4, Org. Lett. 2000, 2, 2717-2719 (도전 문제, p.339)

- SM: bicyclo[3.3.0]octanone (diquinane) bearing gem-dimethyl next to ketone, angular methyl at ring fusion, and a side chain CH(CH₃)-CH₂CH₂Br on the carbon adjacent to the α-ring-fusion CH; exact SMILES uncertain.
- Reagent: KOt-Bu → angular triquinane ketone (new cyclopentane ring; the ketone α-carbon at the ring fusion becomes quaternary).
- Asks: plausible mechanism.
- Answer [H]: t-BuO⁻ deprotonates the α-C-H (ring-fusion carbon α to C=O; the other α-carbon is quaternary gem-dimethyl so no competition) → enolate; intramolecular SN₂ (C-alkylation) on the primary C-Br → 5-membered ring, Br⁻ leaves.
- Concept: enolate as nucleophile, intramolecular SN₂ forming 5-ring.

7.81 — thienamycin (from *Streptomyces cattleya*), fn 5, J. Am. Chem. Soc. 1980, 102, 6161-6163

- SM 1: (S)-4-(iodomethyl)-1-(trimethylsilyl)azetidin-2-one (CH₂I on wedge) C[Si](C)(C)N1C(=O)C[C@H]1CI [M on stereo label]. Nucleophile 2: 2-lithio-2-(trimethylsilyl)-1,3-dithiane [Li]C1([Si](C)(C)C)SCCCS1.
- SN₂ → 3. Asks: (a) draw 3; (b) why the dithiane ring in 3 exists as two chair conformers in comparable amounts while in 2 it is essentially one.
- Answer: (a) [H] 3 = 4-[[2-(trimethylsilyl)-1,3-dithian-2-yl]methyl]-1-TMS-azetidin-2-one C[Si](C)(C)N1C(=O)C[C@H]1CC1([Si](C)(C)C)SCCCS1 (C₁₄H₂₉NOS₂Si₂; configuration retained—substitution at CH₂, not the stereocentre). (b) [M] In 2, C₂ bears SiMe₃ vs Li (small) → SiMe₃ strongly prefers equatorial, one chair; in 3, C₂ bears two large groups (SiMe₃ and CH₂-azetidinone) of similar size → either chair puts one bulky group axial; two conformers of similar energy.

7.82 — 2-octyl sulfonate hydrolysis in water/dioxane, fn 6, J. Am. Chem. Soc. 1965, 87, 287-291

- SM: optically pure (R)-2-octyl sulfonate (ROSO₂-O-CH(CH₃)C₆H₁₃); H₂O/dioxane → (S)-2-octanol CCCCC[C@H](C)O with optical purity 77% (25:75 water:dioxane), 88% (50:50), 95% (75:25), 100% (100:0).
- Asks: complete mechanism for formation of (R)-2-octanol in the presence of dioxane.
- Answer [H]: two successive SN₂: dioxane O attacks C₂ (inversion) → oxonium (dioxanyl-alkyl cation); water then attacks C₂ displacing dioxane (second inversion) → net retention → (R)-2-octanol, then deprotonation. Direct SN₂ by water gives (S).

7.83 — cyclopropanol synthesis, fn 7, Tetrahedron Lett. 1967, 8, 4941-4944

- 1 cyclopropyl chloride **X** (NaOH) → 4 cyclopropanol. Route: 1 + Mg → 2 cyclopropylmagnesium chloride; 2 + tert-butyl peroxyacetate (t-BuO-O-C(O)CH₃) → 3 cyclopropyl tert-butyl ether CC(C)(C)OC1CC1; H₃O⁺ → 4 OC1CC1.
- Asks: (a) mechanism 2→3 (SN₂-type on O) and identify LG; (b) why irreversible; (c) mechanisms 3→4 via SN₁ and via SN₂; (d) the other alcohol by-product.
- Answer [H]: (a) carbanion attacks the t-BuO oxygen of the O-O bond; LG = acetate (CH₃CO₂⁻). (b) acetate is a far weaker base/nucleophile than the cyclopropyl carbanion (pK_a ~4.8 vs ~46) → equilibrium hugely favours products. (c) protonate ether O. SN₁: C(t-Bu)-O cleavage releases cyclopropanol and the tert-butyl cation; water traps it, deprotonation → t-BuOH. "SN₂" alternative: water attacks (concerted)

the carbon of the protonated ether with cyclopropanol as LG (book wants the cleavage on the t-Bu side so cyclopropanol is released intact; SN1 is chemically the realistic path). (d) tert-butanol (CH₃)₃COH [H].

7.84 — 1-deoxynojirimycin (HIV chemotherapy agent) synthesis, fn 8, Org. Lett. 2010, 12, 136-139

- SM 1: N-benzyl azepane (7-ring) with fused epoxide (3,4-epoxy) bearing OR and adjacent OMs (mesylate, wedge); Cl⁻ → 2: N-benzyl piperidine (fused epoxide) with exocyclic CH₂Cl on the carbon adjacent to N (ring contraction). SMILES uncertain.
- Asks: plausible mechanism with stereochemical evidence.
- Answer [H]: neighbouring-group participation: ring N attacks the C-OMs carbon intramolecularly (SN2, inversion) → bicyclic aziridinium ion; Cl⁻ opens aziridinium at the former ring CH₂ (less hindered, SN2) → piperidine + CH₂Cl; net double inversion at the carbinol carbon → retention-type outcome consistent with drawn stereochemistry.

7.85 — solvolysis of toluene halogen derivatives (PhCXYZ), fn 9, J. Am. Chem. Soc. 1951, 73, 22-23

- Hydrolysis (SN1) in 50% aqueous acetone, 30 °C; rates (×10⁴ min⁻¹): X=H,Y=Cl: Z=H 0.22, Z=Cl 2.21, Z=Br 31.1; X=Cl,Y=Cl: Z=H 2.21, Cl 110.5, Br 2122; X=Br,Y=Br: Z=H 6.85, Cl 1803, Br 1131.
- Asks: (a) which is better LG, Cl⁻ or Br⁻ (use Fig. 7.28), support with data; (b) is a carbocation stabilized by adjacent Cl (resonance structures); (c) by adjacent Br; (d) Cl vs Br which stabilizes more; (e) which factor matters more, LG ability or cation stability.
- Table decoded: row X=H,Y=Cl: PhCH₂Cl 0.22 / PhCHCl₂ 2.21 / PhCHClBr 31.1; row X=Y=Cl: PhCHCl₂ 2.21 / PhCCl₃ 110.5 / PhCCl₂Br 2122; row X=Y=Br: PhCHBr₂ 6.85 / PhCBr₂Cl 1803 / PhCBr₃ 1131.
- Answer [M]: (a) Br⁻ is the better LG: same cation, different LG — PhCHCl₂ (2.21) vs PhCHClBr (31.1); PhCCl₃ (110.5) vs PhCCl₂Br (2122). (b) α-Cl stabilizes the cation by lone-pair donation (C=Cl⁺ resonance): PhCH₂Cl 0.22 → PhCHCl₂ 2.21 → PhCCl₃ 110.5 (same LG Cl). (c) α-Br also stabilizes: PhCHBr₂ 6.85 and PhCBr₂Cl 1803 ≫ PhCH₂Cl/PhCH₂-type. (d) Data are mixed: with Cl leaving, PhCBr₂⁺ formation (PhCBr₂Cl, 1803) is faster than PhCCl₂⁺ (PhCCl₃, 110.5), but with Br leaving PhCCl₂Br (2122) > PhCBr₃ (1131). Textbook expected answer likely: Cl stabilizes better (2p-3p overlap) — low confidence; check original paper. (e) Most likely "leaving-group ability" dominates in the H-series, cation stabilization dominates overall (rate spans 10⁴ with α-halogens) — [L].

7.86 — vinylic SN2 forming 2-alkylidene aziridine, fn 10, J. Am. Chem. Soc. 2004, 126, 6868-6869

- SM 1: (E)-2-bromobut-2-en-1-ol OC/C(Br)=C\C (CH₂OH cis to CH₃) [M on geometry]. 1) TsCl, pyridine → 2; 2) BnNH₂ → 3; 3) NaNH₂, NH₃, -78 °C → 4: 1-benzyl-2-ethylideneaziridine C/C=C1/CN1Cc1ccccc1 (CH₃ cis to N, as drawn) [M].
- Asks: (a) structures 2, 3; (b) mechanism/TS for 3→4 (NaNH₂ base); (c) how stereochemistry matches SN2-type process.
- Answer: 2 = allylic tosylate CC=C(Br)C(=O)S(=O)(=O)C1CCC(C)CC1 (E); 3 = N-benzyl-2-bromobut-2-en-1-amine (SN2 at sp³ allylic C; vinylic C-Br untouched) [H]. NaNH₂ deprotonates N-H → amide anion attacks the vinylic C-Br carbon in-plane from the back side (σ* C-Br), Br⁻ leaves → 3-ring (in-plane SNV σ). Alkene geometry inverted: Br was trans to CH₃, N ends up cis to CH₃ [M].

7.87 — aziridine → azetidine ring expansion, fn 11, J. Org. Chem. 2012, 77, 3181-3190

- SM: 1-(arylmethyl)-2-(bromomethyl)-2-methylaziridine (R = CH₂Ar) CC1(CBr)CN1Cc1ccccc1 (aryl generic); NaOAc, CH₃CN → minor: (1-R-2-methylaziridin-2-yl)methyl acetate; major: 1-R-3-methylazetidin-3-yl

acetate CC(=O)OC1(C)CN(Cc2ccccc2)C1 (Ar=Ph example).

- Asks: mechanism explaining both products (hint: chapter's medicinal chemistry - mustard/aziridinium).
- Answer [H]: aziridine N displaces Br⁻ intramolecularly → bicyclic 1-azoniabicyclo[1.1.0]butane ion; acetate opens it: attack at the quaternary bridgehead C (cleaving the more strained central bond, SN1-like/SN2) → azetidine (major); attack at the CH₂ → regenerates aziridine with OAc (minor). (Minor product may also arise by direct SN2 on CH₂Br.)

7.88 — biotin (Hoffmann-La Roche total synthesis), fn 12, J. Am. Chem. Soc. 1982, 104, 6460-6462

- SM 1: biotin precursor: cis-fused tetrahydrothieno[3,4-d]imidazol-2-one with side chain CH(OH)-(CH₂)₃-COOH (OH wedge) at C next to S. SOCl₂, pyridine → [2: acid chloride + chlorosulfite ROS(O)Cl] → 3 (acyl chloride + secondary alkyl chloride, retention by X-ray) → 4 biotin OC(=O)CCCC[C@@H]1SC[C@@H]2NC(=O)N[C@H]12 (biotin SMILES standard [H]; precursor stereo uncertain).
- Asks: propose mechanism explaining overall retention of configuration in 2 → 3 (hint: Section 7.9 medicinal chemistry = sulfur mustards).
- Answer [H]: neighbouring-group participation by ring sulfur: S attacks the carbon bearing OS(O)Cl (intramolecular SN2, inversion; SO₂ + Cl⁻ released) → bicyclic sulfonium ion; Cl⁻ then attacks that carbon (SN2, second inversion) opening the sulfonium → net retention.

Chapter 8 — 알켄: 구조 및 제거 반응을 이용한 제법 (Alkenes: Structure and Preparation via Elimination Reactions) (book pp. 395-404; PDF p-10 … p-19)

Key reactions / concepts from summary (p-10-p-14)

- **E1 dehydration of alcohols:** conc. H₂SO₄, heat → most substituted (Zaitsev) alkene (e.g. 1,2-dimethylcyclohexanol → 1,2-dimethylcyclohexene); carbocation, possible rearrangement; regioselective, stereoselective (trans), not stereospecific.
- **E2 dehydrohalogenation, Zaitsev product:** NaOEt (unhindered strong base) → more substituted alkene.
- **E2 dehydrohalogenation, Hofmann product:** t-BuOK (bulky base) → less substituted alkene.
- E2: concerted, rate = k[substrate][base], 3° fastest; requires anti-periplanar H/LG (on cyclohexane both axial); stereospecific when β-carbon has one H.
- E1: rate = k[substrate]; competes with SN1. Alkene naming, E/Z, Bredt's rule, stability vs substitution.
- Reagent roles: Nu only (halides, HS⁻, RS⁻, H₂S, RSH); base only (H⁻, DBN, DBU); strong Nu/strong base (HO⁻, MeO⁻, EtO⁻, t-BuO⁻ — t-BuO⁻ mostly base); weak Nu/weak base (H₂O, MeOH, EtOH). 1° + strong base → SN2 major (t-BuOK → E2), 2° → E2 major, 3° → E2 only.
- 8.14 retrosynthesis example: ethyl acetate ⇌ NaOAc + EtBr (SN2).
- Note: 8.53 Dactylone (marine natural product, E/Z assignment) has NO footnote.

Literature-based problems (참고문헌 on p-19, book p. 404)

fn	Problem (book page)	Citation
1	8.89 (p.402)	Org. Lett. 2012, 14, 4730-4733.
2	8.90 (p.403)	Tetrahedron Lett. 2010, 51, 6948-6950.

fn	Problem (book page)	Citation
3	8.91 (p.403)	Tetrahedron Lett. 2007, 48, 8353-8355.
4	8.92 (p.403)	J. Org. Chem. 1991, 56, 2582-2584.
5	8.93 (p.403)	J. Org. Chem. 2009, 74, 2433-2437.
6	8.94 (p.403)	J. Am. Chem. Soc. 2010, 132, 2530-2531.
7	8.95 (p.403-404)	J. Am. Chem. Soc. 1973, 95, 3405-3407.
8	8.96 (p.404)	J. Am. Chem. Soc. 1965, 87, 5517-5518.

8.89 — Pladienolide B (macrolide from *Streptomyces platensis* mutant strain), fn 1, Org. Lett. 2012, 14, 4730-4733 (종합 문제, p.402)

- 31-step stereoselective synthesis reported in 2012 (vs earlier 59-step). Structure only (no reaction).
- Constitution SMILES (no stereo) [H for constitution, checked C30H48O8]: CCC(O)C(C)C10C1CC(C)C=CC=C(C)C2OC(=O)CC(O)CCC(C)(O)C(OC(C)=O)C=CC2C — 12-membered lactone with 3-OH, 6-OH/6-Me (tertiary), 7-OAc, 8,9-ring E-alkene, 10-Me, side chain with trisubstituted C12=C13(Me) and C14=C15 diene, 16-Me, 18,19-epoxide, 20-Me, 21-OH.
- Asks: (a) identify trisubstituted π bond and its E/Z; (b) number of possible stereoisomers; (c) draw enantiomer; (d) diastereomer differing only at chiral centres bonded directly to ester oxygens (C7 bearing OAc and C11 lactone carbon); (e) diastereomer differing only in configuration of the disubstituted π bond outside the ring (C14=C15).
- Answers: (a) C12=C13 (bearing CH3), E [M — natural pladienolide B is (8E,12E,14E)]. (b) 10 chirality centres + 3 stereogenic C=C = $13 \rightarrow 2^{13} = 8192$ [H on counting from drawn structure]. (c) invert all centres, keep alkene geometry. (d) flip C7 and C11. (e) make C14=C15 Z.

8.90 — steroid derivative (cholic-acid-type), fn 2, Tetrahedron Lett. 2010, 51, 6948-6950 (도전 문제)

- SM 1a: methyl 3,7-diacetoxy-12-hydroxy-cholanoate (5 β -steroid; 3-OAc, 7-OAc (axial, down), 12-OH axial down; C11 substituents R (equatorial, drawn toward C10-Me) and R' (axial, up/ β); side chain CH(CH3)CH2CH2CO2CH3). 1a R=R'=H; 1b R=H, R'=D; 1c R=D, R'=H. SMILES uncertain (likely methyl 3 α ,7 α -diacetoxy-12 α -hydroxy-5 β -cholan-24-oate).
- Conditions: TsCl, pyridine; then NaOAc (as base) $\rightarrow$ 2a.
- Asks: (a) structure of 2a; (b) which of 2b/2c retains D.
- Answer: (a) 12-OTs then E2 $\rightarrow$ Δ^{11} alkene (C11=C12; C13 is quaternary so only C11 H's available): methyl 3,7-diacetoxy-5 β -chol-11-enoate [H]. (b) E2 requires anti-periplanar (trans-diaxial) H; 12 α -OTs is axial (down) so the 11 β axial H (R') is removed. 1b (R'=D) loses D $\rightarrow$ 2b has no D; 1c (R=D, equatorial) keeps D $\rightarrow$ **2c contains deuterium** [M-H].

8.91 — oxetane-containing diastereomers (oxetane as in taxol), fn 3, Tetrahedron Lett. 2007, 48, 8353-8355

- SM 1: Ph-CH2CH2-CH(OH)-(oxetan-2-yl), OH hashed, oxetane O hashed; SM 2: same with OH wedged (diastereomer). Constitution OC(CCC1CCCC1)C1CCO1.
- Conditions: TsCl, pyridine; then t-BuOK $\rightarrow$ Zaitsev product predominates despite bulky base.
- Asks: stereochemical outcome for each, draw products, justify with Newman projections.

- Answer [M]: tosylation (retention), anti-periplanar E2 removing the single oxetane C2-H → trisubstituted exocyclic enol-ether alkene 2-(3-phenylpropylidene)oxetane C(=C1CCO1)CCc1ccccc1. Because the β-carbon has only one H, E2 is stereospecific: 1 and 2 give opposite geometric isomers. By my Newman analysis of the drawn configurations: 1 → (Z) (CH₂CH₂Ph cis to ring O), 2 → (E). Geometry assignment medium confidence.

8.92 — debromination of 1,2-dibromo-1,2-diphenylethane in DMF, fn 4, J. Org. Chem. 1991, 56, 2582-2584

- SM 1: meso-1,2-dibromo-1,2-diphenylethane (as drawn, anti-periplanar Br/Br gives E) Br[C@@H](c1ccccc1)[C@H](Br)c1ccccc1 (meso, 1R,2S) [M]; DMF → only (E)-stilbene 2 C(=C/c1ccccc1)\c1ccccc1. Proposed concerted mechanism: DMF carbonyl O attacks Br (anti-periplanar Br-C-C-Br, E2-like).
- Asks: (a) why proposed concerted mechanism conflicts with observations; (b) stereoselective or stereospecific?
- Answer [H]: concerted anti elimination from the other diastereomer (dl / (R,R)/(S,S)) would have to give (Z)-stilbene, but it also gives only E → mechanism is stepwise (e.g. loss of Br⁻/carbocation or bromonium-type intermediate with free rotation, then loss of Br⁺ to DMF) → product determined by stability (trans). (b) stereoselective, not stereospecific.

8.93 — (+)-coronafacic acid (component of phytotoxin coronatine), fn 5, J. Org. Chem. 2009, 74, 2433-2437

- SM: cis-hydrindane: cyclopentane ring fused, bearing spiro 5,5-dimethyl-1,3-dioxane ketal at C1 (ketone protected); six-membered ring carries ethyl (wedge), adjacent secondary OH (hashed), adjacent CH-CO₂CH₃ (CO₂Me hashed). SMILES uncertain.
- Conditions: 1) TsCl, pyridine; 2) DBU → ?
- Answer [M-H]: OTs formed with retention; DBU E2 removes H α to the ester (anti-periplanar: OTs and CO₂Me both hashed → α-H up, trans/anti-diaxial possible; on the ethyl-bearing carbon the H is cis to OTs, cannot be anti-periplanar) → α,β-unsaturated methyl ester (C=C between ester carbon and former carbinol carbon, conjugated with C=O). Regiochemistry justified by anti-periplanar requirement + acidity/conjugation.

8.94 — intramolecular Schmidt-type reaction of azido ketal, fn 6, J. Am. Chem. Soc. 2010, 132, 2530-2531

- SM 1: 1,4-dioxaspiro[4.5]decane (ethylene ketal of cyclohexanone) with C₆ quaternary (R + CH₂CH₂CH₂N₃) and C₈-R. Under acid → intermediate 2 (bicyclic aminodiazonium: azide N attached to the former ketal carbon, which bears O-CH₂CH₂OH ether; N-N₂⁺). Steps x,y,z to product 3.
- Asks: draw mechanism for last three steps as described: (x) alkyl group anti-periplanar to N₂⁺ undergoes 1,2-shift to N with N₂ loss (backside displacement); (y) alcohol O attacks resulting carbocation (oxocarbenium / iminium-type carbon); (z) oxonium deprotonated.
- Answer [M]: 1,2-C→N migration (ring expansion, Schmidt) gives a carbocation at the former ketal carbon stabilized by O (and N) → pendant 2-hydroxyethoxy OH closes back to a 1,3-dioxolane → after deprotonation 3 = bridged/fused bicyclic amine (ring-expanded azacycle) bearing a 2,2-(ethylenedioxy) N,O,O-orthoamide carbon. Exact structure of 3 uncertain (depends on which C-C bond is anti to N₂⁺; the analogy is E2 anti-periplanar).

8.95 — base strength vs. regioselectivity (2-iodobutane), fn 7, J. Am. Chem. Soc. 1973, 95, 3405-3407

- 2-iodobutane, DMSO, 50 °C, oxyanion bases → % 1-butene of total butenes: potassium benzoate 7.2; potassium phenoxide 11.4; sodium 2,2,2-trifluoroethoxide 14.3; sodium ethoxide 17.1.
- Asks: (a) trend vs base strength and relation between reactivity and selectivity; (b) predict % 1-butene with potassium 4-nitrophenoxide (pKa 4-nitrophenol 7.1).
- Answer: (a) [H] conjugate-acid pKa: benzoic ~4.2 < 4-nitrophenol 7.1 < phenol ~10 < CF₃CH₂OH ~12.4 < EtOH ~16; stronger base → more Hofmann (1-butene), i.e. more reactive base is less selective (earlier, more carbanion-like TS / less alkene character). (b) [M] between benzoate and phenoxide: ≈ 9% (8-10%).

8.96 — 2-butyl-Y + t-BuOK/t-BuOH, 50 °C, fn 8, J. Am. Chem. Soc. 1965, 87, 5517-5518

- trans:cis 2-butene ratio: Y = I 2.2, Br 1.4, Cl 1.3, OTs 0.6 (1-butene major in all).
- Asks: transition-state argument with Newman projections for why tosylate favours cis.
- Answer [L-M]: In anti-periplanar Newman projections along C2-C3, the C3-methyl is gauche to the LG in both 2-butene TSs; trans TS places the C3-CH₃ between LG and C2-H, cis TS places it between LG and C2-CH₃. For halides, CH₃/CH₃ gauche strain dominates → trans. For the large, oxygen-bonded tosylate (O-SO₂Ar rotates over C2-H side / the bulky Ts group), the dominant repulsion is between the C3-methyl and the tosyl group, which is larger in the trans-forming conformer where the methyl sits next to the region occupied by the sulfonyl group → cis favoured. (Also bulky base t-BuO⁻ interacts with the C2 substituent flanking the H removed.) Confidence low — verify against the paper.

Chapter 9 — 알켄의 첨가 반응 (Addition Reactions of Alkenes) (book pp. 452-461; PDF p-20 ... p-29)

Note: chapter 9's end-of-chapter problems run past p-26 (to p-28) and its 참고문헌 is on p-29; all footnoted ch. 9 problems are covered here (overlaps possible with whoever extracts p-27+). Chapter 10 (알카인, book p.489) starts on p-30 — not covered here.

Key reactions from summary (반응의 복습 wheel, p-20, shown on 1-methylcyclohexene)

1. HX → Markovnikov hydrohalogenation (3° halide); carbocation, possible rearrangement.
2. HBr, ROOR → anti-Markovnikov hydrobromination (radical).
3. H₃O⁺ → acid-catalyzed hydration (Markovnikov, carbocation, rearrangement possible); 1) Hg(OAc)₂, H₂O 2) NaBH₄ → oxymercuration-demercuration (Markovnikov, no rearrangement, mercurinium ion).
4. 1) BH₃·THF 2) H₂O₂, NaOH → hydroboration-oxidation (anti-Markovnikov, syn addition; trans-2-methylcyclohexanol + En).
5. H₂, Pt → catalytic hydrogenation (syn addition).
6. Br₂ → bromination via bromonium ion, anti addition (trans-1,2-dibromide + En).
7. Br₂, H₂O → halohydrin (OH at more substituted C, anti).
8. 1) RCO₃H 2) H₃O⁺ → anti dihydroxylation (via epoxide).
9. 1) OsO₄ 2) NaHSO₃, H₂O, or KMnO₄/NaOH cold → syn dihydroxylation.

10. 1) O3 2) DMS → ozonolysis (6-oxoheptanal from 1-methylcyclohexene). - Also: asymmetric hydrogenation with chiral catalysts; changing LG position (t-BuOK then HBr/ROOR) and π -bond position (HBr then t-BuOK).

Literature-based problems (참고문헌 on p-29, book p. 461)

fn	Problem (book page, PDF page)	Citation
1	9.68 (p.458, p-26)	Aust. J. Chem. 1987, 40, 1321-1325.
2	9.82 (p.459, p-27)	Nature 1994, 367, 630-634.
3	9.83 (p.459, p-27)	J. Am. Chem. Soc. 1985, 107, 4504-4508.
4	9.84 (p.459, p-27)	J. Org. Chem. 1984, 49, 945-947.
5	9.85 (p.460, p-28)	Angew. Chem. Int. Ed. 1994, 33, 2187-2190.
6	9.86 (p.460, p-28)	J. Am. Chem. Soc. 1980, 102, 2118-2120.
7	9.87 (p.460, p-28)	J. Am. Chem. Soc. 1989, 111, 1414-1418.
8	9.88 (p.460, p-28)	J. Am. Chem. Soc. 1989, 111, 1414-1418. (printed identically to fn 7 in the book)
9	9.89 (p.460, p-28)	Org. Lett. 2012, 14, 2929-2931.
10	9.90 (p.460, p-28)	Tetrahedron 1958, 2, 1-57.

9.68 — bicyclo[3.1.0]hexane skeleton (as in sabinene, nutmeg/pepper aroma), fn 1, Aust. J. Chem. 1987, 40, 1321-1325

- SM 1: 1-methoxy-4-methylcyclohexa-1,4-diene COC1=CCC(C)=CC1 (C₈H₁₂O). Excess HCl → 2 (C₈H₁₄Cl₂O); "two steps" → 3: 5-methylbicyclo[3.1.0]hexan-2-one (drawn as bicyclic ketone with bridgehead methyl) CC12CC1C(=O)CC2 [M on exact drawing].
- Asks: structure of 2 and mechanism of its formation.
- Answer [H]: 2 = 1,4-dichloro-1-methoxy-4-methylcyclohexane COC1(C1)CCC(C)(C1)CC1 (formula verified). Mechanism: two Markovnikov HCl additions — protonation of enol-ether C=C gives O-stabilized oxocarbenium → Cl⁻ (α -chloro ether); protonation of trisubstituted C=C gives 3° cation → Cl⁻. (Then: hydrolysis of α -chloro ether to 4-chloro-4-methylcyclohexanone; base → enolate intramolecular S_N2 at C4 → cyclopropane = 3.)

9.82 — Taxol (from *Taxus brevifolia*), K. C. Nicolaou total synthesis, fn 2, Nature 1994, 367, 630-634 (도전 문제)

- SM 1: Nicolaou tricyclic taxane intermediate: A-ring tetrasubstituted alkene, C₉ ketone, C₁₀-OAc, C₁-C₂ cyclic carbonate, C₇-OBn (wedge), C-ring disubstituted alkene C₅=C₆, C₄ quaternary bearing acetonide (with C₂₀ CH₂O). SMILES uncertain. Conditions: 1) BH₃·THF 2) H₂O₂, basic → alcohol 2 (regio- and stereoselective) → several steps → Taxol 3.
- Asks: (a) which functional group reacts with BH₃ and why; (b) predict regio/stereochem considering bulky benzyl group; draw 2.
- Answer [M]: (a) only the disubstituted C-ring C₅=C₆ alkene; the A-ring alkene is tetrasubstituted and very hindered (bridgehead-like), ketone/esters/carbonate react much slower with BH₃. (b) B adds to C₅ (remote from the bulky C₇-OBn, adjacent to C₄), H to C₆ → C₅-OH (needed for taxol's oxetane at C₅); syn addition from the face opposite to the benzyl ether (α face, since OBn is β -wedge) → C₅ α -OH (hashed), H's cis. Stereo face assignment medium/low confidence.

9.83 — adamantylideneadamantane bromonium ion, fn 3, J. Am. Chem. Soc. 1985, 107, 4504-4508

- SM: adamantylideneadamantane C1C2CC3CC1CC(C2)C3=C1C2CC3CC(C2)CC1C3; Br₂ → stable bromonium ion (high yield; with Br⁻/Br₃⁻ counterion); no further attack by Br⁻ to give dibromide.
- Asks: draw energy diagram consistent with this and compare with normal bromination (e.g. propylene).
- Answer [H]: Backside attack on either carbon is blocked by the adamantyl cages → second step has prohibitively high barrier; bromonium ion sits in a deep well (stable, isolable). For propylene: bromonium is a shallow high-energy intermediate rapidly opened by Br⁻ → overall exergonic to the vicinal dibromide.

9.84 — diisopinocampheylborane (lpc2BH), fn 4, J. Org. Chem. 1984, 49, 945-947

- SM: (+)-α-pinene (drawn single enantiomer) CC1=CCC2CC1C2(C)C; BH₃·THF 0.5 equiv → lpc2BH (one stereoisomer).
- Asks: why only one stereoisomer of lpc2BH forms.
- Answer [H]: α-pinene is enantiopure; syn hydroboration occurs exclusively from the face anti to the gem-dimethyl bridge (other face blocked), B to less substituted C (anti-Markovnikov) → each pinene gives the same configuration at the three new/existing centres; both lpc groups identical → single stereoisomer (chiral, used in asymmetric hydroboration).

9.85 — zaragozic acid A (squalene synthase inhibitor; isolated 1992 from fungal cultures), fn 5, Angew. Chem. Int. Ed. 1994, 33, 2187-2190 (Nicolaou)

- SM 1: spiro bicyclic: γ-butyrolactone with exocyclic alkylidene =CH-CH₂OH at C3, spiro-fused (C2) to 2,2-dimethyl-1,3-dioxolane bearing CH₂OSEM (SEM = CH₂OCH₂CH₂SiMe₃). Cat. OsO₄, NMO → triol 2 (syn dihydroxylation from top face only) → rapid rearrangement (translactonization) → 3 (lactone now formed to the new secondary OH; free CH₂OH arms).
- Asks: justify the diastereofacial selectivity (top face, not bottom).
- Answer [M]: bottom face of the alkene is shielded by the spiro acetonide (the dioxolane O/gem-dimethyl and the CH₂OSEM substituent are oriented under the π-bond in the rigid spirocycle), so OsO₄ approaches from the less hindered top face; syn addition delivers both OH from that face. Build model.

9.86 — monensin (from *Streptomyces cinnamomensis*), W. C. Still synthesis, fn 6, J. Am. Chem. Soc. 1980, 102, 2118-2120

- SM 1: (Z)-alkenoic acid: HO₂C(1)-CH(2)(CH₃)-CH₂(3)-CH(4)(CH₃)-CH(5)=CH(6)-CH₂-CH(CH₃)-(4-yl-2,2-dimethyl-1,3-dioxolane); Z C₅=C₆. I₂, NaHCO₃ → 2: δ-lactone (6-membered: O₁-C₅), 2,4-dimethyl, with C₆ bearing I (exocyclic CHI-CH₂-...), single diastereomer (5S,6S); (5R,6R) not observed.
- Asks: (a) mechanism of iodolactonization; (b) rationalize via C₄-C₅ Newman projection (lowest-energy conformer).
- Answer [M-H]: (a) I₂ forms iodonium on C₅=C₆; carboxylate (deprotonated by NaHCO₃) attacks C₅ anti (6-exo, Markovnikov-like at the carbon giving 6-ring) → trans addition of O and I. (b) allylic 1,3-strain (A_{1,3}) of the Z-alkene forces C₄-H to eclipse the C=C (C₄-H in plane, syn to C₆ substituent); the C₄-methyl and C₃ chain sit on opposite faces; I⁺ attacks the face anti to the bulkier substituent/from the face allowing the C₃-CO₂⁻ chain to reach C₅ → only (5S,6S).

9.87 — relative rates of oxymercuration with Hg(OAc)₂, fn 7, J. Am. Chem. Soc. 1989, 111, 1414-1418

- Data: CH₂=C(CH₃)CH₂CH₂CH₃ 1000; CH₂=CHCH₂CH₂CH₂CH₃ 100; CH₂=CHCH₂OMe 31.8; CH(CH₃)=C(CH₃)₂ 25.8; CH₂=CHCH₂Cl 2.4.
- Asks: explain trend structurally.
- Answer [M-H]: electrophilic addition → more electron-rich alkene/more stabilized cationic (mercurinium) TS faster: 1,1-disubstituted (2 alkyls, 3°-like) > monosubstituted. Allylic electronegative OMe, Cl withdraw inductively → slower (Cl > OMe withdrawing). Trisubstituted 4 is slower than expected because of steric hindrance to the bulky Hg(OAc)₂ (and mercurinium formation at a more crowded C).

9.88 — relative rates of hydroboration with 9-BBN, fn 8, J. Am. Chem. Soc. 1989, 111, 1414-1418 (same citation printed as fn 7)

- Data: CH₂=CHOBu 1615; CH₂=CHBu 100; CH₂=CHCH₂OMe 32.5; CH₂=CHOAc 22.8; CH₂=CHCH₂OAc 21.9; CH₂=CHCH₂CN 5.9; CH₂=CHCH₂Cl 4.0; cis-2-butene 0.95; trans-CH₃CH₂CH=CHCl 0.003.
- Asks: explain (a) 1-5, (b) 5-7, (c) 2, 8, 9.
- Answer [M]: hydroboration is electrophilic (B electrophile, partial + on carbon in TS). (a) vinyl ether 1: O lone-pair resonance donation → most electron-rich → fastest; 2 alkyl reference; allylic OMe (3), vinyl OAc (4: acyl reduces O donation, induction withdraws), allylic OAc (5) inductively withdrawing → slower. (b) allylic EWG strength: CN and Cl withdraw more than OAc → 5 > 6 > 7. (c) bulky 9-BBN is very sensitive to sterics: monosubstituted 2 ≫ 1,2-disubstituted cis-2-butene 8; 9 is disubstituted AND has an electron-withdrawing vinylic Cl → slowest.

9.89 — heliol (natural product), fn 9, Org. Lett. 2012, 14, 2929-2931

- SM: 2-(2-hydroxy-6-methylhept-6-en-2-yl)phenol CC(O)(CCCC(C)=C)C1CCCCC1O (C₁₄H₂₀O₂). H₃O⁺, CH₃CN → bridged benzoxabicyclo (heliol skeleton) + enantiomer (racemic): CC12CCCC(C)(C1)OC1CCCCC12 (C₁₄H₁₈O) [M].
- Asks: plausible mechanism.
- Answer [M-H]: protonate tertiary benzylic OH, lose H₂O → tertiary benzylic cation; intramolecular attack by the 1,1-disubstituted alkene (terminal CH₂ bonds to cation; Markovnikov) → six-membered carbocycle with new tertiary cation; phenol OH traps that cation (intramolecular) → oxonium; deprotonation → bridged ether. Racemic because cation planar/achiral SM.

9.90 — reserpine (from *Rauwolfia serpentina*, Indian snakeroot), R. B. Woodward 1958 synthesis, fn 10, Tetrahedron 1958, 2, 1-57

- SM 1: tricyclic cis-fused γ-lactone (hydronaphthalene) with an allylic secondary OH (wedge) in the upper ring (adjacent alkene) and an isolated cyclohexene C=C in the lower ring; 1) Br₂ 2) NaOMe → short-lived 2 (ether-bridged, α,β-unsaturated lactone) → Michael addition of MeO⁻ → 3 (OMe added, ether bridge retained).
 - Asks: mechanism 1 → 2.
 - Answer [M-H]: Br₂ adds to the lower-ring alkene forming a bromonium ion; the nearby OH attacks intramolecularly (anti opening) → cyclic ether bridge + C-Br (bromoetherification); NaOMe then removes the acidic H α to the lactone carbonyl and expels Br⁻ (E2/E1cB) → conjugated α,β-unsaturated lactone 2; MeO⁻ then adds conjugately (Michael) to give 3.
-

Klein 유기화학 — Chapters 7-11, second half (PDF pages 27-51)

Book page = printed page number. PDF page N = p-NN.jpg. Confidence: H = high, M = medium, L = low.

Chapter 9 — 알켄의 첨가 반응 (Addition Reactions of Alkenes) — tail end (PDF p-25..p-29, book pp. 457-461)

The chapter starts before p-27, so its summary page is in the other agent's range (p-21 = book p. 453 shows the 개념 summary 9.6-9.14). Key reactions: HX addition (Markovnikov, carbocation and rearrangement); HBr/ROOR (anti-Markovnikov); acid-catalyzed hydration; oxymercuration-demercuration ($\text{Hg}(\text{OAc})_2$, H_2O ; NaBH_4) → Markovnikov alcohol, no rearrangement (mercurinium ion); hydroboration-oxidation ($\text{BH}_3\cdot\text{THF}$; H_2O_2 , NaOH) → anti-Markovnikov, syn; catalytic hydrogenation (H_2 , Pt/Pd/Ni ; syn; asymmetric with chiral catalysts); halogenation X_2 (bromonium ion, anti); halohydrin formation (X_2 , H_2O); anti-dihydroxylation (RCO_3H ; H_3O^+); syn-dihydroxylation (OsO_4/NMO or cold KMnO_4); ozonolysis (O_3 ; DMS).

참고문헌 (book p. 461, PDF p-29), exact text

1. Aust. J. Chem. 1987, 40, 1321-1325.
2. Nature 1994, 367, 630-634.
3. J. Am. Chem. Soc. 1985, 107, 4504-4508.
4. J. Org. Chem. 1984, 49, 945-947.
5. Angew. Chem. Int. Ed. 1994, 33, 2187-2190.
6. J. Am. Chem. Soc. 1980, 102, 2118-2120.
7. J. Am. Chem. Soc. 1989, 111, 1414-1418.
8. J. Am. Chem. Soc. 1989, 111, 1414-1418.
9. Org. Lett. 2012, 14, 2929-2931.
10. Tetrahedron 1958, 2, 1-57.

Problem 9.68 — book p. 458 (PDF p-26) — footnote 1 → Aust. J. Chem. 1987, 40, 1321-1325

- Topic: building a bicyclo[3.1.0]hexane ring, as found in natural flavor compounds such as sabinene (nutmeg).
- Starting material 1: 1-methoxy-4-methylcyclohexa-1,4-diene (Birch product of 4-methylanisole). SMILES COC1=CCC(C)=CC1 ($\text{C}_8\text{H}_{12}\text{O}$) [H].
- Reagents: excess HCl → compound 2, $\text{C}_8\text{H}_{14}\text{Cl}_2\text{O}$; then "two steps" → compound 3, a bicyclo[3.1.0]hexanone with a methyl at the bridgehead. The drawing shows a hexagon with a transannular bond from $\text{C}(\text{Me})$ to the carbon next to $\text{C}=\text{O}$, which reads as 5-methylbicyclo[3.1.0]hexan-2-one, CC12CCC(=O)C1C2 [M].
- Asked: draw 2 and give a mechanism for its formation.
- Answer: Markovnikov addition of HCl to both $\text{C}=\text{C}$ bonds. The enol ether is protonated to give an oxocarbenium ion, which Cl^- traps (α -chloro ether). The trisubstituted alkene is protonated to give a

tertiary cation, which Cl^- traps. 2 = 1,4-dichloro-1-methoxy-4-methylcyclohexane, COC1(C1)CCC(C)(C1)CC1, $\text{C}_8\text{H}_{14}\text{Cl}_2\text{O}$ (formula checked) [H for constitution]. Likely two steps (not asked): hydrolysis to 4-chloro-4-methylcyclohexanone, then base-mediated enolate C-alkylation at C4 (1,3-closure) to make the cyclopropane [M].

- Concept: Markovnikov HX addition, carbocation/oxocarbenium stability.

Problem 9.82 — book p. 459 (PDF p-27) — footnote 2 → Nature 1994, 367, 630-634 (K. C. Nicolaou Taxol total synthesis)

- Natural product: Taxol (paclitaxel, 3), from *Taxus brevifolia*.
- Starting material 1: an advanced taxane ABC tricyclic intermediate. A ring: tetrasubstituted bridgehead $\text{C}_{11}=\text{C}_{12}$ alkene with vinyl CH_3 , $\text{C}_{10}\text{-OAc}$, C_9 ketone, gem-dimethyl C_{15} . C_1/C_2 cyclic carbonate. C ring: $\text{C}_7\text{-OBn}$ (benzyl ether), $\text{C}_5=\text{C}_6$ disubstituted ring alkene, and C_4 quaternary as part of an acetonide ($\text{C}_4\text{-O}$ and $\text{C}_{20}\text{CH}_2\text{-O}$, 2,2-dimethyl-1,3-dioxolane). C_8 has an angular CH_3 , $\text{C}_3\text{-H}$. SMILES uncertain (complex polycycle with stereocenters).
- Reagents: 1) $\text{BH}_3\cdot\text{THF}$; 2) H_2O_2 , mild base → alcohol 2 (regio- and stereoselective). Then many steps → Taxol 3.
- Asked: (a) which functional groups in 1 can react with BH_3 , with reasons; (b) the benzyl group is very bulky and blocks hydroboration, so predict the regio- and stereochemistry and draw 2.
- Expected answer: (a) In principle BH_3 can react with the two $\text{C}=\text{C}$ bonds and the ketone. The tetrasubstituted bridgehead A-ring alkene is too hindered to react, and the C_9 ketone is very hindered. The esters, carbonate, acetal and ether don't react. Only the disubstituted $\text{C}_5=\text{C}_6$ alkene reacts [H]. (b) Boron adds to C_5 , the carbon farther from the bulky $\text{C}_7\text{-OBn}$ (next to quaternary C_4). The addition is syn, from the less hindered face, so 2 is the C_5 secondary alcohol, with H added at C_6 . C_5 -oxygenation matches Taxol's oxetane oxygen at C_5 [M; face assignment L-M].
- Concept: chemoselectivity of BH_3 ; steric control of regiochemistry and of the face of syn addition in hydroboration.

Problem 9.83 — book p. 459 (PDF p-27) — footnote 3 → J. Am. Chem. Soc. 1985, 107, 4504-4508

- Compound: adamantylideneadamantane, C1C2CC3CC1CC(C2)C3=C1C2CC3CC(C2)CC1C3 ($\text{C}_{20}\text{H}_{28}$) [H].
- Reagent: Br_2 → stable bromonium ion. Br^- cannot attack it from the back ($\text{S}_{\text{N}}2$) to give the dibromide.
- Asked: draw an energy diagram consistent with this, and compare it with the diagram for normal bromination (e.g. propylene).
- Answer: for adamantylideneadamantane, the bromonium ion (with Br^- or Br_3^-) sits in a deep well. The backside approach is blocked by the two adamantyl cages, so the barrier from the bromonium ion to the dibromide is very high and no product forms. For propene, the bromonium ion is a high-energy intermediate with a small barrier to anti attack by Br^- , and the dibromide product lies downhill [H].
- Concept: bromonium-ion mechanism; steric hindrance to the $\text{S}_{\text{N}}2$ -like opening step.

Problem 9.84 — book p. 459 (PDF p-27) — footnote 4 → J. Org. Chem. 1984, 49, 945-947

- Compound: diisopinocampheylborane (lpc_2BH), a chiral hydroborating agent made from BH_3 and 2 equiv α -pinene (CC1=CCC2CC1C2(C)C, single enantiomer).
- Reagents: α -pinene + $\text{BH}_3\cdot\text{THF}$ (0.5 equiv) → lpc_2BH (single diastereomer/enantiomer).

- Asked: explain why only one stereoisomer of lpc2BH forms.
- Answer: hydroboration is syn and concerted. Boron adds to the less substituted alkene carbon, anti-Markovnikov. B-H adds only to the face of the alkene opposite the C6 gem-dimethyl bridge (the less hindered face). Because the α -pinene is enantiopure, a single isomer forms. The second hydroboration (by lpcBH₂) also occurs from the same face [H].
- Concept: steric face selectivity; substrate-controlled asymmetric hydroboration.

Problem 9.85 — book p. 460 (PDF p-28) — footnote 5 → Angew. Chem. Int. Ed. 1994, 33, 2187-2190 (zaragozic acid A synthesis)

- Natural product: zaragozic acid A (squalene synthase inhibitor, cholesterol-lowering).
- Starting material 1: an allylic alcohol, (2-hydroxyethylidene)- γ -butyrolactone. Its α -carbon is a spiro center shared with a 2,2-dimethyl-1,3-dioxolane (acetonide) that carries CH₂OSEM (SEM = CH₂OCH₂CH₂Si(CH₃)₃). Constitution SMILES OCC=C1COC(=O)C12OC(C)(C)OC2COC0CC[Si](C)(C)C; alkene geometry and stereo not encoded [M].
- Reagents: OsO₄ (cat.), NMO → triol 2 (diastereoselective), which rapidly rearranges (translactonization) to lactone 3.
- What happens: 2 = syn-dihydroxylated triol (a tertiary OH at the former ring alkene carbon, a secondary OH plus CH₂OH on the side chain). In 3, the lactone carbonyl has migrated to the new secondary OH, forming a new γ -lactone that bears CH₂OH and a quaternary C(OH)CH₂OH, and releasing the former ring CH₂OH.
- Asked: explain the diastereoselectivity (syn addition from the top π face, not the bottom), using a molecular model.
- Answer: the bottom face of the exocyclic alkene is shielded by the spiro-fused acetonide (its gem-dimethyl/ring O) and the CH₂OSEM substituent, so OsO₄ approaches from the open top face [M]. Optional: mechanism of translactonization (intramolecular transesterification via a tetrahedral intermediate) [H].
- Concept: OsO₄ syn dihydroxylation; steric face selectivity in a rigid spiro system.

Problem 9.86 — book p. 460 (PDF p-28) — footnote 6 → J. Am. Chem. Soc. 1980, 102, 2118-2120 (monensin synthesis; book says "W. S. Still")

- Natural product: monensin (antibiotic, Streptomyces cinnamomensis).
- Starting material 1: (Z)-C₅=C₆ unsaturated carboxylic acid with C₂-Me and C₄-Me stereocenters. The chain end carries a CH(CH₃) attached to a 2,2-dimethyl-1,3-dioxolan-4-yl (acetonide). SMILES (no stereocenters) CC(C(C)C(=O)O)/C=C\CC(C)C1COC(C)(C)O1. The alkene reads as Z from the drawing [M].
- Reagents: NaHCO₃, I₂ (iodolactonization) → δ -lactone 2. Ring O is bonded to C₅ and I to C₆; only the (5S,6S) diastereomer forms, and (5R,6R) is not observed. Product constitution CC1CC(C)C(=O)OC1C(I)CC(C)C1COC(C)(C)O1.
- Asked: (a) mechanism (iodonium formation, then anti attack of carboxylate at C₅, 6-membered ring); (b) explain the diastereoselectivity from the conformation of 1 using a C₄-C₅ Newman projection.
- Answer: the lowest-energy conformer about C₄-C₅ avoids A(1,3) strain between the C₄ substituents and the cis C₆ substituent of the Z-alkene, so C₄-H eclipses the alkene. I⁺ then adds to the less hindered face, and carboxylate attacks anti, giving 5S,6S [M].
- Concept: halolactonization, anti addition, allylic (A1,3) strain-controlled diastereoselectivity.

Problem 9.87 (도전 문제) — book p. 460 — footnote 7 → J. Am. Chem. Soc. 1989, 111, 1414-1418

- Data: relative rates of oxymercuration with $\text{Hg}(\text{OAc})_2$. $\text{CH}_2=\text{C}(\text{CH}_3)\text{CH}_2\text{CH}_2\text{CH}_3$ (1) 1000; $\text{CH}_2=\text{CHCH}_2\text{CH}_2\text{CH}_2\text{CH}_3$ (2) 100; $\text{CH}_2=\text{CHCH}_2\text{OMe}$ (3) 31.8; $(\text{CH}_3)\text{CH}=\text{C}(\text{CH}_3)_2$ (2-methyl-2-butene, 4) 25.8; $\text{CH}_2=\text{CHCH}_2\text{Cl}$ (5) 2.4.
- Asked: explain the trends in structural terms.
- Answer: electrophilic addition goes through a mercurinium ion with carbocation character. 1,1-disubstituted 1 > monosubstituted 2 (more alkyl donation stabilizes the cationic TS). Allylic OMe (3) and Cl (5) are inductively electron-withdrawing and slow the reaction, Cl more strongly. The trisubstituted 4 is slower than 1 despite being more electron-rich because steric hindrance at the bulky $\text{Hg}(\text{OAc})_2$ approach dominates [M-H].

Problem 9.88 (도전 문제) — book p. 460 — footnote 8 → J. Am. Chem. Soc. 1989, 111, 1414-1418

- Data: relative rates of hydroboration with 9-BBN. $\text{CH}_2=\text{CHOBu}$ (1) 1615; $\text{CH}_2=\text{CHBu}$ (2) 100; $\text{CH}_2=\text{CHCH}_2\text{OMe}$ (3) 32.5; $\text{CH}_2=\text{CHOAc}$ (4) 22.8; $\text{CH}_2=\text{CHCH}_2\text{OAc}$ (5) 21.9; $\text{CH}_2=\text{CHCH}_2\text{CN}$ (6) 5.9; $\text{CH}_2=\text{CHCH}_2\text{Cl}$ (7) 4.0; cis-2-butene (8) 0.95; trans- $\text{CH}_3\text{CH}_2\text{CH}=\text{CHCl}$ (9) 0.003.
- Asked: (a) explain 1-5; (b) 5, 6, 7; (c) 2, 8, 9.
- Answers: (a) vinyl ether 1 is the fastest because O donates by resonance and makes the π bond more nucleophilic (boron is electrophilic). The vinyl ester 4 donates much less because the O lone pair is delocalized into $\text{C}=\text{O}$. Allylic OMe (3) and OAc (5) are inductively withdrawing, so slower than 2. (b) The rate falls with the inductive withdrawal of the allylic group: $\text{OAc} < \text{CN} \approx \text{Cl}$ (Cl and CN strongest). (c) Internal disubstituted 8 is $\sim 100\times$ slower than terminal 2 because bulky 9-BBN is very sensitive to steric hindrance. 9 (internal plus a vinylic Cl that withdraws inductively) is slowest [M].

Problem 9.89 (도전 문제) — book p. 460 — footnote 9 → Org. Lett. 2012, 14, 2929-2931 (heliol synthesis)

- Starting material: 2-(2-hydroxy-6-methylhept-6-en-2-yl)phenol, C=C(C)CCCC(C)(O)c1ccccc1O [H].
- Reagents: H_3O^+ , CH_3CN → bridged benzoxa-bicyclic (heliol core) + enantiomer (racemic).
- Product: bicyclo[3.3.1] bridged ether fused to benzene, with two bridgehead methyls, CC12CCCC(C)(C1)Oc1c2ccccc12 ($\text{C}_{14}\text{H}_{18}\text{O}$) [M-H].
- Asked: mechanism.
- Answer: protonate the tertiary benzylic OH and lose water to give a tertiary benzylic cation. The 1,1-disubstituted alkene attacks it intramolecularly (terminal CH_2 forms a new C-C bond, closing a 6-membered ring) to give a new tertiary cation. The phenol O traps that cation to form the O-bridge, and deprotonation finishes. Racemic because the cation is planar and achiral [H].
- Concept: carbocation formation, cation-olefin cyclization, intramolecular ether formation.

Problem 9.90 (도전 문제) — book p. 460 — footnote 10 → Tetrahedron 1958, 2, 1-57 (R. B. Woodward reserpine)

- Natural product: reserpine (*Rauwolfia serpentina*).

- Starting material 1: a cis-fused tricyclic γ -lactone (decalin-type) with an allylic secondary OH on a cyclohexene ring and a second C=C in the other ring. SMILES uncertain.
- Reagents: 1) Br₂; 2) NaOMe $\rightarrow$ short-lived intermediate 2 (an ether-bridged α,β -unsaturated lactone), which undergoes Michael addition of MeO⁻ to give 3 (OMe added β to the lactone carbonyl).
- Asked: mechanism for 1 $\rightarrow$ 2.
- Answer: Br₂ adds to the ring alkene near the OH to form a bromonium ion. The OH attacks intramolecularly, anti, to form the cyclic ether (bromoetherification), leaving Br on the adjacent carbon. NaOMe deprotonates the acidic lactone α -C-H, and E2 (or E1cb) eliminates HBr to give the conjugated α,β -unsaturated lactone 2. MeO⁻ then adds 1,4 to give 3 [M].
- Concept: bromonium ion with an intramolecular O-nucleophile; base-induced elimination; conjugate addition.

(Other problems in 9.53-9.81, e.g. 9.76 muscalure and 9.80 iodolactonization, have no literature footnote.)

Chapter 10 — 알카인 (Alkynes) — PDF p-30..p-37, book pp. 489-496

Summary page (반응의 복습, book p. 489, PDF p-30)

1. Elimination: geminal or vicinal dihalide + excess NaNH₂; H₂O $\rightarrow$ terminal alkyne (double E2).
2. Hydrohalogenation, excess HX $\rightarrow$ geminal dihalide (Markovnikov).
3. HX (1 equiv) $\rightarrow$ vinyl halide (Markovnikov).
4. Acid-catalyzed hydration: H₂SO₄, H₂O, HgSO₄ $\rightarrow$ methyl ketone (via enol, then tautomerization).
5. Hydroboration-oxidation: 1) R₂BH (disiamylborane/9-BBN); 2) H₂O₂, NaOH $\rightarrow$ aldehyde (anti-Markovnikov enol, then tautomerization).
6. Halogenation, X₂ (1 equiv), CCl₄ $\rightarrow$ (E)-1,2-dihaloalkene.
7. Excess X₂, CCl₄ $\rightarrow$ 1,1,2,2-tetrahaloalkane.
8. Ozonolysis: 1) O₃; 2) H₂O $\rightarrow$ carboxylic acids (terminal carbon $\rightarrow$ CO₂).
9. Alkylation: 1) NaNH₂; 2) RX (methyl/1°) $\rightarrow$ internal alkyne.
10. Dissolving-metal reduction: Na, NH₃(l) $\rightarrow$ trans alkene (radical anion; fishhook arrows).
11. Hydrogenation: H₂, Pt $\rightarrow$ alkane.
12. Poisoned catalyst: H₂, Lindlar catalyst (or Ni₂B) $\rightarrow$ cis alkene.

Concepts: sp C, acetylide pK_a ~25 vs H₂O 15.7 (NaNH₂ or NaH needed); keto-enol tautomerization under acid/base; vinyl cation vs termolecular HX addition; HBr/ROOR anti-Markovnikov; acetylene double alkylation; alkene $\rightarrow$ alkyne via Br₂ then excess NaNH₂.

참고문헌 (book p. 496, PDF p-37), exact text

1. Org. Lett. 2003, 5, 4017-4020.
2. J. Org. Chem. 1959, 24, 840-842.
3. J. Am. Chem. Soc. 2007, 129, 8968-8969.
4. Tetrahedron Lett. 1996, 37, 1571-1574.
5. Acc. Chem. Res. 1988, 21, 442-449.
6. J. Am. Chem. Soc. 1966, 88, 4525-4526.

7. J. Am. Chem. Soc. 2008, 130, 6281-6287.
8. Tetrahedron Lett. 2001, 42, 7277-7280.
9. J. Phys. Chem. A 2007, 111, 3513-3518.
10. J. Am. Chem. Soc. 1976, 98, 3295-3300.

Problem 10.69 — book p. 494 (PDF p-35) — footnote 1 → Org. Lett. 2003, 5, 4017-4020

- Topic: synthesis of α,α -disubstituted α -amino acids (Ch. 25 targets).
- Starting material: a quaternary carbon bearing N(Boc)₂, a terminal ethynyl (C≡CH), CH₂-O-CH₂-(3,4-dimethoxyphenyl) (wedge), and CH₂-O-TBS (hash). Model SMILES with Boc and TBS in full: C#CC(N(C(=O)OC(C)(C)C(=O)OC(C)(C)C)(COCc1ccc(OC)c(OC)c1)C0[Si](C)(C)C(C)(C)C, stereo omitted [M].
- Product: the C≡CH becomes CH₂CH₂CH₂-O-CH₂CH₂-OCH₃ (three CH₂ groups, then the ether; verified at 600-dpi zoom). Everything else is unchanged.
- Asked: (a) reagents; (b) configuration at the chiral center in the starting material and product.
- Answer: (a) 1) NaNH₂ (or n-BuLi), to form the acetylide; 2) CH₃OCH₂CH₂OCH₂Cl (MEM chloride) for S_N2 alkylation, giving C≡C-CH₂-O-CH₂CH₂OMe; 3) H₂, Pt (or Pd/C) to reduce the alkyne fully to the alkane. This accounts for the extra carbon [M-H]. (b) CIP priority: N > CH₂OTBS (O-Si beats O-C) > CH₂O-CH₂Ar > the C≡CH or CH₂CH₂ chain. The priorities don't change, so both are S as drawn (RDKit on the drawn wedge/hash: N up-left, alkyne up-right, CH₂OAr wedge, CH₂OTBS hash) [M].
- Concept: acetylide alkylation followed by complete hydrogenation; CIP ranking.

Problem 10.70 — book p. 494 (PDF p-35) — footnote 2 → J. Org. Chem. 1959, 24, 840-842

- Reaction: (CH₃O)₂SO₂ (1 mol) + HC≡CNa (2 mol) → 2 mol propyne (CH₃C≡CH) + ?
- Asked: (a) the ionic inorganic byproduct and its mechanism; (b) why 2-butyne is a byproduct, consistent with acetylene being observed; (c) products with diethyl sulfate.
- Answer: (a) Na₂SO₄. The first S_N2 transfers a methyl and gives CH₃OSO₃⁻Na⁺. Methyl sulfate is still an alkylating agent and methylates the second acetylide, leaving SO₄²⁻ (2 Na⁺) [H]. (b) Acetylide (a stronger base) deprotonates the propyne product (acid-base exchange). This gives acetylene and CH₃C≡C⁻, which is methylated to CH₃C≡CCH₃. Acetylene shows up among the products [H]. (c) Main product 1-butyne (2 mol), plus Na₂SO₄; byproducts 3-hexyne and acetylene [H].

Problem 10.71 — book pp. 494-495 (PDF p-35/36) — footnote 3 → J. Am. Chem. Soc. 2007, 129, 8968-8969 (salvinorin A synthesis)

- Natural product: salvinorin A (*Salvia divinorum*, κ -opioid agonist, hallucinogen).
- Starting material 1: 1-(furan-3-yl)but-2-yn-1-ol, CC#CC(O)c1ccoc1 [H].
- Reagents: 1) NaNH₂; 2) H₂O → 2: 1-(furan-3-yl)but-3-yn-1-ol, C#CCC(O)c1ccoc1 [H].
- Asked: mechanism (hint: problem 10.8).
- Answer: alkyne "zipper" isomerization. Repeated deprotonation and reprotonation moves the triple bond through an allene intermediate (internal alkyne → allene → terminal alkyne). Irreversible deprotonation of the terminal alkyne to the acetylide (pK_a ~25, NH₃ ~38) drives the equilibrium. H₂O workup then protonates the acetylide. The alkoxide is also formed and is reprotonated on workup [H].

Problem 10.72 — book p. 495 (PDF p-36) — footnote 4 → Tetrahedron Lett. 1996, 37, 1571-1574

- Starting material: cyclodeca-1,6-diyne, C1CC#CCCCC#CC1 [H].
- Reagent: I₂ → diiododiene (94%), a transannular cyclization. Product: bicyclic (two fused 6-membered rings), conjugated diene with two vinyl iodides, IC1=C2CCCC(I)=C2CCC1 (C₁₀H₁₂I₂) [M-H].
- Asked: (a) draw the bridged iodonium ion (iodirenium) from attack of the alkyne on I₂; (b) draw the vinyl carbocation that would form if the other alkyne attacked as a nucleophile, and explain why it is high in energy (sp-hybridized vinyl cation, positive charge on an electronegative sp carbon, little stabilization); (c) draw the concerted step: the second alkyne attacks the iodonium while iodide attacks the developing cation (iodonium opening and transannular C-C bond formation together, avoiding a free vinyl cation), so the full mechanism has two steps [M-H].

Problem 10.73 — book p. 495 (PDF p-36) — footnote 5 → Acc. Chem. Res. 1988, 21, 442-449 (stable simple enols)

- Compounds: 1 = diphenylacetaldehyde, O=CC(c1ccccc1)c1ccccc1 (9.1% enol); 2 = dimesitylacetaldehyde, O=CC(c1c(C)cc(C)cc1C)c1c(C)cc(C)cc1C (95% enol). Compare (CH₃)₂CHCHO at 0.014% enol.
- Asked: (a) hybridization and bond angles at the carbon carrying both rings (sp³, ~109.5°); (b) the enol of 1 and the same for that carbon in the enol (sp², ~120°); (c) whether the angle increases or decreases on enolization, and why sterics favor the enol; (d) why enol 2 is more favored than enol 1.
- Answer: on enolization the angle widens from 109.5° to 120°, which relieves steric repulsion between the two bulky aryls. The enol C=C is also conjugated with the aryl rings. Mesityl groups (ortho-methyls) are much bulkier, so relief of steric strain in 2 is larger and the keto form of 2 is destabilized [H].

Problem 10.74 — book p. 495 (PDF p-36) — footnote 6 → J. Am. Chem. Soc. 1966, 88, 4525-4526

- Starting material 1: 1,2-bis(phenylethynyl)benzene, C(#Cc1ccccc1)c1ccccc1C#Cc1ccccc1 [H].
- Reactions:
- H₂SO₄, H₂O → 3-benzoyl-2-phenyl-1H-indene, O=C(c1ccccc1)C1=C(c2ccccc2)Cc2ccccc21 [H].
- Br₂ → 3-bromo-1-[bromo(phenyl)methylene]-2-phenyl-1H-indene, BrC(c1ccccc1)=C1C(c2ccccc2)=C(Br)c2ccccc21 (exocyclic alkene). As drawn, the exocyclic Br points toward the benzo ring and that carbon's Ph points toward C2 [M].
- Asked: arrow-pushing mechanisms for both, and an explanation of the geometry of the exocyclic alkene in the bromination product.
- Answer:
- Hydration: H⁺ adds to the arene-bound alkyne carbon (Ca), giving a Ph-stabilized vinyl cation at Cb. The other alkyne's arene-bound carbon Cc attacks Cb (5-membered ring closure), giving a new Ph-stabilized vinyl cation at Cd. Water attacks Cd, then deprotonation gives the enol, which tautomerizes to the ketone. A final acid-catalyzed tautomerization/H shift gives the conjugated indene [M-H].
- Bromination: Br⁺ adds to Ca (bromonium or vinyl cation), and the same cascade gives an exocyclic vinyl cation at Cd. Its empty p orbital lies in the ring plane. Br⁻ attacks in plane from the less hindered side, away from the C2-phenyl and toward the benzo/peri H side, which sets the geometry [M].

Problem 10.75 (도전 문제) — book p. 495 (PDF p-36) — footnote 7 → J. Am. Chem. Soc. 2008, 130, 6281-6287 (roquefortine C)

- Natural product: roquefortine C (*Penicillium roqueforti*, blue cheese; antibacterial). Indoline-fused diketopiperazine with a reverse-prenyl ($\text{C}(\text{CH}_3)_2\text{CH}=\text{CH}_2$) group, a dehydrohistidine exocyclic alkylidene, and an imidazole. It exists as two imidazole tautomers: the 1H ($\text{N}\pi\text{-H}$) form on the left and the 3H ($\text{N}\pi\text{-H}$, NH adjacent to the vinyl-substituted carbon) form on the right. Full SMILES uncertain (complex stereo).
- Asked: (a) base-catalyzed tautomerization mechanism; (b) acid-catalyzed mechanism; (c) which tautomer is more stable, and why.
- Answer: (a) base removes the imidazole N-H to give an imidazolid anion (resonance delocalized over both N), which is reprotonated at the other N [H]. (b) Acid protonates the basic imidazole sp^2 N to give an imidazolium cation, and the original N-H is then deprotonated [H]. (c) Likely the right-hand (3H) tautomer: its N-H can form an intramolecular hydrogen bond to the nearby diketopiperazine $\text{C}=\text{O}$, and it is conjugated through the exocyclic alkene. The paper reports a solvent-dependent mixture, and the H-bonded tautomer is favored in nonpolar media [L-M; check against the paper].

Problem 10.76 (도전 문제) — book pp. 495-496 (PDF p-36/37) — footnote 8 → Tetrahedron Lett. 2001, 42, 7277-7280

- Topic: N-reverse-prenylated indoles (N-1,1-dimethyl-2-propenyl), antitumor/antifungal.
- Starting material 1: indoline, C1Cc2ccccc2N1. Reagents: 2-methylbut-3-en-2-yl acetate (CC(C)(C#C)OC(C)=O), CuCl (cat.), *i*-Pr₂NEt → 2 (C13H15N), then → 3 = N-(2-methylbut-3-en-2-yl)indoline, CC(C)(C=C)N1CCc2ccccc21.
- Asked: (a) structure of 2 and mechanism, ignoring the catalyst; (b) reagents for 2 → 3.
- Answer: (a) 2 = N-(2-methylbut-3-en-2-yl)indoline, CC(C)(C#C)N1CCc2ccccc21 (C13H15N, checked). Mechanism treated as $\text{S}_{\text{N}}1$: the acetate leaves to give a resonance-stabilized tertiary propargylic cation (with an allenyl resonance form). The indoline N attacks the tertiary carbon, and *i*-Pr₂NEt removes N-H [H]. (b) H₂, Lindlar catalyst (or Ni₂B/P-2) → terminal alkene [H].

Problem 10.77 (도전 문제) — book p. 496 (PDF p-37) — footnote 9 → J. Phys. Chem. A 2007, 111, 3513-3518

- Compound: 3-heptyne, CCCC#CC, treated as a "stretched" pentane with the alkyne inserted between pentane C2 and C3. Pentane's lowest conformers are anti-anti and anti-gauche.
- Finding: in 3-heptyne's two most stable conformers, C1 and C6 are nearly eclipsed (sighting down the alkyne). C4/C7 are anti about C5-C6 in one conformer and gauche in the other.
- Asked: (a) wedge-dash drawings of pentane's anti-anti and anti-gauche conformers; (b) the 3-heptyne analog of anti-anti; (c) Newman projection sighted down the $\text{C}\equiv\text{C}$ axis ($\text{C}2\cdots\text{C}5$) showing C1 eclipsing C6, which is acceptable because the 4.1 Å separation removes steric strain and there is a slight dispersion attraction; (d) Newman projections about C5-C6 with C4 and C7 anti vs gauche [M-H].

Problem 10.78 (도전 문제) — book p. 496 (PDF p-37) — footnote 10 → J. Am. Chem. Soc. 1976, 98, 3295-3300

- Substrates: $\text{Ph-C}\equiv\text{C-R}$ (1a R = Me, CC#Cc1ccccc1; 1b Et; 1c *i*-Pr; 1d *t*-Bu) + HCl → $\text{PhC}(\text{Cl})=\text{CHR}$, E:Z = 70:30, 80:20, 95:5, 100:0.

- As drawn, E means Cl trans to R (R cis to Ph).
- Asked: (a) the vinyl cation from protonation of a phenyl-substituted acetylene, and the regioselectivity; (b) the stereoselectivity (E:Z) from transition-state stability during capture by chloride.
- Answer: (a) H^+ adds to the R-bearing carbon, giving a linear sp vinyl cation $Ph-C^+=CHR$ stabilized by resonance with phenyl, so Cl ends up on the Ph-bearing carbon [H]. (b) The empty p orbital lies in the plane of the C2 substituents (H and R). Cl^- approaches in plane from the H side (less hindered) rather than the R side, giving Cl cis to H and trans to R (E). E-selectivity rises as R gets bulkier ($Me < Et < i-Pr < t-Bu$) [H].

Chapter 11 — 라디칼 반응 (Radical Reactions) — PDF p-38..p-45, book pp. 533-540

Summary (반응의 복습, book p. 533; 개념 pp. 533-534, PDF p-38/39)

Radical reactions useful in synthesis: - Alkane bromination: $Br_2, h\nu \rightarrow Br$ at the most substituted carbon ($3^\circ \gg 2^\circ > 1^\circ$; much more selective than Cl_2). - Anti-Markovnikov HBr addition: $HBr, ROOR \rightarrow Br$ on the less substituted carbon (radical chain). - Allylic bromination: $NBS, h\nu$ (or heat/peroxide) $\rightarrow$ allylic bromide; the allylic radical can give two regioisomers. - Also: radical stability $3^\circ > 2^\circ > 1^\circ$ (allylic/benzylic resonance stabilized; vinylic not). Six arrow patterns: homolysis, addition to π , H abstraction, halogen abstraction, elimination, coupling. Initiation / propagation / termination. Initiators: peroxides, acyl peroxides, AIBN. Inhibitors: O_2 , hydroquinone. A new stereocenter formed at a radical carbon $\rightarrow$ racemic. Ozone/CFCs. Autoxidation $\rightarrow$ hydroperoxides. Antioxidants (BHT, BHA, vitamins E, C). Radical polymerization (PE, PVC), cracking/hydrocracking. Halogenation installs a functional group on an alkane.

참고문헌 (book p. 540, PDF p-45), exact text

1. J. Am. Chem. Soc. 2012, 134, 6916-6919.
2. J. Phys. Chem. A 2008, 112, 7930-7938.
3. J. Org. Chem. 2009, 74, 5025-5031.
4. J. Am. Chem. Soc. 2000, 122, 3367-3374.
5. J. Am. Chem. Soc. 2006, 128, 6310-6311.
6. J. Am. Chem. Soc. 1986, 108, 1708-1709.
7. Tetrahedron Lett. 2006, 47, 2115-2118.
8. J. Am. Chem. Soc. 2001, 123, 1842-1847.
9. Tetrahedron Lett. 2002, 43, 8587-8590.
10. J. Am. Chem. Soc. 2009, 131, 653-661.

Problem 11.49 — book p. 538 (PDF p-43) — footnote 1 $\rightarrow$ J. Am. Chem. Soc. 2012, 134, 6916-6919 (footnote is on "특별한 효소")

- Reaction (thiol-ene coupling): $R-CH=CH_2 + H-S-R'$, radical initiator (Δ or $h\nu$) $\rightarrow R-CH_2-CH_2-S-R'$ (anti-Markovnikov).
- Asked: (a) mechanism with initiation and propagation; (b) the product expected when a protein bearing an N-allyl amide (protein- $C(=O)NH-CH_2CH=CH_2$) is thiol-ene coupled with a protein thiol (protein-SH,

cysteine).

- Answer: (a) Initiation: the initiator radical abstracts H from R'S-H to give R'S • . Propagation: (1) R'S • adds to the terminal CH₂, giving the more stable 2° radical; (2) that radical abstracts H from R'SH, giving the product and regenerating R'S • [H]. (b) Protein-C(=O)NH-CH₂CH₂CH₂-S-protein, a thioether link made by anti-Markovnikov addition [H].

Problem 11.50 — book p. 538 (PDF p-43) — footnote 2 → J. Phys. Chem. A 2008, 112, 7930-7938

- Starting material: 1-bromopropane, BrCCC (industrial solvent, CFC replacement). Atmospheric oxidation → bromoacetone, BrCC(C)=O (main product) + others.
- Asked: mechanism with 4 given steps.
- Answer: (1) HO • abstracts the C2-H (secondary) to give BrCH₂C • HCH₃ + H₂O. (2) O₂ couples to give the peroxy radical BrCH₂CH(OO •)CH₃. (3) NO abstracts an O atom to give NO₂ + the alkoxy radical BrCH₂CH(O •)CH₃. (4) O₂ abstracts the C2-H α to O • , giving BrCH₂C(=O)CH₃ + HOO • [H].

Problem 11.51 — book p. 538 (PDF p-43) — footnote 3 → J. Org. Chem. 2009, 74, 5025-5031

- Compound: resveratrol (trans-3,5,4'-trihydroxystilbene; grapes, mulberry, peanuts; antioxidant), Oc1ccc(/C=C/c2cc(O)cc(O)c2)cc1.
- Reaction: alkoxy radical RO • abstracts a phenolic H.
- Asked: which H is abstracted preferentially, and why.
- Answer: the 4'-OH (para to the vinyl group). The resulting phenoxyl radical is delocalized through the para position onto the C=C and into the other ring (extended conjugation; many resonance forms). Radicals from the 3- or 5-OH (meta) cannot delocalize onto the alkene [H].

Problem 11.52 — book p. 538 (PDF p-43) — footnote 4 → J. Am. Chem. Soc. 2000, 122, 3367-3374

- Concept: resonance of a radical next to an atom with a lone pair (P, S can expand the octet).
- Reaction: cinnamyl arenesulfenate PhCH=CHCH₂-O-S-Ar, heat → cinnamyl sulfoxide PhCH=CHCH₂-S(=O)-Ar. Reported to go by C-O homolysis followed by radical recombination (two steps).
- Asked: (a) mechanism; (b) why C-O breaks first rather than O-S.
- Answer: (a) C-O homolysis gives the cinnamyl radical (allylic, conjugated with Ph) + the arylsulfinyl radical ArS-O • ↔ ArS • =O (delocalized over O and S). Recombination at S gives the sulfoxide [M-H]. (b) C-O cleavage gives two resonance-stabilized radicals (allylic/benzylic-type cinnamyl; sulfinyl delocalized onto S through the adjacent lone pair). O-S cleavage would give an unstabilized alkoxyl radical, so C-O is the weaker bond here [M-H].

Problem 11.53 — book p. 539 (PDF p-44) — footnote 5 → J. Am. Chem. Soc. 2006, 128, 6310-6311 (oseltamivir synthesis)

- Target: oseltamivir (Tamiflu, anti-influenza). 2 is a key intermediate.
- Starting material 1: bicyclic iodolactam, a 6-azabicyclo[3.2.1]octan-7-one with N-R. I sits on the ring carbon next to the N-bearing bridgehead (I wedge). Constitution IC1CCC2CC1N(R)C2=O (R generic; shown with R = CH₃ in the RDKit check) [M].

- Reagents: 1) t-BuOK; 2) NBS, hv → 2: allylic bromide BrC1C=CC2CC1N(R)C2=O. Br is on the same carbon and the same face (wedge) as the I was; the C=C has moved to the next two carbons, away from the bridgehead [M].
- Asked: mechanism and stereochemical outcome.
- Answer: E2 with t-BuOK removes the anti-periplanar H from the CH₂ away from the bridgehead (Bredt's rule: no alkene to the bridgehead), giving the alkene. NBS/hv then abstracts the allylic H at the CH₂ next to the other bridgehead (the bridgehead C-H is not usable). The resulting allylic radical is delocalized, and Br₂ delivers Br to the other allylic terminus, moving the alkene. Br adds from the less hindered (exo) face, away from the lactam bridge, so a single diastereomer forms [M].

Problem 11.54 (도전 문제) — book p. 539 (PDF p-44) — footnote 6 → J. Am. Chem. Soc. 1986, 108, 1708-1709

- Starting radical: $\cdot C(CN)(CO_2Et)-CH_2CH_2-(\text{cycloocta-2,6-dien-1-yl})$, modeled as [C](C#N)(C(=O)OCC)CCC1C=CC=CC1. Diene positions inferred from the products [L-M].
- Products: two tricyclic radicals, each formed by two new ring closures. (i) A linear triquinane radical (three fused cyclopentanes; C(CO₂Et)(CN) in the terminal ring; radical in the far ring). (ii) A bridged tricycle (5-ring + bicyclic containing C(CN)(CO₂Et)) with the radical in the cyclopentane.
- Asked: mechanisms for each two-ring formation.
- Answer: (i) 5-exo-trig: the α-cyano-ester radical attacks ring C2 (forming a cyclopentane with CH₂CH₂ and C1), leaving a radical at C3. Transannular C3→C7 addition then forms the bicyclo[3.3.0]octane, leaving a radical at C6 (linear triquinane) [M]. (ii) 6-endo addition at C3 first (radical at C2), then transannular C2→C6, leaving a radical at C7 (bridged product) [L].

Problem 11.55 (도전 문제) — book p. 539 (PDF p-44) — footnote 7 → Tetrahedron Lett. 2006, 47, 2115-2118

- Starting material: bisphenyl carbodiazone PhN=N-C(=O)-N=NPh, O=C(N=Nc1ccccc1)N=Nc1ccccc1.
- Conditions: hv (350 nm), TEMPO → three products: PhN=N-C(=O)-O-TMP (O=C(N=Nc1ccccc1)ON1C(C)(C)CCCC1(C)C), Ph-O-TMP (CC1(C)CCCC(C)(C)N1Oc1ccccc1), and N₂.
- Asked: a mechanism explaining all three products.
- Answer: homolysis of one C-N bond gives PhN=N-C(=O)· (acyl-type radical) + PhN=N·. PhN=N· loses N₂ to give Ph·. TEMPO traps Ph· (O-C coupling) and traps the acyl radical, giving the carbamoyl-type ester [H].

Problem 11.56 (도전 문제) — book p. 539 (PDF p-44) — footnote 8 → J. Am. Chem. Soc. 2001, 123, 1842-1847

- Reaction: cubane (1, C12C3C4C1C5C2C3C45) + I-Z (2a I₂: no reaction; 2b Cl₄) → iodocubane (3, IC12C3C4C1C5C2C3C45) + H-Z (4a HI; 4b HCl₃, iodoform). Initiated by $\cdot Cl_3$.
- Asked: (a) propagation cycle; (b) three possible termination steps; (c) given BDE(I-Cl₃) = 192 and BDE(H-Cl₃) = 423 kJ/mol, why 2b works and 2a does not.
- Answer: (a) $\cdot Cl_3 + \text{cubane-H} \rightarrow HCl_3 + \text{cubyl} \cdot$; $\text{cubyl} \cdot + Cl_4 \rightarrow \text{cubyl-I} + \cdot Cl_3$ [H]. (b) $\text{cubyl} \cdot + \cdot Cl_3 \rightarrow \text{cubyl-Cl}_3$; $2 \cdot Cl_3 \rightarrow I_3C-Cl_3$; $2 \text{ cubyl} \cdot \rightarrow \text{bicubyl}$ [H]. (c) H abstraction by I $\cdot$ forms H-I (~297 kJ/mol), while the cubane C-H costs ~400+ kJ/mol, so it is strongly endothermic. $\cdot Cl_3$ forms H-Cl₃ (423 kJ/mol),

so H abstraction is roughly thermoneutral or favorable. The weak I-Cl₃ bond (192) makes iodine transfer to cubyl • exothermic. Both propagation steps are thermodynamically feasible only with Cl₄ [H].

Problem 11.57 (도전 문제) — book p. 539 (PDF p-44) — footnote 9 → Tetrahedron Lett. 2002, 43, 8587-8590 (paclitaxel fragment)

- (a) Kharasch reaction: $R-CH=CH_2 + CCl_4, (PhCO_2)_2, \text{heat} \rightarrow R-CH(Cl)-CH_2-CCl_3$.
- Answer: initiation: $PhCO_2-O_2CPh \rightarrow 2 PhCO_2 \cdot$, then $Ph \cdot + CO_2$; $Ph \cdot$ (or $PhCO_2 \cdot$) abstracts Cl from CCl_4 to give $\cdot CCl_3$. Propagation: $\cdot CCl_3$ adds to the terminal CH_2 (less hindered, gives the 2° radical); the radical abstracts Cl from CCl_4 to give the product and regenerate $\cdot CCl_3$. Regiochemistry follows the more stable radical intermediate [H].
- (b) Substrate: a furanose acetonide (1,2-O-isopropylidene) with a C3 quaternary carbinol bearing a vinyl group, and C4 bearing $CH_2-CH(OH)-C(CH_3)=CH_2$. SMILES C=CC1(O)C2OC(C)(C)OC2OC1CC(O)C(C)=C (C₁₄H₂₂O₅) [M-H].
- Conditions: $CCl_4, (PhCOO)_2, \text{heat} \rightarrow$ cyclohexane fused to the furanose: ring $C(CH_3)(CH_2CCl_3)-CH(OH)-CH_2-CH(O)-C(OH)-CH(CH_2Cl)$. SMILES OC1CC2OC3OC(C)(C)OC3C2(O)C(CCl)C1(C)CC(C1)(Cl)Cl (C₁₅H₂₂Cl₄O₅ = SM + CCl_4 , checked) [M-H].
- Answer: $\cdot CCl_3$ adds to the $=CH_2$ of the isopropenyl group, giving a 3° radical. 6-exo-trig cyclization onto the internal CH of the vinyl group gives a primary $CH_2 \cdot$ radical, which abstracts Cl from CCl_4 (CH_2Cl) and regenerates $\cdot CCl_3$ [H].

Problem 11.58 (도전 문제) — book p. 540 (PDF p-45) — footnote 10 → J. Am. Chem. Soc. 2009, 131, 653-661

- Reaction: phenylacetylene (C#Cc1ccccc1) + nitrosobenzene (O=Nc1ccccc1), heat $\rightarrow$ diradical $Ph-C \cdot =CH-N(O \cdot)-Ph$ (vinyl radical + nitroxide) $\rightarrow$ N-hydroxy-3-phenylindole, ON1C=C(c2ccccc2)c2ccccc21 [H].
- Asked: stepwise mechanism from the diradical to the product.
- Answer: the vinyl radical (carbon bearing Ph) adds to the ortho carbon of the N-phenyl ring (5-membered ring closure), giving a cyclohexadienyl radical alongside the nitroxide. Transfer of the ring ipso/ortho H (rearomatization; intramolecular H-atom transfer, or tautomerization) to the nitroxide O gives the aromatic indole N-OH [M-H].

Chapter 12 — 합성 (Synthesis) — PDF p-46..p-51, book pp. 561-566 (bonus: in my page range although the file is labeled 7-11)

Summary (개념과 용어의 복습 p. 561 + 학습능력 p. 562, PDF p-46/47)

- Chapter-opening box: Taxol, synthesized in 1994 by Holton (Florida State) and Nicolaou (Scripps). Semisynthesis from 10-deacetylbaccatin III (Taxus baccata needles); Taxotere (docetaxel).
- Moving a halogen: elimination then addition (e.g., NaOEt; then HBr Markovnikov or HBr/ROOR).
- Moving a π bond: addition then elimination (HBr; then t-BuOK or NaOEt).
- Alkane $\rightleftharpoons$ alkene $\rightleftharpoons$ alkyne interconversion:
- alkyne $\rightarrow$ cis-alkene: H₂/Lindlar; alkyne $\rightarrow$ trans-alkene: Na/NH₃.

- alkene → alkane: H₂/Pt.
- alkane → alkene: 1) Br₂, hv; 2) NaOEt.
- alkene → alkyne: 1) Br₂/CCl₄; 2) excess NaNH₂; 3) H₂O.
- C-C bond formation: acetylide + alkyl halide. C-C cleavage: ozonolysis (1) O₃; 2) DMS).
- Two questions for any synthesis problem: (1) does the carbon skeleton change? (2) does the identity or position of the functional group change? Retrosynthetic analysis; green chemistry; multiple correct answers.
- Comprehensive problems 12.27/12.28 introduce acetylide + epoxide → homopropargylic alcohol, and acetylide + ketone/aldehyde → propargylic alcohol (protonation as a separate step).

참고문헌 (book p. 566, PDF p-51), exact text

1. J. Agric. Food Chem. 2003, 51, 4344-4348.
2. Agric. Biol. Chem. 1988, 52, 1053-1055.
3. J. Agric. Food Chem. 2005, 53, 7204-7211.
4. J. Agric. Food Chem. 1983, 31, 1287-1292.
5. J. Nat. Prod. 2012, 75, 1765-1776.
6. Nat. Prod. Rep. 2012, 29, 1288-1303.
7. J. Agric. Food Chem. 2002, 50, 3401-3404.
8. Actinomycetologica 2009, 23, 27-33.
9. J. Agric. Food Chem. 2009, 57, 3818-3830.
10. Tetrahedron 1977, 33, 2447-2450.

Problem 12.32 — book p. 564 (PDF p-49) — footnote 1 → J. Agric. Food Chem. 2003, 51, 4344-4348

- Target: 3-phenylpropyl acetate, CC(=O)OCCc1ccccc1 (one of 27 volatiles in Sri Lankan endemic cinnamon). Carbon sources: toluene (Cc1ccccc1) plus two-carbon reactants.
- Proposed route [M]: 1. Toluene + NBS, hv → benzyl bromide. 2. + HC≡CNa → 3-phenylpropyne. 3. H₂, Lindlar → allylbenzene. 4. HBr, ROOR → 1-bromo-3-phenylpropane. 5. CH₃CO₂⁻Na⁺ (S_N2) → ester.
- Alternative for the last steps: hydroboration-oxidation → 3-phenyl-1-propanol, then acetylation.

Problem 12.33 — book p. 565 (PDF p-50) — footnote 2 → Agric. Biol. Chem. 1988, 52, 1053-1055

- Target: 2,5-hexanedione, CC(=O)CCC(C)=O (one of 33 volatiles of kakkon, Pueraria root). Starting material: 3,4-dimethylcyclobutene, CC1C=CC1C.
- Key point: direct ozonolysis would give 2,3-dimethylsuccinaldehyde, so the C=C must first move to the methyl-bearing carbons.
- Route [M-H]: 1. H₂, Pt → 1,2-dimethylcyclobutane. 2. Br₂, hv → 1-bromo-1,2-dimethylcyclobutane (3°). 3. NaOEt (Zaitsev) → 1,2-dimethylcyclobutene. 4. O₃; DMS → 2,5-hexanedione.

Problem 12.34 — book p. 565 (PDF p-50) — footnote 3 → J. Agric. Food Chem. 2005, 53, 7204-7211

- Target: 1-penten-3-ol, C=CC(O)CC (volatile of indoor- vs outdoor-raised quail feed). Acetylene is the only carbon source; use the acetylide + carbonyl reaction of 12.28.
- Route [M-H]: 1. Make propanal:
 - Acetylene + H₂/Lindlar → ethylene; + HBr → EtBr.
 - HC≡CNa + EtBr → 1-butyne; H₂/Lindlar → 1-butene; O₃/DMS → propanal (+ HCHO). 2. HC≡CNa + propanal, then H₂O → 1-pentyn-3-ol. 3. H₂, Lindlar → 1-penten-3-ol.

Problem 12.35 (도전 문제) — book p. 565 (PDF p-50) — footnote 4 → J. Agric. Food Chem. 1983, 31, 1287-1292

- Target: tetradecane (C₁₄ linear; one of 130 volatiles from fried chicken). Acetylene is the only carbon source.
- Route [M]: 1. Make 1-bromohexane by iterative two-carbon homologation:
 - EtBr (acetylene → H₂/Lindlar → HBr).
 - HC≡CNa + EtBr → 1-butyne; Lindlar → 1-butene; HBr/ROOR → 1-bromobutane.
 - HC≡CNa + 1-bromobutane → 1-hexyne; Lindlar → 1-hexene; HBr/ROOR → 1-bromohexane. 2. Dialkylate acetylene (NaNH₂ / 1-bromohexane, twice) → 7-tetradecyne. 3. H₂, Pt → tetradecane.
- Many equivalent routes are possible.

Problem 12.36 (도전 문제) — book p. 565 (PDF p-50) — footnote 5 → J. Nat. Prod. 2012, 75, 1765-1776

- Targets: 2-decanone (CCCCCCCCC(C)=O) and decanal (CCCCCCCCC=O), structural isomers among 254 microbial volatiles. Carbon sources: only alkenes with fewer than 6 carbons.
- Common intermediate: 1-decyne.
- 2-Decanone: H₂SO₄, H₂O, HgSO₄ (Markovnikov hydration).
- Decanal: 1) R₂BH (disiamylborane/9-BBN); 2) H₂O₂, NaOH.
- Making 1-decyne [M]:
- Acetylide from ethylene: Br₂; excess NaNH₂.
- 1-Bromooctane by iterative homologation: 1-butene → HBr/ROOR → BuBr; HC≡CNa → 1-hexyne; Lindlar → 1-hexene; HBr/ROOR → 1-bromohexane; repeat → 1-bromooctane.
- HC≡CNa + 1-bromooctane → 1-decyne.
- The book's intended route may differ.

Problem 12.37 (도전 문제) — book p. 565 (PDF p-50) — footnote 6 → Nat. Prod. Rep. 2012, 29, 1288-1303

- Target: (Z)-3-hexenyl acetate, CC/C=C\CCCC(C)=O (green-leaf volatile released by wounded Arabidopsis). Carbon sources: only ethylene and acetic acid. Must use the acetylide + epoxide reaction (12.27).
- Route [M-H]: 1. Ethylene → Br₂; excess NaNH₂ → acetylene. 2. Ethylene + HBr → EtBr; HC≡CNa + EtBr → 1-butyne. 3. NaNH₂; then ethylene oxide (from ethylene + RCO₃H); then H₂O → 3-hexyn-1-ol. 4. H₂, Lindlar → (Z)-3-hexen-1-ol. 5. Acetic acid, H⁺ (Fischer esterification) → target.

Problem 12.38 (도전 문제) — book p. 565 (PDF p-50) — footnote 7 → J. Agric. Food Chem. 2002, 50, 3401-3404

- Targets: 3-methylbutanal (CC(C)CC=O) and hexanal (CCCCCC=O) from tomato aroma (367 compounds).
Asked: a single synthesis that gives an equimolar mixture of both, starting from one 1°, one 2° and one 3° alcohol, each with fewer than 6 C. (Hint: 1°/2°/3° alcohols are named like alkyl halides.)
- Key idea [H]: ozonolysis (O3; DMS) of (Z)- or (E)-2-methylundec-4-ene, CC(C)C/C=C\CCCC, gives exactly 1:1 3-methylbutanal + hexanal.
- Alkene from an internal alkyne (CH3)2CHCH2-C#C-(CH2)4CH3, then H2/Lindlar.
- Assembly [L]:
- Isobutyl bromide from 2-methyl-2-propanol (3°): H2SO4/heat → isobutylene; HBr/ROOR.
- 1-Bromopentane from 1-pentanol (1°): conc. H2SO4 → 1-pentene; HBr/ROOR. Or HBr directly.
- The C2 alkyne unit would be acetylene, alkylated twice.
- I could not identify how the 2° alcohol fits; the intended disconnection is unclear (flag).

Problem 12.39 (도전 문제) — book p. 565 (PDF p-50) — footnote 8 → Actinomycetologica 2009, 23, 27-33

- Target: methyl 4-methylpentanoate, CCOC(=O)CCC(C)C (Streptomyces volatile). Starting material: 2-methylbutane, CCC(C)C.
- Route [M]: 1. Br2, hv → 2-bromo-2-methylbutane. 2. NaOEt → 2-methyl-2-butene. 3. HBr, ROOR → 2-bromo-3-methylbutane. 4. t-BuOK (Hofmann) → 3-methyl-1-butene. 5. HBr, ROOR → 1-bromo-3-methylbutane. 6. HC≡CNa → 5-methyl-1-hexyne. 7. 1) O3; 2) H2O → 4-methylpentanoic acid + CO2. 8. CH3OH, H+ (Fischer) → methyl ester.
- Net: +2 C from the acetylide, -1 C lost by ozonolysis.

Problem 12.40 (도전 문제) — book p. 565 (PDF p-50) — footnote 9 → J. Agric. Food Chem. 2009, 57, 3818-3830

- Target: (E)-2-hexenal, CCC/C=C/C=O (Cabernet Sauvignon grape volatile). Starting material: 1,1-dibromopentane, CCCC(Br)Br. Use the 12.28 chemistry (acetylide + carbonyl).
- Route [M]: 1. Excess NaNH2 → pentynide (1-pentyne anion). 2. H2C=O; then H2O → 2-hexyn-1-ol. 3. Na, NH3(l) → (E)-2-hexen-1-ol. 4. Oxidation (PCC; next chapter) → (E)-2-hexenal.

Problem 12.41 (도전 문제) — book pp. 565-566 (PDF p-50/51) — footnote 10 → Tetrahedron 1977, 33, 2447-2450 (dermestid beetle sex pheromone)

- Starting material 1: 5-methylhept-1-ene, C=CCC[C@@H](C)CC. The stereocenter is drawn with a hashed methyl and computes as S as drawn (RDKit).
- Sequence [H for constitution]:
- 1 → 2 (BH3·THF): trialkylborane (5-methylheptyl)3B (anti-Markovnikov).
- 2 → 3 (1) Br2; 2) NaOMe): alkyl bromide 3 = 1-bromo-5-methylheptane, BrCCCC[C@@H](C)CC (given).
- 3 → 4 (HC≡CLi, DMSO): SN2 → 7-methylnon-1-yne, C#CCCC[C@@H](C)CC.
- 4 → 5 (1) BuLi; 2) I(CH2)7OTHP): THPO(CH2)7C≡C(CH2)4CH(CH3)CH2CH3.
- 5 → 6 (TsOH, MeOH; THP removal): 14-methylhexadec-8-yn-1-ol, CCCCCCCC#CCCC[C@@H](C)CC (C17H32O).

- 6 → 7 (H₂, Lindlar): (Z)-14-methylhexadec-8-en-1-ol, CCCCCCCC/C=C\CCCC[C@@H](C)CC (C₁₇H₃₄O), a Trogoderma pheromone.
- The stereocenter is untouched throughout, and its CIP label stays the same in every compound.
- Caveat: the literature natural pheromone is often cited as (R,Z), while the book's hashed drawing computes as S. Check the drawing direction [M].

Klein 유기화학 — PDF "12-14" (p-01 … p-20) extraction

Source: /root/.claude/uploads/e196aa5e-9821-56b2-b877-2f07b790959e/71ffacf4-12-14.pdf (20 pages)

Page map (no chapter 12 content present)

PDF page	Book page	Content
p-01	609	Ch.13 반응의 복습 (end), 개념과 용어의 복습 13.1-13.10
p-02 - p-04	610-612	Ch.13 개념 복습 (end), 학습능력 기르기 복습 13.1-13.9, 연습 문제 13.28-
p-05 - p-07	613-615	Ch.13 연습 문제 to 13.50; 종합 문제 13.51-13.63
p-08 - p-09	616-617	Ch.13 도전 문제 13.64-13.73 + Ch.13 참고문헌 1-10
p-10	653	Ch.14 학습능력 응용 14.25-14.26 (body, end of §14.12), 반응의 복습 begins
p-11 - p-13	654-656	Ch.14 반응의 복습, 개념과 용어의 복습, 학습능력 기르기 복습 14.1-14.4
p-14 - p-16	657-659	Ch.14 학습능력 14.5-14.7, 연습 문제 14.27-14.46, 종합 문제 14.47-
p-17 - p-18	660-661	Ch.14 종합 문제 14.49k-14.64
p-19 - p-20	662-663	Ch.14 14.63-14.64 (cont.), 도전 문제 14.65-14.70 + Ch.14 참고문헌 1-11

The file name says 12-14, but the pages hold only the end of Ch.13 (pp. 609-617) and the end of Ch.14 (pp. 653-663). No Ch.12 pages are present.

제13장 알코올과 Phenol (Alcohols and Phenols)

Key reactions (summary pages 609-612)

- Alcohol → alkyl chloride with inversion: SOCl₂/py (R→S shown); HCl/ZnCl₂ also shown in the summary scheme.
- E1/E2: tertiary alcohol + conc. H₂SO₄, heat → alkene + H₂O; ROH → TsCl/py → ROTs → NaOEt → alkene (E2).
- Oxidation: 2° alcohol + Na₂Cr₂O₇/H₂SO₄/H₂O → ketone. 1° alcohol + Na₂Cr₂O₇ (chromic acid) → carboxylic acid. 1° alcohol + PCC/CH₂Cl₂, DMP/CH₂Cl₂, or Swern (1) DMSO, (COCl)₂; 2) Et₃N → aldehyde. Phenol + Na₂Cr₂O₇/H₂SO₄ → benzoquinone.
- Concept review (13.1-13.13): alkoxide/phenoxide, pK_a 15-18; NaBH₄ vs LiAlH₄ (LiAlH₄ reduces acids/esters); Grignard + ketone/aldehyde/ester (ester adds 2 R); TMS protecting group for Grignard;

SN1 (3°) vs SN2 (1°/2°: HX, SOCl₂, PBr₃, or TsCl then nucleophile); oxidation of 1°/2°/3° alcohols; NADH/NAD⁺; phenol → quinone; synthesis = carbon skeleton change + functional-group change.

- Skill builders 13.1-13.9: naming, acidity (resonance/induction/solvation), oxidation state, LiAlH₄ mechanism, Grignard retrosynthesis, ROH→RCl (TsCl, py then NaCl; or SOCl₂/py, inversion), oxidation products, functional-group interconversion map, synthesis design.

Literature-footnoted problems, Ch.13

Ch.13 참고문헌 (p.617), verbatim

1. J. Org. Chem. **2012**, 77, 3390-3400.
2. J. Org. Chem. **2000**, 65, 9249-9251.
3. J. Agric. Food Chem. **1986**, 34, 140-144.
4. Tetrahedron Lett. **2007**, 48, 8479-8481.
5. J. Org. Chem. **2009**, 74, 5458-5470.
6. J. Nat. Prod. **2011**, 74, 1262-1267.
7. J. Nat. Prod. **1987**, 50, 990-991.
8. J. Nat. Prod. **1996**, 59, 49-51.
9. J. Chem. Ecol. **2012**, 38, 814-824.
10. J. Am. Chem. Soc. **1989**, 111, 6682-6690.

Refs 1-3 are not cited in any problem on these pages. No end-of-chapter problem from 13.28 to 13.73 has footnote marks 1, 2 or 3. The first footnote mark in the problems is "4", in 13.63. Footnotes 1-3 must belong to the Ch.13 body text (book pp. ~560-608), which is not in this PDF. I did not attach them to any compound.

13.63 — (S)-Gizzerosine synthesis (footnote 4) — book p.615 (p-07)

- Citation: ref 4, Tetrahedron Lett. 2007, 48, 8479-8481.
- Context: (S)-gizzerosine is an amino acid that causes "black vomit" in chickens; it is a possible drug for osteoporosis and hyperacidity. This step comes from a recent synthesis.
- Starting material: methyl (S,E)-2-(Boc-amino)-6-oxohex-4-enoate, i.e. OHC-CH=CH-CH₂-CH(NHBoc)-CO₂CH₃. SMILES CCOC(=O)C=C[C@H](NC(=O)OC(C)(C)C)C=O (S, confirmed by RDKit; E is assumed from the trans drawing)
- Product: methyl (S)-2-(Boc-amino)-6-hydroxyhexanoate, HO-(CH₂)₄-CH(NHBoc)-CO₂CH₃. SMILES CCCC[C@H](NC(=O)OC(C)(C)C)C(=O)OC
- Asked: propose the 2-step ("2 단계") synthesis.
- Concept: chemoselective reduction. Reduce the C=C and the aldehyde, but keep the methyl ester and the Boc carbamate.
- Expected answer (high): 1) H₂, Pd/C (or Pt, Ni) to hydrogenate the C=C; 2) NaBH₄/MeOH to reduce the aldehyde to the 1° alcohol (NaBH₄ leaves the ester alone; LiAlH₄ must not be used). Doing NaBH₄ first and then H₂/Pd is also acceptable.

13.68 — Briarellin E (footnote 5) — p.616 (p-08)

- Citation: ref 5, J. Org. Chem. 2009, 74, 5458-5470.

- Context: briarellin E (compound 3) is a marine natural product from Caribbean and Mediterranean corals, studied as a potential anticancer agent.
- Reaction: compound 1 (the briarellin core: tricyclic ether-bridged diterpenoid, cyclohexene with methyls, oxacycle, exocyclic methylene, with **two O-TBS ethers** ($\text{Si}(\text{CH}_3)_2\text{C}(\text{CH}_3)_3$) and one free OH) + **TBAF** $\rightarrow$ 2 $\rightarrow$ (further steps) $\rightarrow$ 3 (briarellin E, which carries an octanoate ester and an extra ring OH).
- Asked: draw compound 2.
- Concept: fluoride removes silyl protecting groups (Si-F bond strength).
- Expected answer (high): 2 = 1 with **both OTBS groups turned into OH** (a triol: the two former silyl ethers plus the existing HO). The skeleton and all stereocenters stay the same. No SMILES given because the polycyclic stereostructure cannot be read reliably at this resolution.

13.69 — Duryne (footnote 6) — p.616 (p-08)

- Citation: ref 6, J. Nat. Prod. 2011, 74, 1262-1267.
- Context: duryne is one of many compounds isolated from a marine sponge.
- Target: $\text{HC}\equiv\text{C}-\text{CH}(\text{OH})-\text{CH}=\text{CH}-(\text{CH}_2)_9-\text{CH}=\text{CH}-(\text{CH}_2)_9-\text{CH}=\text{CH}-\text{CH}(\text{OH})-\text{C}\equiv\text{CH}$ (C30). As drawn, the outer alkenes are E (trans) and the central alkene is Z (cis). SMILES (stereocenters unspecified) C#CC(O)/C=C/CCCCCCCC/C=C\CCCCCCCC/C=C/C(O)C#C
- Asked: propose an efficient synthesis from compounds with ≤ 11 carbons.
- Concept: alkyne chemistry (acetylide alkylation; Lindlar $\rightarrow$ cis; Na/NH₃ $\rightarrow$ trans) plus Grignard addition of an acetylide to an aldehyde. The molecule is symmetric, so it can be built from two identical halves.
- Expected answer (medium, design question): build the central cis alkene from acetylene by double alkylation with two $\text{C}\geq 9$ halides (e.g. a protected $\text{X}-(\text{CH}_2)_9-$ fragment), then H₂/Lindlar. At each end: $\text{E}-\text{CH}=\text{CH}-\text{CHO}$ (from an alkyne reduced with Na/NH₃ or built from $\text{C}\leq 11$ pieces) + $\text{HC}\equiv\text{CMgBr}$ (or $\text{HC}\equiv\text{CNa}$), then H₂O, gives the propargylic alcohols. Several routes are acceptable.

13.70 — Estragole (footnote 7) — p.617 (p-09)

- Citation: ref 7, J. Nat. Prod. 1987, 50, 990-991.
- Context: estragole was isolated from the leaves of the *Clausena anisata* tree as an insect repellent.
- Starting material: 4-methylphenol (p-cresol) Cc1ccc(O)cc1
- Target: estragole, 4-allylanisole C=CCc1ccc(OC)cc1 (C10H12O)
- Asked: propose a synthesis.
- Concept: phenol acidity (phenoxide + MeI), benzylic radical bromination, Grignard, functional-group interconversion.
- Expected answer (medium, one acceptable route): 1) NaOH (or NaH), CH₃I → 4-methylanisole; 2) NBS (or Br₂), hν → 4-MeO-C₆H₄CH₂Br; 3) Mg → ArCH₂MgBr; 4) ethylene oxide (or H₂C=O, then repeat the homologation), then H₃O⁺ → ArCH₂CH₂CH₂OH; 5) TsCl/py, then t-BuOK (bulky base, Hofmann-type E2) → ArCH₂CH=CH₂. Alternative: ArCH₂MgBr + CH₂=CH-CH₂Br-type coupling. Any route that adds the allyl unit and methylates the phenol O is acceptable.

13.71 — Ketone isomers from marijuana / hops volatile oil (footnote 8) — p.617 (p-09)

- Citation: ref 8, J. Nat. Prod. 1996, 59, 49-51.
- Context: two ketone isomers are among 68 compounds in the volatile oil from dried marijuana and hops.

- Targets: (i) **2-methylheptan-4-one** CC(C)CC(=O)CCC; (ii) **3-methylheptan-4-one** CCC(C)C(=O)CCC (both C₈H₁₆O).
- Asked: propose a synthesis of each from alkenes with (no more than) 4 carbons. The Korean text reads "4개의 탄소원자를 가지는 ... 알켄을 이용하여".
- Concept: alkene → alkyl halide or alcohol, Grignard + aldehyde, then oxidation of the 2° alcohol.
- Expected answer (high):
- Shared piece: butanal from 1-butene: 1) BH₃-THF; 2) H₂O₂, NaOH → 1-butanol; then PCC (or DMP/Swern) → butanal.
- (i) 2-methylpropene + HBr/ROOR → isobutyl bromide → Mg → i-BuMgBr + butanal → H₂O → 2-methylheptan-4-ol → Na₂Cr₂O₇/H₂SO₄ (or PCC) → 2-methylheptan-4-one.
- (ii) 2-butene + HBr → 2-bromobutane → Mg → sec-BuMgBr + butanal → 3-methylheptan-4-ol → oxidize → 3-methylheptan-4-one.

13.72 — 5,9,12,16-Tetramethyleicosane, Brazilian (tomato) bug pheromone (footnote 9) — p.617 (p-09)

- Citation: ref 9, J. Chem. Ecol. 2012, 38, 814-824.
- Context: this is a male sex pheromone of a Brazilian bug that eats tomato leaves and fruit. Building blocks: **A** 1-bromobutane CCCCBr; **B** cyclopropyl methyl ketone CC(=O)C1CC1; **C** hexane-2,5-dione CC(=O)CCC(C)=O.
- Key step: 2-cyclopropylhexan-2-ol CCCCC(C)(O)C1CC1 + HBr → (E/Z)-1-bromo-4-methyloct-3-ene CCCC/C(C)=C/CCBr (E+Z, C₉H₁₇Br).
- Asked: (a) mechanism of the HBr step; (b) efficient synthesis of the target CCCCC(C)CCCC(C)CCC(C)CCCC(C)CCCC (C₂₄H₅₀, numbering checked) using A, B and C.
- Concept: SN1-type ring opening of a cyclopropylcarbinyl cation to a homoallylic bromide (Julia-type), then Grignard, dehydration and hydrogenation.
- Expected answer (high):
- (a) Protonate the OH and lose H₂O to get a tertiary cyclopropylcarbinyl cation. Br⁻ attacks a cyclopropane CH₂ and the ring C-C bond breaks (cyclopropylcarbinyl → homoallyl, relief of ring strain), giving C=C at the former carbinol carbon. The trisubstituted alkene forms as an E/Z mixture.
- (b) A + Mg → BuMgBr. BuMgBr + B → 2-cyclopropylhexan-2-ol. HBr → homoallylic bromide (C₉). Mg → Grignard. 2 equiv of that Grignard + C (hexanedione), then H₃O⁺ → C₂₄ tertiary diol. Conc. H₂SO₄/heat (dehydration) → polyene. H₂, Pd/C → 5,9,12,16-tetramethyleicosane. Carbon count: 9 + 6 + 9 = 24, and the methyls land at 5, 9, 12, 16.

13.73 — Brevetoxin B, K-ring fragment (Nicolaou) (footnote 10) — p.617 (p-09)

- Citation: ref 10, J. Am. Chem. Soc. 1989, 111, 6682-6690.
- Context: brevetoxin B (2) is a red-tide neurotoxin from *Ptychodiscus brevis* Davis. Nicolaou's synthesis builds the K ring from compound 1. In 1, the syn arrangement of the hydroxyl and the silicon-based protecting group makes the conversion 1 → 5 possible. The step 3 → 4 is a selective tosylation of the primary OH.
- Starting material 1: a tetrahydropyran (K ring). Going round the ring: O1; C2 (H, allyl CH₂CH=CH₂); C3 (CH₃, tertiary OH); C4 (OCH₂Ph); C5H₂; C6 (CH₂-O-SiR₃). SMILES (stereo omitted) C=CCC1OC(CO[Si](R)(R)R)CC(OCc2ccccc2)C1(C)O

- Sequence: 1 + TBAF → 3; TsCl, pyridine → 4; NaOMe → 5 (a bridged bicyclic ether, drawn as 2,6-dioxabicyclo[2.2.2]octane with OBn, Me, allyl).
- Asked: (a) structures of 3 and 4; (b) mechanism 4 → 5 with NaOMe; (c) how the conversion 1 → 5 relates to the syn arrangement of OH and the silyl group in 1.
- Expected answer (high):
- (a) 3 = the diol (CH₂OSiR₃ → CH₂OH). SMILES C=CCC1OC(CO)CC(OCc2ccccc2)C1(C)O. 4 = the primary tosylate (only the 1° OH reacts; the 3° OH is too hindered). SMILES C=CCC1OC(COS(=O)(=O)c2ccc(C)cc2)CC(OCc2ccccc2)C1(C)O.
- (b) NaOMe deprotonates the 3° OH. The alkoxide does an intramolecular S_N2 (Williamson) on CH₂-OTs and gives the bridged ether 5. The ring flips to a boat so that C3-O⁻ and C6-CH₂OTs reach each other across the ring.
- (c) Ring closure is only possible because the C3-OH and the C6-CH₂OSiR₃/CH₂OTs are cis (syn, same face) on the K ring. In the trans isomer the two groups cannot reach each other.

Note: 13.64-13.67 are spectroscopy problems with no footnotes. No other Ch.13 problem has a literature mark.

제14장 에터와 에폭사이드; 싸이올과 설파이드 (Ethers and Epoxides; Thiols and Sulfides)

Key reactions (summary pages 653-655)

- Williamson ether synthesis: R-OH → 1) NaH 2) RX → R-O-R (1° RX; poor with 2°, fails with 3°).
- Alkoxymercuration-demercuration: alkene + 1) Hg(OAc)₂, ROH 2) NaBH₄ → Markovnikov ether.
- Acidic cleavage: R-O-R + excess HX, heat → 2 RX + H₂O. Aryl alkyl ether gives phenol + RX.
- Autoxidation: Et₂O + O₂ (slow) → hydroperoxide.
- Epoxide formation: RCO₃H, or 1) Br₂/H₂O 2) NaOH (stereospecific: cis alkene → cis epoxide).
- Sharpless asymmetric epoxidation: allylic alcohol + (CH₃)₃COOH, Ti[OCH(CH₃)₂]₄, (+)-DET or (-)-DET → enantiomeric epoxy alcohols.
- Epoxide ring opening. Strong nucleophiles (RONa, NaCN, NaSR, RMgBr, LiAlH₄; then H₂O) attack the less substituted C (S_N2, inversion). Acid catalysis (HX, H₃O⁺, ROH/H⁺) favors the more substituted C (electronic effect wins at a 3° center).
- Thiols: RBr + NaSH → RSH (S_N2, inversion). 2 RSH ⇌ RSSR (NaOH/H₂O, Br₂ oxidizes; HCl/Zn reduces).
- Sulfides: RSH 1) NaOH 2) RX → RSR. R₂S + MeX → sulfonium. R₂S → NaIO₄ → sulfoxide → H₂O₂ → sulfone.
- Concept review 14.1-14.12: naming; crown ethers / polyethers; oxirane ring strain; anti-Markovnikov vs Markovnikov in ring opening; Grignard + epoxide puts the OH on the 2nd carbon of the new chain (Grignard + C=O puts it on the 1st).
- Skill builders 14.1-14.7: naming ethers; Williamson design (phenoxide + EtI); making epoxides with stereo retention; strong-nucleophile and acid-catalyzed opening (regio- and stereochemistry); two adjacent functional groups; Grignard + epoxide vs aldehyde.

Ch.14 참고문헌 (p.663), verbatim

1. Tetrahedron Lett. **2009**, 50, 5012-5014.
2. Tetrahedron Lett. **2012**, 53, 3233-3236.
3. J. Org. Chem. **2009**, 74, 6151-6156.
4. Org. Proc. Res. & Devel. **2007**, 11, 354-358.
5. J. Org. Chem. **2002**, 67, 4856-4859.
6. Synth. Commun. **2003**, 33, 2135-2143.
7. J. Am. Chem. Soc. **1982**, 104, 3515-3516.
8. J. Org. Chem. **2012**, 77, 7883-7890.
9. J. Org. Chem. **1981**, 46, 930-939.
10. J. Org. Chem. **2008**, 73, 5592-5594.
11. J. Org. Chem. **2002**, 67, 1253-1260.

All 11 refs are cited in problems 14.60-14.70 (one each, in order).

14.60 — Reidispongolide A synthesis (footnote 1) — p.661 (p-18)

- Citation: ref 1, Tetrahedron Lett. 2009, 50, 5012-5014.
- Starting material: trityl glycidyl ether. The drawing puts a wedge on the ring C-O bond, which reads as **(R)**-2-[(trityloxy)methyl]oxirane (RDKit from a 2D wedge model; medium confidence). SMILES c1ccccc1C(c1ccccc1)(c1ccccc1)OC[C@H]1C01
- Sequence: 1) CH₂=CHMgBr 2) H₂O → **A**; 1) NaH 2) CH₃I → **B**; cat. OsO₄, NMO → **C + D**.
- Asked: draw A, B, C, D.
- Concept: Grignard attacks the less hindered epoxide CH₂ (S_N2, no change at the stereocenter); Williamson methylation; syn-dihydroxylation that gives two diastereomers.
- Expected answer (high on constitution, medium on CIP):
- A = (R)-1-(trityloxy)pent-4-en-2-ol C=CC[C@H](O)COC(c1ccccc1)(c1ccccc1)c1ccccc1
- B = (R)-2-methoxy-1-(trityloxy)pent-4-ene C=CC[C@H](OC)COC(c1ccccc1)(c1ccccc1)c1ccccc1
- C + D = the two diastereomers of 4-methoxy-5-(trityloxy)pentane-1,2-diol (new stereocenter at C2, configuration not controlled). SMILES OC(O)C[C@H](OC)COC(c1ccccc1)(c1ccccc1)c1ccccc1, giving (2R,4R) and (2S,4R).

14.61 — Fumonisin B1, C7-C9 installation (footnote 2) — p.661 (p-18)

- Citation: ref 2, Tetrahedron Lett. 2012, 53, 3233-3236.
- Context: fumonisin B1 is a potential sphingolipid biosynthesis inhibitor. To add C7-C9 in a recent synthesis, epoxide 1 was opened to give alkyne 2.
- 1: cyclohexylidene-ketal (1,4-dioxaspiro[4.5]decane)-CH(OR)-CH₂-(terminal epoxide, C5/C6). SMILES (no stereo) C1CCC2(CC1)OCC(O2)C(OR)CC1C01
- 2: ...CH(OR)-CH₂-CH(OH)(C5)-CH₂(C6)-CH₂(C7)-C₈≡C₉H. SMILES C1CCC2(CC1)OCC(O2)C(OR)CC(O)CCC#C
- Asked: propose 1 → 2 starting from 3-bromo-1-propyne (propargyl bromide).
- Concept: a carbon nucleophile opens the epoxide at the less substituted C6 (S_N2).
- Expected answer (medium): turn HC≡C-CH₂Br into a propargyl organometallic (Mg → propargyl/allenyl-MgBr). It attacks C6 of 1, then H₂O gives 2 (C6-C7 bond formed; C5-OH). The acidic alkyne H and

propargyl/allenyl isomerism are real-world problems; the literature probably used a protected or modified version (not verified).

14.62 — (+)-Myrrhanol A (guggul component) (footnote 3) — p.661 (p-18)

- Citation: ref 3, J. Org. Chem. 2009, 74, 6151-6156.
- Context: guggul is the resin extract of the mukul myrrh tree, with potential effects on high blood pressure. The step comes from a recent synthesis of (+)-myrrhanol A.
- 1: trans-decalin, labdane-type: 3-OTMS, gem-dimethyl, angular methyl, CH₂OH on C₉, and an **exocyclic C₈=CH₂**.
- Reaction: 1) RCO₃H (MCPBA); 2) LiAlH₄; 3) H₂O → 2 (C₈ becomes a tertiary alcohol C(OH)CH₃) as the only diastereomer.
- Asked: mechanism of 1 → 2, and explain the stereochemistry at the new chiral center.
- Expected answer (high): the peracid epoxidizes the exocyclic alkene from the less hindered face (concerted "butterfly"), giving a spiro epoxide. LiAlH₄ hydride does S_N2 at the less hindered epoxide CH₂, giving a tertiary alkoxide; H₃O⁺ protonates it. The C₈-O bond is never broken, so the configuration of the C₈-OH is set entirely by the face of epoxidation. That is why only one diastereomer forms.

14.63 — (S,S)-Reboxetine, Pfizer process (footnote 4) — pp.661-662 (p-18/p-19)

- Citation: ref 4, Org. Proc. Res. & Devel. 2007, 11, 354-358.
- Context: reboxetine mesylate is an antidepressant sold as a racemate. The (S,S)-enantiomer works against neuropathic pain. Pfizer route.
- 1: (E)-cinnamyl alcohol OC/C=C/c1ccccc1
- 1 → 2: t-BuOOH, Ti[OCH(CH₃)₂]₄, (-)-DIPT (Sharpless). 2 + 2-ethoxyphenol, NaOH → 3 (drawn diol: ArO-CH(Ph)-CH(OH)-CH₂OH). 3 → 4: 1) TMSCl, Et₃N 2) MsCl, Et₃N 3) H₃O⁺. 4 + NaOH → 5 (terminal epoxide).
- Asked: (a) structure of 2 ((-)-DIPT acts like (-)-DET); (b) mechanism 2 → 3 and the configuration at each chiral center of 3; (c) structure of 4 (TMS protects the less hindered 1° OH, MsCl mesylates the 2° OH, acid removes TMS); (d) mechanism 4 → 5.
- Expected answers (RDKit CIP; high on mechanism, medium-high on absolute configuration):
- 2 = (2R,3R)-3-phenylglycidol OC[C@H]1O[C@@H]1c1ccccc1. (+)-tartrate gives (2S,3S) with E-cinnamyl alcohol, so (-) gives (2R,3R).
- 3: phenoxide does S_N2 at the benzylic C₃ with inversion → (2R,3S)-3-(2-ethoxyphenoxy)-3-phenylpropane-1,2-diol OC[C@@H](O)[C@@H](OC2ccccc2OCC)c2ccccc2
- 4 = 2° mesylate with a free 1° OH OC[C@@H](OS(C)(=O)=O)[C@@H](OC2ccccc2OCC)c2ccccc2
- 5: the 1° alkoxide does an intramolecular S_N2 on C₂-OMs (inversion) → (S)-2-[(S)-(2-ethoxyphenoxy)(phenyl)methyl]oxirane CCOC1ccccc1O[C@@H](c1ccccc1)[C@@H]1CO1. This matches the (S,S)-reboxetine target, which supports the assignment.

14.64 — 2-Methyl-D-erythritol (isoprenoid biosynthesis) (footnote 5) — p.662 (p-19)

- Citation: ref 5, J. Org. Chem. 2002, 67, 4856-4859.
- 1: (E)-4-(benzyloxy)-2-methylbut-2-en-1-ol OC/C(C)=C/COc1ccccc1
- 2 (as drawn): (2R,3R)-3-[(benzyloxy)methyl]-2-methyloxiranyl-methanol. CIP from a 2D wedge model in RDKit: C[C@]1(CO)O[C@@H]1COc1ccccc1

- $2 \rightarrow 3$: acetylation of the 1° OH (OAc). $3 + [\text{H}_3\text{O}^+] \rightarrow 4$, a bicyclic orthoester (1-methyl-2,6,7-trioxabicyclo[2.2.1]heptane carrying CH_2OBn and a bridgehead CH_3).
- Asked: (a) reagents for $1 \rightarrow 2$; (b) mechanism $3 \rightarrow 4$ (hint: the ester $\text{C}=\text{O}$ oxygen acts as the nucleophile).
- Expected answer:
- (a) (medium-high) Sharpless asymmetric epoxidation: t-BuOOH , $\text{Ti}[\text{OCH}(\text{CH}_3)_2]_4$, **(-)-DET** (D-tartrate). By the same CIP mapping as (E)-2-hexenol / L-(+)-DET $\rightarrow$ (2S,3S), the (2R,3R) epoxide needs (-)-DET.
- (b) (high) Protonate the epoxide O. The acetate carbonyl O attacks the **tertiary** epoxide carbon (the one carrying CH_3 and CH_2OAc). This is a 5-membered ring closure with $\text{S}_\text{N}1$ -like character, with inversion, and gives a 1,3-dioxolan-2-ylum cation. The freed C3-OH then adds to that cation carbon, and loss of H^+ gives orthoester 4. Check: inversion at C2 and retention at C3 give, after hydrolysis, (2S,3R)-2-methylbutane-1,2,3,4-tetraol = 2-C-methyl-D-erythritol (RDKit). That agrees with the target.

14.65 — Corey-Chaykovsky epoxidation (footnote 6) — p.662 (p-19)

- Citation: ref 6, Synth. Commun. 2003, 33, 2135-2143.
- Reaction: trimethylsulfonium (1) C[S+](C)C + 1) NaH, 2) acetophenone CC(=O)c1ccccc1 → 2-methyl-2-phenyloxirane CC1(c2ccccc2)CO1 + dimethyl sulfide CSC.
- Asked: mechanism.
- Expected answer (high): NaH deprotonates a methyl to give the sulfur ylide (dimethylsulfonium methylide). The ylide carbon attacks the ketone C=O and gives a betaine alkoxide. The alkoxide then does an intramolecular SN2 on the CH2-S+Me2 carbon, with Me2S leaving, and closes the epoxide.

14.66 — Payne rearrangement / carbohydrate synthesis strategy (footnote 7) — p.662 (p-19)

- Citation: ref 7, J. Am. Chem. Soc. 1982, 104, 3515-3516.
- (a) Substrate: 2,3-epoxy alcohol carrying a 2,2-dimethyl-1,3-dioxolane (acetone) acetonide. With NaOH/H₂O, PhSH it gives the acetonide-CH(OH)-CH(OH)-CH₂SPh (1-phenylthio-2,3-diol). SMILES (no stereo) CC1(C)OC(=O)C1OC2C(C)C(C)C2 → CC1(C)OC(=O)C1C(O)C(O)CS2CCCC2
- Asked (a): full mechanism with stereochemistry. The hint: NaOH makes an alkoxide that opens the epoxide intramolecularly and forms a new epoxide.
- Expected (high): **Payne rearrangement**. The 1° alkoxide attacks C2 (inversion) and makes a terminal 1,2-epoxide plus a C3 alkoxide. This equilibrium is pulled forward because PhS⁻ opens the terminal epoxide at the unhindered CH₂ (S_N2). Net result: C2 inverted, C3 retained.
- (b) Substrate: (E)-4-(benzhydryloxy)but-2-en-1-ol OC/C=C/COC(c1ccccc1)c1ccccc1 (drawn trans). Reagents: 1) t-BuOOH, Ti(OiPr)₄, (+)-DET; 2) NaOH/H₂O, PhSH → ?
- Expected (medium): the (+)-DET epoxide is (2S,3S)-3-[(benzhydryloxy)methyl]oxiranyl-methanol OC[C@H]1OC[C@H]1COC(c1ccccc1)c1ccccc1. Payne rearrangement + PhS⁻ then gives (2S,3S)-4-(benzhydryloxy)-1-(phenylthio)butane-2,3-diol OC[C@H](CS2CCCC2)[C@H](O)COC(c1ccccc1)c1ccccc1 (RDKit, inversion at C2).

14.67 — Laureatin skeletal rearrangement (footnote 8) — p.662 (p-19), structures p-19 right column

- Citation: ref 8, J. Org. Chem. 2012, 77, 7883-7890.
- Context: laureatin is a natural product from the red alga *Laurencia nipponica* with strong insecticidal activity against mosquitoes.

- 1: a 2,4-disubstituted oxetane. C2 carries CH(OSiR₃)-CH₂-CH=CH-CH₂CH₃ (E); C4 carries CH(OH)-CH₂CH₂CH₂-OR. SMILES (no stereo) CC/C=C/CC(O[Si](R)(R)R)C1CC(O1)C(O)CCOR
- Br₂ → 2: a 2,3,5-trisubstituted tetrahydrofuran. It has CHBr-CH₂CH₃ on C2 (α to O), OSiR₃ on a ring carbon, and CH₂-(epoxide)-CH₂CH₂CH₂OR on the other α carbon. Constitution as interpreted: CCC(Br)C1OC(CC2OC2CCCOR)CC1O[Si](R)(R)R. The OSiR₃ position on the THF ring was read from a partial crop; confidence medium.
- Asked: mechanism.
- Expected (medium-high): Br₂ forms a bromonium ion. The oxetane O attacks the proximal bromonium carbon (5-exo), giving a fused bicyclic oxonium (THF + oxetanium) with Br on the adjacent carbon (anti). The neighboring C-OH then attacks the oxetanium carbon C4 intramolecularly (inversion), opening the strained 4-membered ring and forming the epoxide. After loss of H⁺, the result is the THF + epoxide (oxetane → THF ring expansion).

14.68 — Oxymercuration of 1,2-epoxyalkenes (footnote 9) — p.662 (p-19)

- Citation: ref 9, J. Org. Chem. 1981, 46, 930-939.
- 1: 2-allyloxirane C=CCC1C01 + 1) Hg(OAc)₂, H₂O 2) NaBH₄ → 2 (96 %).
- 3: 2-(but-3-enyl)oxirane C=CCCC1C01 + Hg(OAc)₂, H₂O → [4] → 5 + 6 → NaBH₄ → 7 + 8. Only a little normal hydration product forms.
- 7 = 6-methyltetrahydro-2H-pyran-3-ol CC1CCC(O)C01; 8 = (5-methyltetrahydrofuran-2-yl)methanol CC1CCC(CO)O1
- Asked: (a) structure of 2; (b) structures of 4, 5, 6 and mechanisms 4 → 5 and 4 → 6; (c) why the product distributions for epoxyalkenes 1 and 3 differ.
- Expected (high):
- (a) 2 = Markovnikov hydration product, 1-(oxiran-2-yl)propan-2-ol CC(O)CC1C01.
- (b) 4 = bicyclic oxonium: the epoxide O has attacked the mercurinium ion at the internal carbon (5-exo), making a 1-oxonia-bicyclo[3.1.0] system with CH₂HgOAc. Water opens 4 at the epoxide CH₂ → 5 = 2-(hydroxymethyl)-5-(acetoxymethyl)THF (gives 8 after NaBH₄). Water opens 4 at the epoxide CH → 6 = 6-(acetoxymethyl)tetrahydropyran-3-ol (gives 7). The ring-size outcome depends on which C-O bond of the oxonium breaks.
- (c) In 3, the epoxide O can reach the mercurinium carbon through a favorable 5-membered transition state, so it out-competes water. In 1 that would need a strained 4-membered ring, so water attacks normally.

14.69 — Halonium-induced ether (alkoxy) transfer (footnote 10) — p.662-663 (p-19/p-20)

- Citation: ref 10, J. Org. Chem. 2008, 73, 5592-5594.
- Context: this electrophile-induced ether transfer was used to make several structurally related natural products.
- Starting material: (S)- or (R)-4-(methoxymethoxy)-5-phenylpent-1-ene, PhCH₂-CH(OCH₂OCH₃)-CH₂-CH=CH₂. SMILES C=CC[C@@H](Cc1ccccc1)OCOC (stereo as drawn, wedge; not CIP-assigned)
- Reagents: 1) I₂; 2) PhSH, Et₃N.
- Product: PhCH₂-CH(OCH₂SPh)-CH₂-CH(OCH₃)-CH₂I, a 1,3-syn arrangement as drawn. SMILES (no stereo) ICC(OC)CC(Cc1ccccc1)OCSc1ccccc1
- Asked: mechanism.

- Expected (high): I₂ forms an iodonium on the terminal alkene. The distal MOM oxygen (the OCH₃ oxygen) attacks the internal iodonium carbon (6-membered TS) and gives a cyclic 1,3-dioxan-ium oxonium (anti to I). The ring then fragments at the O-CH₂ bond, leaving OCH₃ on the former alkene carbon and producing an oxocarbenium R-O=CH₂⁺ on the original oxygen. PhSH/Et₃N traps it to give the O-CH₂-SPh (phenylthiomethyl ether). Net: OMe moves to C₄, I goes to C₅, and the 1,3-syn stereochemistry is set through the chair-like 6-membered ring.

14.70 — Artemisinin derivative epoxide opening with HFIP (footnote 11) — p.663 (p-20)

- Citation: ref 11, J. Org. Chem. 2002, 67, 1253-1260.
- Context: artemisinin (qinghaosu) is an endoperoxide from traditional Chinese medicine. Its synthetic derivatives are the standard treatment for *Plasmodium falciparum* malaria.
- 1: the 9,10-epoxide of anhydrodihydroartemisinin. The epoxide is fused at C₉ (quaternary, CH₃) and C₁₀ (CH, the anomeric carbon next to the ring O). The 1,2,4-trioxane endoperoxide is kept.
- Reagents: (CF₃)₂CHOH (HFIP), cat. H₃O⁺.
- Product 2: 10-O-[CH(CF₃)₂] acetal at C₁₀ and a tertiary OH at C₉ (9-hydroxy-10-(hexafluoroisopropoxy) dihydroartemisinin). As drawn, OH and OCH(CF₃)₂ are cis (same face).
- Asked: explain mechanistically why both the regiochemistry and the stereochemistry are opposite to what you'd expect (expected: attack at the more substituted 3° carbon, with anti, S_N2-like inversion).
- Expected (medium-high): protonating the epoxide breaks the C₁₀-O bond to give an oxocarbenium ion at C₁₀. The adjacent ring oxygen stabilizes it more than a tertiary carbocation at C₉ would be stabilized, which is why C₁₀ is attacked. The weakly nucleophilic HFIP then adds to the planar oxocarbenium from the less hindered face (or under anomeric/thermodynamic control). The C₉-OH and the new acetal end up cis, i.e. net retention/syn rather than anti S_N2 inversion.

Note: 14.25/14.26 (p-10), 14.27-14.59 have no literature footnotes. 14.50-14.53 are spectroscopy problems.

Klein 유기화학 — Chapters 15-19: literature-based problems

Source: 1d12609b-15-19.pdf (PDF page N = p-NN.jpg). Book page numbers given as printed. Confidence tags: [H] high, [M] medium, [L] low.

Chapter 15 — 적외선 분광법과 질량 분석법 (IR Spectroscopy and Mass Spectrometry)

Summary pages: book p.703-705 (PDF p1-3). Problems p.705-710 (PDF p3-8). References: book p.710 (PDF p8).

Key concepts (summary 15.1-15.16)

- Light: wavelength/frequency/photon energy; vibrational excitation when photon energy = ΔE between vibrational levels.
- IR: stretching & bending; wavenumber; diagnostic region (double/triple bonds, X-H) vs fingerprint region; frequency depends on bond strength and atomic masses; conjugated C=O absorbs lower than

unconjugated.

- Intensity depends on dipole moment change: C=O strong, C=C weak, symmetric C=C/C≡C silent.
- Shape: H-bonded (concentrated) alcohol O-H broad; dilute alcohol narrow free O-H. 1° amines: symmetric + asymmetric N-H stretch (2 bands).
- Analyzing IR: check 1600-1850 (C=O, C=C), 2100-2300 (C≡C, C≡N), 2750-4000 (X-H; line at 3000 cm⁻¹ separates sp² vs sp³ C-H).
- MS: EI → radical cation M⁺•, fragmentation, m/z, base peak; nitrogen rule; (M+1) gives # C (÷1.1%); (M+2) 1/3 of M → Cl, equal → Br; M-15 (methyl loss), M-29 (ethyl loss); fragmentation favored for stable carbocations; HRMS gives molecular formula (exact masses); GC-MS (retention time); ESI for large biomolecules.
- HDI (hydrogen deficiency index) / degrees of unsaturation: add H for halogen, ignore O, subtract H for N.

Literature-footnoted problems (refs 1-10)

15.69 — book p.708 — footnote 1

- Citation: *Anal. Chem.* **2013**, *85*, 831-836.
- Compound: anthraquinone acid dye (blue textile fiber dye) — 1-amino-4-(4-methoxyanilino)-9,10-dioxoanthracene-2-sulfonate (anion, as drawn). SMILES [H as drawn]: Nc1c(S(=O)(=O)[O-])cc(Nc2ccc(OC)cc2)c2C(=O)c3ccccc3C(=O)c12 — C₂₁H₁₅N₂O₆S⁻, calc. 423.0656.
- Data: dye extracted directly from dyed fiber by pulsed laser, negative-ion mode HRMS: m/z = 423.0647 (100%), 424.0681 (22.5%), 425.0605 (4.21%). Given: ³²S 31.9721 (95.02%), ³⁴S 33.9679 (4.21%).
- Asked: (a) explain origin and relative abundances of each peak; (b) the spectrum also shows m/z 425.0715 — explain its origin.
- Answer [H]: 423.0647 = M⁻ (all ¹²C, ³²S). 424.0681 = M+1 mostly ¹³C (21 C × 1.1% ≈ 23% → 22.5%). 425.0605 = ³⁴S isotopologue (4.21% matches ³⁴S abundance; mass +1.9958). (b) 425.0715 = ion containing two ¹³C atoms (M+2 from ¹³C₂; +2×1.00335 = 425.0714), distinguishable from ³⁴S peak by HRMS.
- Concept: isotope patterns, HRMS mass defects (¹³C vs ³⁴S).

15.70 — book p.708 — footnote 2

- Citation: *J. Am. Chem. Soc.* **1952**, *74*, 4070-4073.
- Compound: ethyl 2-oxocyclohexanecarboxylate (A, keto) ⇌ enol tautomer B (ethyl 2-hydroxycyclohex-1-enecarboxylate). A: CCOC(=O)C1CCCC1=O; B: CCOC(=O)C1=C(O)CCCC1 [H].
- Data: IR of equilibrium mixture shows 4 peaks in 1600-1850 cm⁻¹: 1620, 1660, 1720, 1740.
- Asked: (a) assign the 4 peaks, justify position of 1660; (b) for most simple ketones enol concentration is negligible; here B is significant — why?
- Answer [H]: 1740 = ester C=O of A; 1720 = ketone C=O of A; 1660 = ester C=O of B, lowered by conjugation with C=C and intramolecular H-bond with enol OH; 1620 = enol C=C (conjugated, polarized → strong). (b) Enol stabilized by conjugation (C=C-C=O) and intramolecular H-bond (6-membered chelate) of the β-keto ester.
- Concept: keto-enol tautomerism, conjugation lowering C=O frequency.

15.71 — book p.708 — footnote 3

- Citation: *J. Chem. Ed.* **2008**, *85*, 832-833.
- Compounds: A = 2-(4-bromophenyl)ethanol OCCc1ccc(Br)cc1; B = 1-(4-bromophenyl)ethanol CC(O)c1ccc(Br)cc1 (both C₈H₉BrO, M 200/202).
- Data: Sample X: m/z 202, 200, 187, 185, 159, 157, 121. Sample Y: m/z 202, 200, 171, 169, 121.
- Asked: match A/B with X/Y, give structures for all peaks.
- Answer [H]: X = B: 202/200 M⁺ • (⁸¹Br/⁷⁹Br); 187/185 = M-CH₃ (α-cleavage → ArCH=OH⁺); 159/157 = loss of CO from 187/185 (BrC₆H₄⁺... C₆H₆Br⁺); 121 = M-Br (C₈H₉O⁺). Y = A: 202/200 M⁺ • ; 171/169 = M-CH₂OH (31) → 4-bromobenzyl (tropylium) cation; 121 = M-Br.
- Concept: α-cleavage of alcohols, benzylic cleavage, Br isotope doublets.

15.72 — book p.709 — footnote 4

- Citation: *Anal. Chem.* **1954**, *26*, 1428-1431.
- Compound: myosmine (tobacco alkaloid related to nicotine). Originally proposed structure (drawn): 3-(4,5-dihydro-1H-pyrrol-2-yl)pyridine (enamine, N-H) C1CNC(=C1)c1ccncc1. Correct tautomer: 3-(3,4-dihydro-2H-pyrrol-5-yl)pyridine (1-pyrroline imine) C1CC(=NC1)c1ccncc1 [H].
- Data: IR spectrum given (no N-H band 3300-3500; bands ~3000-2850; very strong 1621 cm⁻¹).
- Asked: (a) why IR inconsistent with original structure; (b) myosmine exists mainly as a tautomer — draw it, explain fit; (c) source of very strong 1621 cm⁻¹.
- Answer [H]: (a) no N-H stretch. (b) imine tautomer (C=N in ring, no N-H). (c) C=N stretch (conjugated with pyridine).

15.73 — book p.709 — footnote 5

- Citation: *J. Am. Chem. Soc.* **1959**, *81*, 4671-4673.
- Compounds: 1 = 6,7-dihydro-5H-pyrrolo[1,2-d]tetrazole (bicyclic tetrazole) C1CC2=NN=NN2C1; 2 = 5-azido-3,4-dihydro-2H-pyrrole (2-azido-1-pyrroline) [N-]=[N+]=NC1=NCCC1 [H].
- Data: 1 ⇌ 2 rapidly; IR of 1 shows no signal in 2120-2160 cm⁻¹.
- Asked: why does absence of this band show 1 is favored over 2?
- Answer [H]: azide (-N₃) asymmetric stretch is a strong band ~2100-2160 cm⁻¹; its absence means negligible azide tautomer (valence tautomerism azide-tetrazole lies on tetrazole side).

15.74 — book p.709 (도전 문제) — footnote 6

- Citation: *Appl. Spectrosc.* **1960**, *14*, 95-97.
- Compound: cyclohexanone O=C1CCCCC1.
- Data: low-res base peak m/z 55; HRMS shows two peaks 55.0183 and 55.0546 (relative 86.7 : 13.3).
- Asked: molecular formulas of ions for each.
- Answer [H]: 55.0183 = C₃H₃O⁺ (calc. 55.018); 55.0546 = C₄H₇⁺ (calc. 55.054).

15.75 — book p.709 — footnote 7

- Citation: *J. Org. Chem.* **1977**, *42*, 126-129.
- Compound: pyrrolidine amide of a cyclopropane fatty acid (from study of cyclopropane derivatives of fatty acids) — drawn: long alkyl-cyclopropane-(CH₂)_n-C(=O)-N(pyrrolidinyl) (e.g., dihydrosterculic acid

pyrrolidide; chain lengths approx.).

- Data: base peak m/z 113.
- Asked: reasonable structure for m/z 113 ion and mechanism.
- Answer [H]: McLafferty rearrangement (γ -H transfer to $C=O$, β -cleavage) $\rightarrow$ enol radical cation $CH_2=C(OH)-N(C_4H_8)^+ \cdot$, $C_6H_{11}NO^+ \cdot$, m/z 113.

15.76 — book p.709 — footnote 8

- Citation: *Anal. Chem.* **2011**, *83*, 4738-4744.
- Compound: pinolenic acid ($C_{17}H_{29}CO_2H$), all-cis triene from pine nuts (appetite suppressant); = (5Z,9Z,12Z)-octadeca-5,9,12-trienoic acid O=C(=O)CCC/C=C\CC/C=C\C/C=C\CCCC [H, structure from knowledge; consistent with data].
- Data: ESI with ozone present (ozonolysis during ionization); positive-ion mode peaks m/z 279, 211, 171, 117; without ozone only 279.
- Asked: (a) structures for each peak and of pinolenic acid; (b) predict four main peaks in negative mode and assign.
- Answer [H]: 279 = $[M+H]^+$ ($M = 278$). Ozonolytic aldehydes ($+H^+$): 211 = $OHC-C_{11}$ chain retaining $C_{11}-C_{12}$ ($C_{12}H_{18}O_3+H$, cleavage of Δ_{12}); 171 = C_1-C_9 aldehyde acid ($C_9H_{14}O_3+H$, Δ_9); 117 = $OHC(CH_2)_3COOH + H$ ($C_5H_8O_3+H$, Δ_5). Hence double bonds at C_5 , C_9 , C_{12} . (b) Negative mode $[M-H]^-$: 277, 209, 169, 115.

15.77 — book p.709 — footnote 9

- Citation: *J. Am. Chem. Soc.* **1957**, *79*, 243-247.
- Compounds: two steroid chlorohydrins (5 α -steroid A-ring): 1 has OH on C_2 and Cl on C_3 (Cl hashed, drawn diaxial-type trans arrangement); 2 has Cl on C_2 and OH (bold) on C_3 — i.e., trans-2,3-chlorohydrins differing in conformational relationship (1 = trans-diaxial; 2 = trans-diequatorial) [M].
- Data: dilute CS_2 IR: 1 O-H at 3617 cm^{-1} ; 2 at 3594 cm^{-1} .
- Asked: (a) why 2 absorbs lower; (b) which signal is expected broader and why.
- Answer [M]: In 2 OH and Cl are diequatorial (gauche, $\sim 60^\circ$) $\rightarrow$ intramolecular $O-H\cdots Cl$ hydrogen bond weakens O-H $\rightarrow$ lower wavenumber; in 1 diaxial (anti, 180°), no intramolecular H-bond $\rightarrow$ free OH 3617 . Dilution excludes intermolecular H-bonding. (b) Signal of 2 (H-bonded) broader.

15.78 — book p.710 — footnote 10

- Citation: *J. Am. Chem. Soc.* **2006**, *128*, 58-59.
 - Compound: phosphatidic acid (anion) with sn-1 palmitoyl ($C_{15}H_{31}CO$) and sn-2 $C_{17}H_{33}CO$ (one cis alkene) — drawn as $HO-P(=O)(O^-)-O-CH_2-CH(OCOC_{17}H_{33})-CH_2-OCOC_{15}H_{31}$.
 - Data: negative-mode MS $[M]^-$ at m/z 673 (calc. $C_{37}H_{70}O_8P^-$ 673.48); with ozone new peak m/z 563. With MeOH solvent: 563 and 611; with EtOH: 563 stays, 611 $\rightarrow$ 625.
 - Asked: (a) full structure showing alkene position; (b) explain solvent results.
 - Answer [H]: loss of 110 (C_9H_{18} replaced by O) $\rightarrow$ aldehyde from ozonolysis at Δ_9 : sn-2 = oleoyl, (9Z)-octadec-9-enoyl $\rightarrow$ PA(16:0/18:1 Δ_9 Z). SMILES: CCCCCCCCCCCCCCCC(=O)OCC(COP(=O)(O)[O-])OC(=O)CCCCCCC/C=C\CCCCCCC. (b) Criegee carbonyl oxide (aldehyde + O, +16) trapped by solvent alcohol $\rightarrow$ α -alkoxy hydroperoxide: +16+32 (MeOH) = 611; +16+46 (EtOH) = 625; aldehyde 563 independent of solvent.
-

Chapter 16 — 핵자기공명 분광법 (Nuclear Magnetic Resonance Spectroscopy)

Summary pages: book p.753-755 (PDF p9-11). Problems p.756-761 (PDF p12-17). References: book p.761 (PDF p17).

Key concepts (summary 16.1-16.13)

- NMR basics: spin, magnetic moment aligned with/against field, rf $\rightarrow$ spin flip (resonance); shielding/deshielding by diamagnetism; FT-NMR (pulse, FID); deuterated solvent + TMS.
- Chemical equivalence: homotopic (rotation) / enantiotopic (reflection) = equivalent; diastereotopic = not; substitution (replacement) test.
- Chemical shift δ : downfield/upfield; methyl 0.9, methylene 1.2, methine 1.7 ppm; corrections: +2.5 for O of alcohol/ether, +3.0 for O of ester, +1.0 for carbonyl; diamagnetic anisotropy deshields aromatic H.
- Integration (step curve); multiplicity $n+1$ rule, coupling constant J (Hz); equivalent protons don't split each other; no coupling across >3 σ bonds; OH protons labile (no coupling); complex splitting $\rightarrow$ multiplets.
- ^1H NMR analysis workflow: HDI $\rightarrow$ number of signals/integration $\rightarrow$ (δ , m , l) $\rightarrow$ fragments $\rightarrow$ assemble $\rightarrow$ verify.
- ^{13}C NMR: 1.1% natural abundance; broadband decoupling (singlets); off-resonance decoupling; δ 0-220 ppm (C=O 160-220, sp^2 100-150, sp 65-90/C-N, C-O 50-100, sp^3 0-50); number of signals by symmetry.
- DEPT: DEPT-90 shows only CH; DEPT-135 CH/CH₃ positive, CH₂ negative, quaternary C absent.

Literature-footnoted problems (refs 1-7)

Footnote 1 — not located in the supplied pages

- Citation: *Food Chem.* **2016**, *196*, 211-219.
- No superscript 1 was found in problems 16.32-16.76 on PDF p12-17 (headers checked at 220 dpi). It is presumably attached to an in-chapter problem/text (before book p.753) not included in this PDF.

16.66 — book p.759 — footnote 2

- Citation: *Angew. Chem. Int. Ed.* **2011**, *50*, 8387-8390.
- Problem: deduce structure of C₆H₁₀O₄ from IR, ^1H NMR, ^{13}C NMR.
- Data (read from spectra): IR strong 1747 cm^{-1} (ester C=O), no O-H. ^1H : ~5.1 ppm (q, 1H), ~3.75 (s, 3H), ~2.1 (s, 3H), ~1.5 (d, 3H). ^{13}C : ~171, ~170 (two C=O near 170), ~69, ~52, ~21, ~17 (solvent CDCl₃ at 77).
- HDI = 2 (two C=O).
- Answer [H]: methyl 2-acetoxypropanoate (methyl O-acetylactate), CC(=O)OC(C)C(=O)OC.
- Concept: combined IR/ ^1H / ^{13}C structure elucidation; deshielded CH (O-acyl, α to ester) ~5.1 ppm.

16.72 — book p.761 (도전 문제) — footnote 3

- Citation: *Org. Lett.* **2012**, *14*, 4770-4773.
- Compound: brevianamide S (antitubercular natural product, Chinese Bohai Sea marine sediment-derived fungus) — dimeric diketopiperazine alkaloid: two (Z)-3-ylidene-diketopiperazines bearing 2-(1,1-

dimethylallyl = reverse prenyl)indol-3-yl groups; the two dehydropyrroline-fused units are linked C-C through their pyrroline carbons (as drawn).

- Given shifts: indole N-H 7.7 ppm; diketopiperazine amide N-H 7.4 ppm; N-CH₂ (pyrroline, highlighted) 4.1 ppm.
- Asked: predict chemical shifts and multiplicity of every signal in the ¹H NMR.
- Expected answer [M]: reverse-prenyl CH₃ singlets ~1.5 (6H, s); vinyl CH= ~6.1 (dd), =CH₂ ~5.1-5.2 (2 × dd); aromatic indole H 7.0-7.5 (d/t); exocyclic vinyl CH= (ylidene) ~7.0 (s); N-CH₂ 4.1 (t); ring CH₂ ~3.0 (t); NH 7.4 & 7.7 (s, broad). Concept: prediction of δ/multiplicity in complex molecules.

16.73 — book p.761 — footnote 4

- Citation: *J. Am. Chem. Soc.* **1973**, *95*, 3437-3438.
- Reaction: 1,3,6-cyclononatriene (1, R = H; C1=CC=CCC=CC1 - ring numbering C1=C2, C3=C4, C6=C7, sp³ C5, C8, C9) or its 8,8-dimethyl derivative (2, R = CH₃) + KNH₂ / liq. NH₃ → anion 1a / 2a (deprotonation at the doubly allylic C5 → cyclic heptatrienyl anion spanning 7 carbons, C8/C9 sp³ remain).
- Asked: (a) draw four resonance structures of 1a; (b) number of ¹H signals for 1a and 2a; (c) 1a: signals 3.74 and 3.39 ppm (H on C1 and C3), H on C2 and C4 at 5.63 and 5.52 ppm — explain the two separated groups; (d) for 2a: how many signals in 3-4 ppm and in 5-6 ppm?
- Answer [M]: (a) negative charge on C1, C3, C5, C7 of the heptatrienyl (anion numbered C1-C7 symmetrically, C4 central). (b) 1a (mirror symmetry): H1/H7, H2/H6, H3/H5, H4, C8/C9-H2 → 5 signals (ignoring possible diastereotopicity); 2a (gem-dimethyl at C8 only, no symmetry): 7 vinylic H all distinct + CH₂ + 2 CH₃ (diastereotopic) → ~10. (c) charge-bearing carbons (odd positions) are shielded (3.4-3.7 ppm), nodal positions (C2, C4, C6) are normal vinylic (5.5-5.6). (d) 2a: 4 signals in 3-4 ppm (H1, H3, H5, H7) and 3 signals in 5-6 ppm (H2, H4, H6).

16.74 — book p.761 — footnote 5

- Citation: *J. Chem. Ed.* **2003**, *80*, 1439-1443.
- Reaction: 4-nitro-3-(trifluoromethyl)aniline (1) Nc1ccc([N+](=O)[O-])c(C(F)(F)F)c1 + isobutyryl chloride (2-methylpropanoyl chloride), pyridine → flutamide (2) CC(C)C(=O)Nc1ccc([N+](=O)[O-])c(C(F)(F)F)c1 (prostate-cancer drug) [H].
- Protons: Ha = ortho to NO₂ (C5), Hb = ortho to NH₂ (C6), Hc = between CF₃ and NH₂ (C2).
- Asked: (a) which aromatic H of 1 is most downfield? justify by resonance & inductive effects; (b) do these H shift upfield or downfield on going 1 → 2, and which is affected most?
- Answer [M]: (a) Ha (ortho to strongly withdrawing NO₂, and not ortho/para to the donating NH₂; Hb and Hc are ortho to NH₂ → shielded by resonance donation). (b) All shift downfield because the amide N lone pair is delocalized into C=O (weaker donor); largest change for the protons ortho to N (Hb and Hc; Hb typically largest, also deshielded by C=O anisotropy) — Ha (meta to N) least affected.

16.75 — book p.761 — footnote 6

- Citation: *J. Chem. Ed.* **2006**, *83*, 287-289.
- Compounds: coumarin (1) O=c1ccc2ccccc2o1; 2 = 3-acetyl-7-(diethylamino)coumarin CCN(CC)c1ccc2cc(C(=O)C)oc2c1.
- Data: two lowest-field ¹H signals of 2 at 7.38 and 8.42 ppm.
- Asked: (a) identify the two protons, justify using resonance; (b) is IR C3=C4 π-bond stretch stronger in 1 or 2?

- Answer [M-H]: (a) 8.42 = H4 (β -H of an α,β -unsaturated dicarbonyl — δ^+ at C4 by resonance with both lactone C=O and acetyl C=O); 7.38 = H5 (ortho to C4, peri; the only aromatic H not ortho/para to the NEt₂ donor — H6 and H8 are shielded). (b) Stronger in 2: push-pull polarization (Et₂N donor $\rightarrow$ carbonyl acceptors) gives a much larger dipole change for the C3=C4 vibration.

16.76 — book p.761 — footnote 7

- Citation: *Tetrahedron Lett.* **2011**, 52, 4182-4185.
- Reaction: 2,6-bis((trimethylsilyl)ethynyl)benzaldehyde (1) O=Cc1c(C#C[Si](C)(C)C)ccc1C#C[Si](C)(C)C + Fischer carbene 2 = (CO)₅Cr=C(OMe)CH₂CH₂CH=CH₂ (pentacarbonyl[1-methoxypent-4-enylidene]chromium) $\rightarrow$ hydronaphthacene (five fused six-membered rings) bis(cyclohexenone) with two TMS groups and a tertiary OH, formed as diastereomers 3 (R) and 4 (S) differing at one stereocenter (highlighted ring-fusion CH).
- Asked: how to distinguish 3 and 4 using only ¹³C NMR.
- Answer [M]: the pentacycle is pseudo-symmetric about the plane through C-OH and the central arene; in one diastereomer the two ring-fusion H's have the relative configuration making the halves mirror-related (meso-like, C_s) $\rightarrow$ about half the number of ¹³C signals; in the other diastereomer the halves are inequivalent $\rightarrow$ every carbon gives its own signal (roughly twice as many). Count signals.

Chapter 17 — 콘쥬게이션 파이계와 페리고리 협동반응 (Conjugated Pi Systems and Pericyclic Reactions)

Summary pages: book p.802-806 (PDF p18-22: reaction review p.802-803, concept review p.803-804, skill review p.805-806). Problems p.806-810 (PDF p22-26). References: book p.810 (PDF p26).

Key reactions / concepts

- Diene preparation: allylic halide or vicinal dihalide + t-BuOK (elimination).
- Electrophilic addition to 1,3-butadiene: HBr $\rightarrow$ 1,2-adduct (kinetic; 71% at 0 °C) vs 1,4-adduct (thermodynamic; 85% at 40 °C); Br₂ $\rightarrow$ 1,2- and 1,4-dibromides; allylic carbocation resonance.
- Diels-Alder [4+2]: cyclohexene products; retro-DA at high T; stereospecific (cis-dienophile $\rightarrow$ cis, trans $\rightarrow$ trans, racemic); alkyne dienophiles (DMAD $\rightarrow$ 1,4-cyclohexadiene); cyclopentadiene gives bicyclic, endo preferred; s-cis diene required; EWG on dienophile accelerates; HOMO(diene) $\rightarrow$ LUMO(dienophile).
- MO theory: HOMO/LUMO, frontier orbitals, photo-excited state; orbital symmetry conservation: thermal [4+2] allowed, thermal [2+2] forbidden, photochemical [2+2] allowed.
- Electrocyclic reactions: hexatriene $\rightleftharpoons$ cyclohexadiene; butadiene $\rightleftharpoons$ cyclobutene; 4 π thermal conrotatory/photochemical disrotatory; 6 π thermal disrotatory/photochemical conrotatory.
- Sigmatropic: [3,3] Cope (15:85 equilibrium example) and Claisen (allyl vinyl ether $\rightarrow$ γ,δ -unsaturated carbonyl, 10:90).
- UV-Vis: $\pi \rightarrow \pi^*$, $\lambda_{\max}$, molar absorptivity ϵ , Beer's law; more conjugation $\rightarrow$ longer $\lambda_{\max}$; chromophore/auxochrome; Woodward-Fieser rules (base 217, +30 per extra double bond, +5 per alkyl, +5 exocyclic, +39 homoannular diene); color if $\lambda_{\max}$ 400-700 nm; vision: rhodopsin, 11-cis-retinal $\rightarrow$ all-trans photoisomerization (Nakanishi locked analog).

Literature-footnoted problems (refs 1-10)

17.71 — book p.809 — footnote 1

- Citation: *J. Am. Chem. Soc.* **2011**, *133*, 17634-17637.
- Context: total synthesis of the toxin gelsemoxonine (*Gelsemium elegans*, SE Asia).
- Substrate (as drawn): a cis-divinylcyclopropane — cyclopropane bearing (i) a trimethylsilyl enol ether (Me₃SiO-C=) whose alkene is part of a small O-containing ring (oxetane/ether bridge drawn above), (ii) an exocyclic (E)-alkylidene at C3 of an N-methoxy-oxindole (N-OMe, 2-oxindole), and (iii) a CH₂OR (R = protecting group) on the quaternary cyclopropane carbon. Heated at 70 °C.
- Asked: predict product and mechanism.
- Answer [M]: divinylcyclopropane → cycloheptadiene rearrangement ([3,3] sigmatropic, boat TS, cis-divinyl): the cyclopropane C-C σ bond between the two vinyl-bearing carbons breaks, new σ bond forms between the vinyl termini (enol-ether carbon terminus and oxindole C3) → a spiro-oxindole fused 1,4-cycloheptadiene bearing the silyl enol ether (azepane-free; spiro at oxindole C3), setting the gelsemoxonine core with relative stereochemistry from the boat TS.

17.72 — book p.809 — footnote 2

- Citation: *J. Photochem. Photobiol. A* **2006**, *184*, 177-183.
- Compound 1: 3,4-bis(2,4,5-trimethylthiophen-3-yl)furan-2,5-dione (dithienylethene photochromic maleic anhydride) O=C1OC(=O)C(c2c(C)sc(C)c2C)=C1c1c(C)sc(C)c1C [H].
- Reaction: UV light → 6π electrocyclic ring closure → 2; visible light → ring-opening back to 1 (photochromism).
- Asked: draw 2, chiral or not, mechanism.
- Answer [H]: photochemical 6π → conrotatory; new C-C bond between the two thiophene C2 carbons, each now sp² bearing CH₃, the two methyls trans (anti). 2 = CC1=C(C)SC2(C)C1=C1C(=O)OC(=O)C1=C1C(C)=C(C)SC12 (closed, colored ring isomer, C₁₈H₁₈O₃S₂). 2 is chiral (two stereocenters, C₂-symmetric trans, no mirror plane) and forms as a racemate (conrotation either way).

17.73 — book p.809 — footnote 3

- Citation: *J. Am. Chem. Soc.* **1971**, *93*, 3836-3837.
- Context: first total synthesis of chelidonine (cytotoxic, *Chelidonium majus* roots).
- Substrate: benzocyclobutene (methylenedioxy-substituted) bearing N(R)-CH₂-Ar on the cyclobutene sp³ carbon; Ar = methylenedioxyphenyl with an ortho ethynyl group. Heat 120 °C.
- Product (drawn): tetracyclic B/C-hexahydrobenzo[c]phenanthridine skeleton (N-R, two methylenedioxy groups, ring C dihydro with C=C).
- Asked: mechanism.
- Answer [H]: (1) thermal 4π conrotatory electrocyclic ring opening of benzocyclobutene → o-quinodimethane (o-xylylene); (2) intramolecular Diels-Alder of the o-quinodimethane (diene) with the alkyne (dienophile) → ring re-aromatizes; gives the 1,4-cyclohexadiene ring of product (trans/cis-fusion from conrotation sense).

17.74 — book p.809 — footnote 4

- Citation: *Org. Lett.* **2010**, *12*, 1152-1155.

- Context: synthesis studies toward atropurpuran (diterpene).
- Substrate: acyclic aminotetraene $R_2N-CH=CH-CH=CH-CH=CH-CH_2-N(Ts)-CH_2CH=CH_2$ (triene conjugated with amino donor; N-allyl sulfonamide tether). Heat 135 °C.
- Product: tricyclic — bicyclo[2.2.2]octene fused to a N-Ts piperidine, with R_2N on the bicyclooctene alkene.
- Answer [H]: (1) thermal 6π disrotatory electrocycloization of the (Z-central) triene $\rightarrow$ 1,3-cyclohexadiene (amino-substituted); (2) intramolecular Diels-Alder between the cyclohexadiene (diene) and the tethered allyl alkene (dienophile) $\rightarrow$ bicyclo[2.2.2]octene + piperidine ring (tricyclic).

17.75 — book p.809 (도전 문제) — footnote 5

- Citation: *Org. Lett.* **2001**, *3*, 723-726.
- Reaction: 2 equiv. 4-(vinylxy)butyl (E)-2-oxo-4-phenylbut-3-enoate O=C(C(=O)OCCCCO=C)/C=C/c1cccc1 heated at 195 °C $\rightarrow$ macrocyclic dimer containing two 3,4-dihydro-2H-pyran rings (each: 6-carboxylate on the $C_5=C_6$ enol-ether double bond, 4-Ph, 2-alkoxy acetal), joined head-to-tail by two $-O(CH_2)_4O-$ / ester linkers.
- Asked: mechanism.
- Answer [H]: two sequential intermolecular hetero-Diels-Alder (inverse-electron-demand oxa-DA) reactions: the β,γ -unsaturated α -ketoester $O=C-C=C$ acts as 1-oxadiene; vinyl ether of a second molecule = electron-rich dienophile $\rightarrow$ 2-alkoxy-3,4-dihydro-2H-pyran; the first DA gives a dimer which then cyclizes by an intramolecular second hetero-DA $\rightarrow$ macrocycle (head-to-tail).

17.76 — book p.809 — footnote 6

- Citation: *Org. Lett.* **2012**, *14*, 502-505.
- Reaction: 2,2':5',2''-terfuran (trifuran) + maleimide $\rightleftharpoons$ only adduct at a terminal furan (7-oxabicyclo[2.2.1]heptene with bifuranyl at bridgehead); adduct at the central furan (two furyl groups on both bridgeheads) not observed.
- Asked: rational explanation.
- Answer [M]: furan DA is reversible (thermodynamic control). Reaction at the terminal furan retains a conjugated bifuran π system; reaction at the central ring would destroy conjugation across the whole oligomer (leaving two isolated furans) and place substituents at both bridgeheads (steric), so it is thermodynamically disfavored. (Also: the terminal ring's diene has an unsubstituted, more accessible terminus.)

17.77 — book p.810 — footnote 7

- Citation: *J. Am. Chem. Soc.* **2012**, *134*, 9199-9208.
- Reaction: tricyclic cyclooctyne A (dibenzo-azacyclooctynone, N-methyl lactam; CN1C(=O)c2ccccc2C#Cc2ccccc21) + benzyl azide $\rightarrow$ triazoles B + C (strain-promoted azide-alkyne [3+2] cycloaddition, copper-free click).
- Asked: (a) mechanism for B; (b) C is a constitutional (regio)isomer formed in ~equal amount — draw it; (c) why alkynes D (2-butyne, no reaction), E (cyclooctyne with $OCH_2-C_6H_4-COOH$ ether, 375× slower than A), F (A with two aryl methyls, 1000× slower) are slower.
- Answer [H/M]: (a) concerted 1,3-dipolar cycloaddition (4π dipole + 2π), driven by relief of ring strain in the bent cyclooctyne. (b) C = regioisomeric triazole with N-benzyl on the other triazole carbon side (adjacent to the other aryl ring, i.e., the benzyl N next to the lactam-N-bearing arene vs. carbonyl-bearing arene). (c) D: linear unstrained alkyne — no strain release, high barrier. E: less strained (no fused benzo

rings / fewer sp^2 centers raising strain) and lacks the electronic activation of A. F: the methyl ortho to the alkyne sterically hinders azide approach (and alters ring strain/electronics).

17.78 — book p.810 — footnote 8

- Citation: *Aust. J. Chem.* **1982**, *35*, 2247-2256.
- Context: endiandric acids (*Endiandra introrsa*, Australia) are isolated as racemates; proposed biosynthesis (Black cascade) from achiral polyene 1: an acyclic (E,Z,Z,E)-conjugated tetraene with chains $CH_2CH=CH-CH_2CO_2R$ and $CH_2CH=CH-CH=CH-Ph$.
- Asked: (a) first step = thermal electrocyclic reaction forming 8-membered ring 2; draw 2, explain racemic; (b) 2 undergoes another electrocyclization to the endiandric acid G skeleton — mechanism; (c) endiandric acid C forms from G by intramolecular DA — draw C and mechanism.
- Answer: (a) [H] 8π thermal conrotatory electrocyclization $\rightarrow$ cycloocta-1,3,5-triene 2 with two trans side chains at the new sp^2 carbons; achiral substrate, no chiral catalyst $\rightarrow$ both conrotatory senses equally likely $\rightarrow$ racemate. (b) [H] 6π thermal disrotatory electrocyclization of the cyclooctatriene $\rightarrow$ cis-bicyclo[4.2.0]octa-2,4-diene (endiandric acid G framework). (c) [L-M] intramolecular [4+2]: cyclohexadiene of the bicyclo[4.2.0]octadiene = diene, a side-chain $C=C$ (tethered on the cyclobutane) = dienophile $\rightarrow$ tetracyclic endiandric acid C (exact dienophile choice not verified from page).

17.79 — book p.810 — footnote 9

- Citation: *J. Am. Chem. Soc.* **2008**, *130*, 6281-6287.
- Compounds: roquefortine C (*Penicillium roqueforti*; blue cheese) and isoroquefortine C (its diastereomer = geometric isomer about the dehydrohistidine exocyclic $C=C$; drawn with the imidazole rotated so the ring $N=$ faces the diketopiperazine $N-H$). Isoroquefortine C is thermodynamically more stable yet not found in nature.
- Asked: explain stability difference.
- Answer [M]: in isoroquefortine C the diketopiperazine amide $N-H$ forms an intramolecular hydrogen bond to the basic imidazole sp^2 nitrogen (planar conjugated 6/7-membered H-bonded array), and reduced A(1,3)-type strain between the imidazole and the $C=O$; roquefortine C lacks this H-bond. Nature's (enzymatic, kinetically controlled) biosynthesis gives roquefortine C, and isomerization does not occur under physiological conditions.

17.80 — book p.810 — footnote 10

- Citation: *J. Am. Chem. Soc.* **2000**, *122*, 190-191.
- Reaction: thioamide 1 = 1-propanethiioyl-(2R,5R)-trans-2,5-diphenylpyrrolidine (C2-symmetric) CCC(=S)N1[C@@H](c2ccccc2)CC[C@H]1c1ccccc1 (stereodescriptors as drawn, [M]); 1) BuLi; 2) 2-cyclohexylideneethyl bromide $\rightarrow$ ketene N,S-acetal 2 $\rightarrow$ thio-Claisen [3,3] $\rightarrow$ 3: thioamide with α -CH(CH_3) attached to a quaternary cyclohexyl carbon bearing vinyl (2-methyl-3-(1-vinylcyclohexyl)propanethiioyl... i.e. α -stereocenter created).
- Asked: (a) structure of 2 with Z $C=C$ (use retro of 3); (b) mechanism 1 $\rightarrow$ 2; (c) explain stereochemical outcome at new stereocenter.
- Answer [H/M]: (a) 2 = (Z)-S-(2-cyclohexylideneethyl) ketene N,S-acetal: $CH_3CH=C(N\text{-pyrrolidiny}) (S\text{-}CH_2CH=C(CH_2)_5)$, C/C=C(/SCC=C1CCCCC1)N1[C@@H](c2ccccc2)CC[C@H]1c1ccccc1. (b) BuLi deprotonates α -CH₂ $\rightarrow$ (Z)-thioenolate (A1,3 strain with the bulky pyrrolidine); S-alkylation (S_N2) by the allylic bromide $\rightarrow$ 2. (c) [3,3] via chair TS; the C2-symmetric 2,5-diphenylpyrrolidine shields one face of the ketene N,S-acetal, so

the allyl group is delivered to one face → single configuration at the new α -stereocenter (Z geometry + chair → defined relative/absolute configuration).

Chapter 18 — 방향족 화합물 (Aromatic Compounds)

Summary pages: book p.842-844 (PDF p27-29: reaction review + concept review p.842-843, skill review p.843-844). Problems p.844-849 (PDF p29-34). References: book p.849 (PDF p34).

Key reactions / concepts

- Benzylic reactions: oxidation ($\text{Na}_2\text{Cr}_2\text{O}_7/\text{H}_2\text{SO}_4$ or KMnO_4 , heat → benzoic acid, needs benzylic H); NBS/heat radical benzylic bromination; benzylic halides undergo $\text{S}_\text{N}1$, $\text{S}_\text{N}2$, E1, E2; dehydration of benzylic alcohols.
- Reduction: catalytic hydrogenation of benzene (Ni , 100 atm, 150 °C) → cyclohexane; vinyl group selectively hydrogenated; Birch reduction (Na , CH_3OH , NH_3) → 1,4-cyclohexadiene — EDG (alkyl) carbons not reduced, EWG ($\text{C}=\text{O}$) carbons reduced.
- Nomenclature: monosubstituted (common names toluene, phenol, aniline, anisole...), ortho/meta/para, locants, alphabetical order.
- Benzene structure (bond order 1.5), resonance; stability (heats of hydrogenation, stabilization energy); MO picture; Hückel $4n+2$ rule; Frost circle; antiaromatic cyclobutadiene, tub-shaped nonaromatic COT; annulenes ([10]annulene nonaromatic due to steric strain); cyclopentadienyl anion & tropylium cation aromatic; heterocycles: pyridine lone pair localized (sp^2), pyrrole lone pair delocalized (part of 6π); lone-pair participation analysis (histamine example); PAHs.
- Spectroscopy: aromatic IR bands; ^1H benzylic 2-3 ppm, aryl H ~7 ppm; ^{13}C 100-150 ppm.

Literature-footnoted problems (refs 1-10)

18.66 — book p.848 — footnote 1

- Citation: *Tetrahedron Lett.* **2012**, 53, 5385-5388.
- Compound A: benzo-fused 14-membered dehydroannulene-type macrocycle with quinoidal π system: ring contains o-phenylene, four $\text{C}\equiv\text{C}$, one cis $\text{CH}=\text{CH}$ and two ring sp^2 carbons each bearing exocyclic $=\text{C}(\text{CN})_2$ (dicyanomethylene, quinoidal). 2e^- reduction → B: dianion; drawn with negative charge on each $\text{C}(\text{CN})_2$ and an unpaired electron (dot) on each ring carbon.
- Asked: (a) is the 14-membered ring of B aromatic? (b) resonance structures showing delocalization of negative charge; (c) at least three resonance structures showing delocalization of unpaired (odd) electrons.
- Answer [M]: (a) Yes — pairing the two ring "radicals" across the ring gives a closed-shell 14- π -electron perimeter (benzo $\text{C}=\text{C}$ 2e^- + 4 alkynes' out-of-plane π 8e^- + $\text{CH}=\text{CH}$ 2e^- + two ring carbons 2e^- = 14 = $4n+2$, $n = 3$) → aromatic dehydro[14]annulene; aromatization is the driving force for facile reduction of the quinoidal A. (b) $\text{C}(\text{CN})_2^-$ charge delocalized onto both nitrile N atoms (ketenimine-type resonance, $^-\text{N}=\text{C}=\text{C}$). (c) unpaired electrons move along the conjugated perimeter: onto alkyne carbons (allenyl-type forms), across $\text{CH}=\text{CH}$, into the benzo ring.

18.67 — book p.848 (도전 문제) — footnote 2

- Citation: *J. Am. Chem. Soc.* **1959**, *81*, 4671-4673. (same reference as ch.15 fn 5)
- Compounds: 1 = tetrazolo[5,1-?]-1,2-benzisothiazole (fused tetrazole, S-N bridgehead) c1ccc2c(c1)sn1nnnc12; 2 = 3-azido-1,2-benzisothiazole [N-]=[N+]=Nc1nsc2ccccc12; rapid equilibrium $1 \rightleftharpoons 2$.
- Asked: in 2 only one of the four N remains in a ring; does that N's hybridization change on isomerization? explain.
- Answer [M-H]: That N (bonded to S) is sp^2 in both. In 1 it is a 3-connected (pyrrole-like) N whose lone pair occupies a p orbital contributing to the aromatic sextet of the tetrazole; in 2 it is a 2-connected (pyridine-like) N with its lone pair in an sp^2 orbital (isothiazole ring aromaticity uses one p electron). Hybridization unchanged; lone-pair orbital location changes.

18.68 — book p.848 — footnote 3

- Citation: *J. Am. Chem. Soc.* **2005**, *127*, 13070-13078.
- Context: Titan atmosphere chemistry modelling: 1,2,5,6-tetracyanocyclooctatetraene N#CC1=C(C#N)C=CC(C#N)=C(C#N)C=C1 formed by gas-phase $Mg^+ \cdot$ -mediated reactions; all four nitrile N lone pairs bind one $Mg^+ \cdot$.
- Asked: (a) draw the complex (dashed lines for $N \cdots Mg$); (b) why 1,2,4,5-tetracyanobenzene cannot form a similar complex.
- Answer [H]: (a) COT is non-planar tub-shaped; the four CN groups at 1,2,5,6 point to the same face (convergent), so Mg^+ sits over the tub with four $N \rightarrow Mg$ dative bonds. (b) Tetracyanobenzene is planar (aromatic); its nitriles are linear and diverge in the ring plane, so the four N lone pairs cannot converge on one metal ion.

18.69(c) — book p.848 — footnote 4

- Citation: *J. Am. Chem. Soc.* **1985**, *107*, 6678-6683.
- Context: (a) naphthalene resonance $\rightarrow$ C1-C2 (1.36 Å) shorter than C2-C3 (1.42 Å) (C1-C2 double in 2 of 3 structures); (b) phenanthrene's 5 resonance structures $\rightarrow$ C9-C10 double in 4 of 5 $\rightarrow$ shortest ("bond fixation"); (c) phenanthrene + $Br_2 \rightarrow C_{14}H_{10}Br_2$ — draw product.
- Answer [H]: trans-9,10-dibromo-9,10-dihydrophenanthrene BrC1c2ccccc2-c2ccccc2C1Br — addition across the alkene-like C9=C10 bond (retains two benzene sextets).

18.70 — book p.848 — footnote 5

- Citation: *Angew. Chem. Int. Ed.* **1988**, *27*, 1437-1455.
- Topic: stable cyclobutadienes (normally dimerize by Diels-Alder).
- Asked: (a) why tri-tert-butylcyclobutadiene ($R_1=R_2=R_3=t\text{-Bu}$, $R_4=H$) is stable; (b) why the push-pull derivative $R_1=R_3=NEt_2$, $R_2=R_4=CO_2Et$ is stable; (c) why the same donor/acceptor pattern destabilizes $4n+2$ systems.
- Answer [M-H]: (a) kinetic (steric) stabilization: bulky t-Bu groups block approach for dimerization (DA). (b) captodative push-pull: zwitterionic resonance forms (iminium/enolate) delocalize charge and give the ring 2π -cation/allyl anion character, reducing cyclic 4π antiaromatic delocalization (thermodynamic stabilization) + sterics. (c) In aromatic $4n+2$ rings, the polarization/charge localization disrupts the stabilizing cyclic delocalization (loss of aromaticity), so it is destabilizing; in antiaromatic rings breaking up cyclic delocalization is stabilizing.

18.71 — book p.848 — footnote 6

- Citation: *J. Org. Chem.* **2004**, *69*, 549-554.
- Reaction: cyclopentadiene-fused [14]annulene = 2,7-di-tert-butyl-trans-10b,10c-dimethyl-10b,10c-dihydropyrene with a fused cyclopentadiene ring; + KH → blue, stable cyclopentadienide anion. Internal CH₃ ¹H shift moves from -3.9 to -1.8 ppm.
- Asked: explain.
- Answer [H]: the internal methyls sit inside the diatropic [14]annulene ring current (strongly shielded, -3.9). On deprotonation, the five-membered ring becomes an aromatic 6π cyclopentadienide; to share π electrons with this new aromatic ring, bond alternation in the macrocycle increases (benzannelation-type competition), weakening the [14]annulene ring current → methyls less shielded (-1.8 ppm).

18.72 — book p.848 — footnote 7

- Citation: *J. Nat. Prod.* **2011**, *74*, 1972-1979.
- Compound: antimalarial (vs. drug-resistant Plasmodium) bromo-β-carboline from a New Zealand ascidian (sea squirt): 7-bromo-1-(4-guanidinobutyl)-9H-pyrido[3,4-b]indole NC(=N)NCCCC1NCCC2C1[nH]c1cc(Br)ccc12 [H as drawn].
- Asked: rank basicity of each N; product with 2 equiv HCl.
- Answer [H]: guanidine (most basic; resonance-stabilized guanidinium) > pyridine-type N of β-carboline (sp² lone pair) > indole N-H (lone pair in aromatic sextet, non-basic). 2 HCl → dihydrochloride: guanidinium chloride + pyridinium (N²-H⁺) chloride.

18.73 — book p.849 — footnote 8

- Citation: *J. Org. Chem.* **1964**, *29*, 2065-2066.
- Data: m/z 39 peak in 17 of 20 N-heteroaromatics (5-84% of base peak), including 2-methylpyrazine, 2-aminopyrimidine, pyrazole, pyrrole; absent in imidazole, 3-amino-1,2,4-triazole, 4-amino-1,2,4-triazole.
- Asked: structure of m/z 39 cation and consistency with data.
- Answer [H]: C₃H₃⁺ cyclopropenyl (cyclopropenium) cation — aromatic (2π, Hückel n = 0). Requires three contiguous carbons in the parent; imidazole (C4-C5 only adjacent pair) and the triazoles lack a C-C-C unit, so no m/z 39.

18.74 — book p.849 — footnote 9

- Citation: *J. Org. Chem.* **2002**, *67*, 7029-7036.
- Reaction: 9,9'-bifluorenylidene (1, highly conjugated, planar) → -2e⁻ → dication 2 (two 9-fluorenyl cations joined by C9-C9', drawn twisted/perpendicular).
- Asked: draw ≥4 resonance structures of 2; explain why the one with adjacent like charges (both + at C9/C9') is the most important contributor.
- Answer [M]: other structures place + on ring carbons (ortho/para positions of the benzo rings), each of which disrupts a benzene aromatic sextet; the C9/C9' structure keeps all four benzene rings aromatic, and the two fluorenyl units are orthogonal (twisted), minimizing charge-charge/π interaction. (Note: the fluorenyl cation five-membered ring is formally 4π antiaromatic; delocalization into it is disfavored.)

18.75 — book p.849 — footnote 10

- Citation: *Bioorg. Med. Chem.* **2009**, *17*, 4047-4063.

- Reaction: 5-ethyl-2-phenyl-6-(trifluoromethyl)pyrimidin-4(3H)-one CCc1c(C(F)(F)F)nc(-c2ccccc2)[nH]c1=O + NaOMe, propargyl bromide (HC#CCCH2Br) → A (herbicide; inhibits chlorophyll/β-carotene accumulation) = N3-propargyl pyrimidinone CCc1c(C(F)(F)F)nc(-c2ccccc2)n(CC#C)c1=O + B (inactive constitutional isomer, also acetylenic); A:B = 1:40.
- Asked: (a) mechanism for A; (b) structure and mechanism of B; (c) why B favored.
- Answer [H]: (a) NaOMe deprotonates N-H → ambident amidate (charge on N and O); N attacks propargyl bromide (S_N2). (b) B = O-alkylation product, 5-ethyl-2-phenyl-4-(prop-2-yn-1-yloxy)-6-(trifluoromethyl)pyrimidine CCc1c(OC#C)nc(-c2ccccc2)nc1C(F)(F)F, S_N2 by O⁻. (c) O-alkylation gives a fully aromatic pyrimidine (aromatic stabilization), and steric hindrance at N3 (flanked by C=O and 2-phenyl) disfavors N-alkylation (CF₃ also makes ring electron-poor, favoring charge on O).

Chapter 19 — 방향족 치환반응 (Aromatic Substitution Reactions)

Summary pages: book p.893-897 (PDF p35-39: reaction review p.893-894, concept review p.894-895, skill review p.895-897). Problems p.897-903 (PDF p39-45). References: book p.903 (PDF p45).

Key reactions / concepts

- EAS of benzene: Br₂/AlBr₃ (or FeBr₃) bromination; Cl₂/AlCl₃ chlorination; HNO₃/H₂SO₄ nitration (NO₂⁺); fuming H₂SO₄ sulfonation (SO₃; reversible — dilute H₂SO₄ desulfonation); Friedel-Crafts alkylation (CH₃Cl/AlCl₃; carbocation rearrangements, polyalkylation, halide C must be sp³); Friedel-Crafts acylation (RCOCl/AlCl₃; acylium ion, no rearrangement, no polyacylation) + Clemmensen reduction Zn(Hg)/HCl → alkyl; nitro → amine (Fe or Zn/HCl then NaOH); benzylic NBS (excess → CBr₃); KMnO₄ side-chain oxidation → benzoic acid.
- Mechanism: sigma complex (arenium ion) formation (rate-determining, endergonic) then deprotonation restoring aromaticity.
- Directing effects: activators (o/p) — strong (-NH₂, -OH, -OR [OR moderate exception]), moderate (-OC(O)R, -NHC(O)R), weak (alkyl); deactivators — weak halogens (o/p directors), moderate (C=O, CN; meta), strong (-NO₂, -NR₃⁺, -CX₃; meta). Multiple substituents: strongest activator dominates; steric effects; blocking groups (SO₃H); retrosynthesis for polysubstituted benzenes; limitations (no nitration of anilines, no FC on moderately/strongly deactivated rings).
- Nucleophilic aromatic substitution (S_NAr): needs strong EWG (NO₂), leaving group, EWG ortho/para to LG; addition-elimination via Meisenheimer complex (e.g., p-bromonitrobenzene + NaOH 70 °C → p-nitrophenol).
- Elimination-addition (benzyne): chlorobenzene + NaOH 350 °C → phenol; p-chlorotoluene + NaNH₂/NH₃ → p- + m-toluidine; isotope labeling evidence.
- Decision tree: electrophile → EAS; nucleophile + 3 S_NAr conditions → S_NAr; otherwise elimination-addition.

Literature-footnoted problems (refs 1-10)

19.83 — book p.901 — footnote 1

- Citation: *J. Org. Chem.* **1973**, *38*, 493-499.

- Reaction: 4-X-4'-Y-stilbenes + Br₂ → 1,2-dibromo-1,2-diarylethanes. X = OMe fixed; rate constants k (×10³ L/mol·min): Y = OMe 220; Me 114; Cl 45; NO₂ 5.2.
- Asked: rational explanation of trend, given that the stilbene C=C acts as nucleophile.
- Answer [H]: rate-determining electrophilic attack of Br₂ on C=C forms a bromonium/benzylic-carbocation-like intermediate with positive charge developing at the benzylic carbons; electron-donating Y (OMe > Me) stabilizes the cationic TS by resonance/hyperconjugation and increases alkene nucleophilicity; electron-withdrawing Y (Cl inductive, NO₂ strongly by resonance+induction) destabilizes it → slower (Hammett ρ < 0).

19.84 — book p.901 — footnote 2

- Citation: *J. Org. Chem.* **2012**, 77, 5503-5514.
- Context: route to aminotetralins (antidepressant candidates).
- Reaction: 2-(2-bromo-5-methylphenyl)acetyl chloride O=C(Cl)Cc1cc(C)ccc1Br + H₂C=CH₂, AlCl₃ → 8-bromo-5-methyl-3,4-dihydronaphthalen-2(1H)-one (8-bromo-5-methyl-2-tetralone) O=C1Cc2c(C)ccc(Br)c2C1 [H].
- Answer [H]: AlCl₃ + acid chloride → acylium ion; ethylene attacks acylium (aliphatic Friedel-Crafts acylation) → primary carbocation ArCH₂C(=O)CH₂CH₂⁺ (or its chloride-trapped β-chloroketone, re-ionized by AlCl₃) → intramolecular Friedel-Crafts alkylation at the ring carbon ortho to the CH₂ (and ortho to CH₃, activated) via arenium ion → deprotonation restores aromaticity → 2-tetralone.

19.85 — book p.901 — footnote 3

- Citation: *Org. Lett.* **2007**, 9, 2741-2743.
- Reaction: 1-chloro-3-fluoro-2-nitrobenzene O=[N+]([O-])c1c(F)cccc1Cl + sodium (4-methoxyphenyl)acetylide → 1-chloro-3-[(4-methoxyphenyl)ethynyl]-2-nitrobenzene COc1ccc(C#Cc2cccc(Cl)c2[N+](=O)[O-])cc1 (F replaced, not Cl).
- Asked: (a) mechanism; (b) identify RDS from regiochemistry.
- Answer [H]: S_NAr addition-elimination: acetylide adds to C-F carbon (ortho to NO₂) → Meisenheimer complex (charge delocalized onto NO₂) → loss of F⁻. (b) Nucleophilic addition (Meisenheimer complex formation) is rate-determining: highly electronegative F makes the ipso carbon more electrophilic and stabilizes the anionic TS; C-X bond breaking (where Cl⁻ would be the better LG) is not in the RDS, hence F > Cl.

19.86 — book p.901-902 — footnote 4

- Citation: *J. Am. Chem. Soc.* **1967**, 89, 4418-4424.
- Compounds: 1 = 2,4-dimethoxy-1-tritiobenzene [3H]c1ccc(OC)cc1OC; 2 = 2,4,6-trimethoxy-1-tritiobenzene [3H]c1c(OC)cc(OC)cc1OC. Stable in aqueous base; prolonged aqueous acid → lose T, giving 1,3-dimethoxybenzene and 1,3,5-trimethoxybenzene.
- Asked: (a) mechanism of T loss; (b) which loses T faster and why.
- Answer [H]: (a) acid-catalyzed electrophilic aromatic hydrogen exchange (protodetratiation): H₃O⁺ protonates the ring carbon bearing T → arenium ion (σ complex, stabilized by OMe groups ortho/para) → loss of T⁺ restores aromaticity. (b) 2 faster: the protonated carbon is ortho/para to three OMe groups (two ortho, one para) vs two (one ortho, one para) in 1 → more stabilized arenium ion/TS.

19.87 — book p.902 (도전 문제) — footnote 5

- Citation: *Prog. Stereochem.* **1958**, *2*, 125.
- Compound: 2-iodo-1,3,5-trinitrobenzene (picryl iodide) Ic1c([N+](=O)[O-])cc([N+](=O)[O-])cc1[N+](=O)[O-]. Bond a = C-NO₂ ortho to I: 1.45 Å; bond b = C-NO₂ para to I: 1.35 Å.
- Answer [H]: steric inhibition of resonance — the large iodine forces the two ortho nitro groups to twist out of the ring plane, so no π overlap with the ring (pure single C-N bond, longer); the para nitro group is coplanar and conjugated (quinoid resonance gives partial C=N double-bond character → shorter).

19.88 — book p.902 — footnote 6

- Citation: *Org. Lett.* **1999**, *1*, 985-988.
- Context: synthesis of antitumor 7-hydroxynitidine (benzo[c]phenanthridine).
- Reaction: 1 = N-(5-bromo-2-benzyloxy-3,4-dimethoxybenzyl)-amine of 6,7-methylenedioxy-1-naphthylamine (aryl bromide with ArCH₂-NH-naphthyl) + NaNH₂ (2 equiv), then H₂O → 2 = dihydrobenzo[c]phenanthridine (new aryl-aryl bond between benzyl ring and naphthalene C2, N-H retained; OCH₂Ph, two OMe, methylenedioxy).
- Answer [H]: 1st equiv NaNH₂ deprotonates N-H → anilide anion (delocalized onto naphthalene C2, enamine-like carbanion character); 2nd equiv removes the aryl H ortho to Br → aryl anion expels Br⁻ → benzyne (between former C-Br and adjacent C-H). Intramolecular nucleophilic attack of the naphthyl C2 (ortho to N⁻) on the proximal benzyne carbon forms the 6-membered ring (aryne cyclization) → aryl anion; H₂O protonates; tautomerization/rearomatization of naphthalene (loss of H from C2) gives 2.

19.89 — book p.902 — footnote 7

- Citation: *J. Org. Chem.* **2005**, *70*, 3511-3517.
- Reaction: 3'-methoxy-2-(phenylethynyl)biphenyl COc1cccc(-c2ccccc2C#Cc2ccccc2)c1 + ICl (I⁺ source) → 9-iodo-10-phenylphenanthrenes A (OMe para to new C-C bond; Ic1c(-c2ccccc2)c2ccc(OC)cc2c2ccccc12) + B (OMe ortho to new C-C bond, peri to C10-Ph; Ic1c(-c2ccccc2)c2c(OC)cccc2c2ccccc12), A:B ≈ 3:1.
- Asked: (a) mechanisms; (b) why A favored.
- Answer [H]: I⁺ activates alkyne (iodonium) → methoxyphenyl ring attacks the Ph-bearing alkyne carbon (6-endo intramolecular SEAr, vinyl-cation-like), either at C para to OMe (→ A) or C ortho to OMe (between OMe and the biaryl bond, → B); deprotonation restores aromaticity. (b) Both positions are activated by OMe (para/ortho), but the ortho site is sterically hindered (flanked by OMe; product B has OMe/phenyl peri-type clash), so para attack (A) dominates.

19.90 — book p.902 — footnote 8

- Citation: *J. Org. Chem.* **1999**, *64*, 6702-6705.
- Reaction: quinuclidin-3-one (1) O=C1CN2CCC1CC2 + benzene, CF₃SO₃H (pK_a ≈ -14) → 3,3-diphenylquinuclidinium (3) [NH+]12CCC(CC1)C(c1ccccc1)(c1ccccc1)C2, via superelectrophilic dication 2 (N-protonated + O-protonated).
- Answer [H]: TfOH protonates N then O → dication 2 (superelectrophile, carbocation adjacent to ammonium); benzene attacks carbonyl carbon (SEAr, arenium ion, deprotonation) → 3-hydroxy-3-phenylquinuclidinium; OH protonated, loss of H₂O → benzylic tertiary carbocation (dication); second benzene attacks (SEAr), deprotonation → 3.

19.91 — book p.902-903 — footnote 9

- Citation: *J. Am. Chem. Soc.* **2011**, *133*, 18614-18617.
- Reaction: bis(biphenyl-2-yl)amine (1) c1ccc(-c2ccccc2Nc3ccccc2-c2ccccc2)cc1 + BuLi, then BCl₃ → N-BCl₂ aminoborane (2); AlCl₃ → 3 = B-N-doped dibenzo[g,p]chrysene analog (BN replaces one central C=C) B12c3ccccc3-c3ccccc3N1c1ccccc1-c1ccccc12 (C₂₄H₁₆BN).
- Asked: (a) mechanism 1→3; (b) are all six-membered rings aromatic (extended aromatic system)? (c) X-ray shows central two rings not planar, slightly twisted — why?
- Answer [M-H]: (a) BuLi deprotonates N-H; amide N attacks BCl₃, Cl⁻ lost → 2. AlCl₃ abstracts Cl⁻ → borenium cation (strong electrophile) → intramolecular electrophilic borylation (SEAr) at ortho carbon of a pendant phenyl → arenium → deprotonation (HCl); repeat for second Cl/phenyl (double intramolecular Friedel-Crafts-type borylation). (b) Yes: each BN-containing ring has 6 π electrons (N lone pair 2 + two fused benzo C=C 4; B contributes empty p) — 1,2-azaborine-type, isoelectronic with benzene; molecule is isoelectronic/isosteric to dibenzo[g,p]chrysene. (c) Steric repulsion between the H atoms in the fjord/bay regions of the four peripheral benzo rings forces a twist (as in the all-carbon dibenzo[g,p]chrysene).

19.92 — book p.903 — footnote 10

- Citation: *Org. Lett.* **2004**, *6*, 2381-2383.
- Reaction: bis(2'-bromo-5'-methoxybiphenyl-2-yl) ketone (1) + catalytic aqueous acid (H₃O⁺) → 4,4'-dibromo-1,1'-dimethoxy-9,9'-spirobifluorene (Br and OMe positions as drawn: OMe adjacent to spiro-fused ring carbon, Br para to OMe).
- Answer [H]: protonation of C=O → carbocation (very electrophilic, can effect cyclization spontaneously); intramolecular SEAr by a methoxy-activated aryl ring (attack ortho to OMe) → fluoren-9-ol (9-aryl) after deprotonation; protonation of OH, loss of H₂O → 9-arylfluorenyl cation; second intramolecular SEAr by the other methoxy-aryl ring → spiro center, deprotonation → spirobifluorene. Double intramolecular Friedel-Crafts.

Notes / caveats

- Ch.16 footnote 1 (*Food Chem.* 2016, 196, 211-219) is listed but its superscript was not found in the pages provided (PDF p12-17 problem pages checked); likely attached to an in-chapter problem outside this PDF.
- Ch.15 fn5 and Ch.18 fn2 are the same printed citation (*J. Am. Chem. Soc.* 1959, 81, 4671-4673) — both azide/tetrazole valence-tautomerism problems.
- Problem answers are my own (not from book solutions); confidence tags indicate reliability. Structures and SMILES were checked for validity with RDKit; exact masses for 15.69/15.74/15.78 computed with RDKit.

Klein 유기화학 — Chapters 20-23: literature-based problems

Source PDF: 287e3d22-20-23.pdf (42 scanned pages, no text layer). PDF page N = p-NN.jpg. Confidence tags: [H] high, [M] medium, [L] low. SMILES checked with RDKit for validity/formula; stereo given only where drawn

clearly.

Chapter 20 — 알데하이드와 케톤 (Aldehydes and Ketones)

PDF pp. 1-10 = book pp. 948-957. (Chapter body not included; only summary + problems.)

Key reactions (반응의 복습, book p. 948)

1. Hydrate formation (H_2O , $[\text{H}^+]$)
2. Acetal formation (ROH , $[\text{H}^+]$, $-\text{H}_2\text{O}$)
3. Cyclic acetal ($\text{HOCH}_2\text{CH}_2\text{OH}$)
4. Cyclic thioacetal ($\text{HSCH}_2\text{CH}_2\text{SH}$)
5. Desulfurization (Raney Ni $\rightarrow$ CH_2)
6. Imine (RNH_2)
7. Enamine (R_2NH)
8. Oxime (NH_2OH)
9. Hydrazone (NH_2NH_2)
10. Wolff-Kishner reduction (NaOH , H_2O , heat)
11. Ketone reduction (LiAlH_4 ; H_2O)
12. Grignard (RMgBr ; H_2O)
13. Cyanohydrin (HCN , KCN)
14. Wittig ($\text{H}_2\text{C}=\text{PPh}_3$)
15. Baeyer-Villiger (RCO_3H). Summary text also stresses: nucleophilic addition (attack, then proton transfer); hemiacetal $\rightarrow$ acetal; acetals as protecting groups (stable to base); unstabilized ylides $\rightarrow$ (Z)-alkenes, stabilized $\rightarrow$ (E); B-V migratory aptitude; IR $\text{C}=\text{O}$ $\sim 1715\text{ cm}^{-1}$, aldehyde C-H 2700/2850; ^1H CHO $\sim 10\text{ ppm}$; ^{13}C $\text{C}=\text{O}$ $\sim 200\text{ ppm}$.

References (참고문헌, book p. 957 / PDF p. 10)

1. Tetrahedron 2012, 68, 9289-9292. (footnote 1 not located in the problem pages — presumably cited in chapter body)
2. Tetrahedron Lett. 2009, 50, 4587-4591.
3. J. Org. Chem. 2001, 66, 2072-2077.
4. Tetrahedron 2014, 43, 8161-8167. (volume printed as "43" — as printed; Tetrahedron 2014 is actually vol. 70, likely a book typo)
5. Tetrahedron Lett. 1969, 10, 4233-4236.
6. J. Org. Chem. 2003, 68, 6455-6458.
7. Tetrahedron Lett. 2010, 51, 4494-4496.
8. Chem. Sci. 2010, 1, 37-42.
9. Chem. Ber. 1962, 95, 2514-2525.
10. J. Am. Chem. Soc. 1998, 120, 4039-4040.
11. J. Am. Chem. Soc. 2011, 133, 9956-9959.
12. J. Am. Chem. Soc. 2006, 128, 6499-6507.

Literature problems

20.83 — book p. 955 (PDF p. 8) — footnote 2 — Tetrahedron Lett. 2009, 50, 4587-4591

- Context: synthetic studies toward **cyclodidemniserinol trisulfate** (HIV-1 integrase inhibitor; the superscript 2 is attached to "HIV-1 인테그라제").
- SM: ethyl β -keto ester, $\text{EtO}_2\text{C}-\text{CH}_2-\text{C}(=\text{O})-(\text{CH}_2)_n-\text{OR}$ ($n \approx 7$ from the zigzag; R = protecting group). SMILES ($n=7$ as drawn, R left generic): CCOC(=O)CC(=O)CCCCCOC[R] [M on chain length].
- Product: 2-[(2-substituted-1,3-dioxolan-2-yl)methyl]-1,3-dithiane, i.e. ketone $\rightarrow$ ethylene acetal; ester carbon $\rightarrow$ 1,3-dithiane C2 (aldehyde oxidation level, thioacetal).
- Asked: a **four-step** synthesis.

- Concept: chemoselective protection (ketone acetal), ester reduction, oxidation to aldehyde, thioacetal formation.
- Expected answer [M]: (1) HOCH₂CH₂OH, [TsOH] (–H₂O) → ketal; (2) xs LiAlH₄, then H₂O → primary alcohol; (3) PCC → aldehyde; (4) HS(CH₂)₃SH, [H⁺] (or BF₃) → 1,3-dithiane. (DIBAH route would be 3 steps; book chapter-20 toolkit implies LAH/PCC.)

20.84 — book p. 955 (PDF p. 8) — footnote 3 — J. Org. Chem. 2001, 66, 2072-2077

- SM 1: rigid polycyclic cage ketone (bridging C=O on a tetracyclic "trishomocubane-type" cage; no name given). 1 →? lactone 2 →? diol 3 (CH₂OH + secondary OH) → (TsCl, py) primary monotosylate 4 →? cyclic ether 5 (one-carbon-expanded oxa-bridge).
- Asked: (a) reagents for each "?" step; (b) why ¹H NMR of 5 shows many more signals than 1.
- Concept: Baeyer-Villiger, LiAlH₄ reduction of lactone, selective tosylation of 1° OH, intramolecular Williamson etherification; molecular symmetry and NMR.
- Expected answer [H]: 1→2 RCO₃H (e.g. MCPBA/CH₃CO₃H) Baeyer-Villiger (O inserted next to more-substituted carbon); 2→3 xs LiAlH₄ then H₂O; 4→5 base (NaH, or heat/py) → alkoxide displaces OTs intramolecularly (S_N2). (b) 1 has a mirror plane (symmetric cage, pairs of equivalent H); ring expansion makes the bridge unsymmetrical (–CH₂–O– vs C=O), destroying the symmetry → all H's inequivalent.

20.85 — book p. 955-956 (PDF pp. 8-9) — footnote 4 — Tetrahedron 2014, "43" [sic], 8161-8167

- Compound: **Scorzocreticin (S)-1** (isolated from a plant used in traditional Cretan cooking); 3-aryl-dihydroisocoumarin.
- SM 1: 3,5-bis(OR)benzyl alcohol, ArCH₂OH (Ar = 3,5-(RO)₂C₆H₃). Product 2: ArCH₂CH₂OH (one-carbon homologation).
- Asked: route from 1 to 2 that includes a Wittig reaction.
- Expected answer [H]: (1) PCC → ArCHO; (2) Ph₃P=CHOCH₃ (Wittig) → ArCH=CHOMe; (3) H₃O⁺ → ArCH₂CHO (enol-ether hydrolysis); (4) NaBH₄ (or LiAlH₄), then H₂O → ArCH₂CH₂OH. (Alternative: Ph₃P=CH₂ then BH₃·THF/H₂O₂, NaOH — also valid, 3 steps.)

20.86 — book p. 956 (PDF p. 9) — footnote 5 — Tetrahedron Lett. 1969, 10, 4233-4236

- Reaction: cyclohexanone + optically pure ethyl prolinatate (pyrrolidine-2-CO₂Et, wedge drawn; L/(S) presumed), TsOH, –H₂O → enamine. SMILES product: CCOC(=O)[C@H]1CCCN1C1=CCCC1 [M on (S)].
- Asked: draw the two **conformational (rotameric) enamines** formed, with wedges/dashes; explain why their interconversion at rt is very slow.
- Concept: enamine formation; N lone pair conjugation with C=C gives the N-C(vinyl) bond partial double-bond character → restricted rotation (s-cis/s-trans rotamers with the ring alkene syn or anti to the CO₂Et-bearing α-carbon).
- Expected answer [M]: two rotamers about the N-C(sp²) bond; slow rotation because resonance (N=C–C[–]) makes it partially double; rotation would break the conjugation plus steric clash with the CO₂Et group.

20.87 — book p. 956 (PDF p. 9) — footnote 6 — J. Org. Chem. 2003, 68, 6455-6458

- SM A: cyclooctenone (cis-cyclooct-4-en-1-one as drawn; ring C=C + C=O). Product B: cyclooctadiene bearing –CH₂CH₂CHO on the former carbonyl carbon (new ring C=C at that carbon).

- Sequences: A = (1) BrMgCH₂CH₂-(1,3-dioxolan-2-yl), (2) H₂SO₄, heat, (3) AcOH, H₂O; B = (1) homoallyl-MgBr (CH₂CH₂CH=CH₂), (2) H₂SO₄ heat, (3) O₃, DMS; C = (1) allyl-MgBr, (2) H₂SO₄ heat, (3) BH₃·THF; H₂O₂, NaOH, (4) PCC.
- Asked: which sequence correctly gives B; explain.
- Concept: chemoselectivity — reagents that attack C=C (O₃, BH₃) cannot discriminate the ring alkenes; acetal as masked aldehyde survives Grignard.
- Expected answer [H]: **Sequence A**. B fails: ozonolysis also cleaves both ring C=C bonds. C fails: hydroboration also attacks the ring alkenes (trisubstituted/cis alkenes), giving mixtures.

20.88 — book p. 956 (PDF p. 9) — footnote 7 — Tetrahedron Lett. 2010, 51, 4494-4496

- Context: synthesis of furo[2,3-b]indoles.
- SM 1: 2-nitrobenzaldehyde O=Cc1ccccc1[N+](=O)[O-]. Product 2: 2-(2-nitrophenyl)pent-4-enal O=CC(CC=C)c1ccccc1[N+](=O)[O-] (C₁₁H₁₁NO₃).
- Asked: two-step synthesis.
- Expected answer [M]: (1) Wittig with allyloxymethylene ylide Ph₃P=CH-O-CH₂CH=CH₂ → allyl vinyl ether ArCH=CH-O-CH₂CH=CH₂; (2) heat → [3,3] Claisen rearrangement → 2. (Claisen is beyond ch. 20; the key ch-20 idea is the alkoxyethylene Wittig homologation.)

20.89 — book p. 956 (PDF p. 9) — footnote 8 — Chem. Sci. 2010, 1, 37-42

- Context: Bamford-Stevens-Shapiro reaction used in total synthesis of **(+)-frondosin B** (marine natural product); vinyl lithium 2 adds to aldehyde 3 (5-methoxybenzofuran-2-yl, (R)-3-substituted butanal).
- SM 1: 2,2-dimethylcyclohexanone 2,4,6-trialkylbenzenesulfonylhydrazide (R = alkyl; trisylhydrazide if R = iPr): CC(C)c1cc(C(C)C)c(S(=O)(=O)NN=C2CCCC2(C)C)c(C(C)C)c1 [M for R = iPr]. Reagent: t-BuLi (2 equiv). Product 2: 6,6-dimethylcyclohex-1-enyllithium [Li]C1=CCCC1(C)C.
- Asked: mechanism 1 → 2 (Wolff-Kishner-like).
- Expected answer [H]: (1) t-BuLi removes N-H; (2) second t-BuLi removes α-C-H (only the CH₂ α-carbon; other side is quaternary) → dianion; (3) loss of ArSO₂⁻ gives vinyl diazenide (C=C-N=N⁻); (4) loss of N₂ → vinyl carbanion = vinyl lithium 2.

20.90 — book p. 956 (PDF p. 9) — footnote 9 — Chem. Ber. 1962, 95, 2514-2525

- Reaction: cyclohexanone + Ph₃P=CHOMe (1); then H₃O⁺ → cyclohexanecarbaldehyde O=CC1CCCCC1.
- Asked: (a) Wittig product; (b) mechanism of acid-catalysed hydrolysis to aldehyde.
- Expected answer [H]: (a) (methoxymethylene)cyclohexane COC=C1CCCCC1. (b) protonation of the enol ether at the ring carbon (C of C=C) → oxocarbenium; H₂O attack; proton transfer → hemiacetal; protonation of OMe, loss of MeOH → protonated aldehyde; deprotonation.

20.91 — book p. 956 (PDF p. 9) — footnote 10 — J. Am. Chem. Soc. 1998, 120, 4039-4040

- Context: synthesis of **hispidospermidin** (phospholipase C inhibitor, fungal). Intramolecular aliphatic Friedel-Crafts acylation.
- SM: hydrindane (cis-fused 5/6) with ring alkene (cyclohexene) and a -CH₂CH₂COCl chain on the angular carbon; methyl on CH next to alkene; AlCl₃. Products: **1** = bridged bicyclic ketone with Cl on carbon next to CH(Me) (β-chloroketone, Cl added); **2** = bridged ketone with trisubstituted alkene bearing the methyl (C(Me)=C at ring junction); ratio 3:2.

- Asked: mechanism for formation of 1 and 2.
- Expected answer [M]: AlCl_3 generates acylium ion; alkene attacks intramolecularly $\rightarrow$ new C-C bond + secondary carbocation adjacent to $\text{CH}(\text{Me})$. Path to 1: chloride (AlCl_4^-) captures the cation. Path to 2: 1,2-hydride shift from $\text{CH}(\text{Me}) \rightarrow$ tertiary cation; loss of H^+ from the adjacent ring-junction C-H gives the trisubstituted alkene.

20.92 — book p. 956 (PDF p. 9) — footnote 11 — J. Am. Chem. Soc. 2011, 133, 9956-9959

- Context: total synthesis of **kibdelone C** (cytotoxic, soil microbe).
- SM 1: 2-aryl-4,6-dihydroxycyclohex-2-en-1-one; aryl = 3-bromo-2,6-dimethoxy... (ring: OMe ortho to $\text{C}=\text{O}$, Br, OMe, OH ortho to $\text{C}=\text{O}$). SMILES (stereo indicative only): COC1CC(Br)C(OC)C(O)C1C(=O)C=C[C@H](O)C[C@H](O)C1=O ($\text{C}_{15}\text{H}_{15}\text{BrO}_7$) [M].
- Reagents: H_3O^+ (cat.), acetone. Product 2: tetrahydroxanthone (9-one, $\text{C}_{4a}=\text{C}_{9a}$) with cis-1,2-acetonide and 4-OH.
- Asked: mechanism.
- Expected answer [M]: phenol OH adds to the ring ketone (C_1) $\rightarrow$ hemiketal; protonation/loss of H_2O gives an oxocarbenium conjugated with the ring $\text{C}=\text{C}$ (allylic/vinylogous cation, positive at C_3); water adds at C_3 (allylic transposition, $\text{C}=\text{C}$ moves to $\text{C}_1=\text{C}_2 = \text{C}_{4a}=\text{C}_{9a}$) $\rightarrow$ vicinal cis-diol (new C-OH next to existing 4-OH); diol + acetone, $\text{H}^+ \rightarrow$ hemiketal $\rightarrow$ oxocarbenium $\rightarrow$ cyclic acetal (acetonide). Former C_6 -OH becomes 4-OH of 2.

20.93 — book p. 957 (PDF p. 10) — footnote 12 — J. Am. Chem. Soc. 2006, 128, 6499-6507

- SM: 6,6-dimethyl-3-oxocyclohex-1-enyl triflate (vinylogous acyl triflate; gem-dimethyl α to $\text{C}=\text{O}$) CC1(C)CCC(=O)(=O)C(F)(F)F=CC1=O. 1) LiAlH_4 2) $\text{H}_2\text{O} \rightarrow$ A (allylic alcohol, CC1(C)CCC(=O)(=O)C(F)(F)F=CC1O) + B ($\text{C}_8\text{H}_{14}\text{O}$; IR 3305 broad, 2117).
- Asked: (a) structure of B from ^1H NMR: 0.89 (6H, s), 1.49 (1H, br s), 1.56 (2H, t), 1.95 (1H, s), 2.19 (2H, t), 3.35 (2H, s); (b) mechanism (hint: triflate leaving group).
- Expected answer [H]: B = **2,2-dimethylhex-5-yn-1-ol** CC(C)(CO)CCC#C (3305 O-H $\equiv$ C-H, 2117 $\text{C}\equiv\text{C}$; 3.35 s CH_2OH on quaternary C; 1.95 $\equiv\text{C}$ -H; 2.19 propargylic CH_2). Mechanism: hydride adds to $\text{C}=\text{O} \rightarrow$ alkoxide; Grob-type fragmentation (Eschenmoser-Tanabe-like): alkoxide re-forms $\text{C}=\text{O}$, C_1 - C_2 bond breaks, electrons form $\text{C}_2\equiv\text{C}_3$ as TfO^- leaves $\rightarrow$ alkynyl aldehyde $\text{OHC}-\text{CMe}_2-\text{CH}_2\text{CH}_2-\text{C}\equiv\text{CH}$; second hydride reduces aldehyde $\rightarrow$ B.

Chapter 21 — 카복실산과 그 유도체 (Carboxylic Acids and Their Derivatives)

PDF pp. 11-20 = book pp. 1004-1013.

Key reactions (반응의 복습, book pp. 1004-1005)

- Preparation of RCOOH : $\text{RCH}_2\text{Br} \rightarrow \text{NaCN} \rightarrow \text{H}_3\text{O}^+/\text{heat}$; $\text{RBr} \rightarrow \text{Mg}, \text{CO}_2, \text{H}_3\text{O}^+$. Reaction: $\text{RCOOH} \rightarrow \text{RCH}_2\text{OH}$ (xs LiAlH_4 or $\text{BH}_3\cdot\text{THF}$).
- Acid chlorides (from $\text{RCOOH} + \text{SOCl}_2$): + $\text{ROH}/\text{py} \rightarrow$ ester; + xs NH_3 / RNH_2 / $\text{R}_2\text{NH} \rightarrow$ amides; + $\text{H}_2\text{O}/\text{py} \rightarrow$ acid; + xs $\text{LiAlH}_4 \rightarrow 1^\circ$ alcohol; + $\text{LiAl}(\text{OtBu})_3\text{H} \rightarrow$ aldehyde; + xs $\text{RMgBr} \rightarrow 3^\circ$ alcohol; + $\text{R}_2\text{CuLi} \rightarrow$ ketone.

- Anhydrides: analogous (acid + heat, or NaOH then CH₃COCl).
- Esters: prep by RCOOH + NaOH then CH₃I; Fischer ([H⁺], MeOH); RCOCl + ROH/py. Reactions: H₃O⁺ or NaOH/heat (saponification) → acid; NH₃ → amide; xs LiAlH₄ → alcohol; DIBAH → aldehyde; xs RMgBr → 3° alcohol.
- Amides: RCOCl + 2 NH₃; hydrolysis (H₃O⁺ or NaOH/heat); xs LiAlH₄ → amine; SOCl₂ → nitrile.
- Nitriles: RBr + NaCN; amide + SOCl₂; hydrolysis → RCOOH; RMgBr then H₃O⁺ → ketone; xs LiAlH₄ → RCH₂NH₂. Summary concepts: nucleophilic acyl substitution (tetrahedral intermediate; no H⁻/C⁻ leaving groups), reactivity order acid halide > anhydride > ester > amide; Henderson-Hasselbalch (carboxylates at physiological pH); IR C=O 1650-1850; ¹³C 160-185 (nitrile 115-130); COOH ¹H ~12 ppm.

References (참고문헌, book p. 1013 / PDF p. 20)

1. Org. Lett. 2000, 2, 791-794.
2. J. Org. Chem. 2012, 77, 5503-5514.
3. Tetrahedron Lett. 2012, 53, 1859-1862.
4. Org. Biomol. Chem. 2006, 4, 2119-2157.
5. Tetrahedron Lett. 2010, 51, 6542-6544.
6. J. Am. Chem. Soc. 1961, 83, 4214-4216.
7. J. Am. Chem. Soc. 1970, 92, 397-398.
8. J. Org. Chem. 1984, 49, 4094-4095.
9. Tetrahedron Lett. 2008, 49, 6853-6855.
10. Angew. Chem. Int. Ed. 2009, 48, 9105-9108.

Literature problems

21.81 — book p. 1012 (PDF p. 19) — footnote 1 — Org. Lett. 2000, 2, 791-794

- Context: olefin metathesis (Grubbs; 2005 Nobel) — RCM to 2,6-disubstituted 3,6-dihydro-2H-pyran.
- SM 1: 1-phenylbut-3-en-1-ol (single enantiomer, Ph wedge) OCC(C=C)c1ccccc1.
- Steps: 1) NaH, then ethyl 2-bromopropanoate (CH₃CHBrCO₂Et, drawn with defined stereocentre); 2) DIBAH; 3) Ph₃P=CH₂ → diene ether 2; then Grubbs catalyst → 3.
- Asked: product of each step.
- Expected answer [H]: step 1: alkoxide SN₂ (inversion at the bromide carbon) → ethyl 2-(1-phenylbut-3-enyloxy)propanoate CCOC(=O)C(C)OC(C=C)c1ccccc1; step 2: DIBAH (1 equiv, low T) → aldehyde O=CC(C)OC(C=C)c1ccccc1; step 3: Wittig → 2 C=CC(C)OC(C=C)c1ccccc1 (C₁₄H₁₈O); RCM → 3, 2-methyl-6-phenyl-3,6-dihydro-2H-pyran CC1OC(C2CCCC2)CC=C1 + ethylene. Stereo: 2,6-cis/trans as drawn (both wedge-type, from inversion of bromide).

21.82 — book p. 1012 (PDF p. 19) — footnote 2 — J. Org. Chem. 2012, 77, 5503-5514

- Context: aminotetralins (antidepressant candidates).
- SM: 1-bromo-3,5-dimethylbenzene Cc1cc(C)cc(Br)c1 → (3-bromo-5-methylphenyl)acetic acid Cc1cc(Br)cc(C(=O)O)c1 in 3 steps.
- Expected answer [H]: (1) NBS, hv (or heat/peroxide) — benzylic monobromination; (2) NaCN (SN₂) → ArCH₂CN; (3) H₃O⁺, heat → ArCH₂COOH.

21.83 — book p. 1012 (PDF p. 19) — footnote 3 — Tetrahedron Lett. 2012, 53, 1859-1862

- Context: steroid derivatives; intermediate for pentacyclic steroids.
- SM 1: cholic-acid-derived methyl ester: 3 α ,7 α -dimethoxy-12 α -hydroxy-5 β -cholan-24-oate methyl ester (12 α -OH, C24 CO₂Me). Product 2: 12 α -O-allyl ether; side chain C24 reduced to CH₂-O-C(=O)CH=CH₂ (acrylate ester); 3 α ,7 α -OMe retained.
- Asked: plausible three-step synthesis.
- Expected answer [M-H]: (1) NaH, then CH₂=CHCH₂Br (Williamson allylation of 12-OH); (2) xs LiAlH₄, then H₂O (ester $\rightarrow$ 1° alcohol C24-OH); (3) CH₂=CHCOCl (acryloyl chloride), pyridine $\rightarrow$ acrylate ester. (Order matters: allylate before reduction so only one OH is free at each stage.)

21.84 — book p. 1012 (PDF p. 19) — footnote 4 — Org. Biomol. Chem. 2006, 4, 2119-2157

- Context: key fragment for total synthesis of an analog of **amphidinolide N** (cytotoxic, marine dinoflagellate Amphidinium sp.).
- SM 1: (E)- α,β -unsaturated (2-methyl-2-enoic) carboxylic acid bearing stereocentres with Me, OPMB (p-methoxybenzyl ether), OTBS (OSiMe₂tBu) and a TMS-capped alkyne (C $\equiv$ C-SiMe₃). Product 2: the corresponding allylic bromide (CO₂H $\rightarrow$ CH₂Br).
- Expected answer [M]: (1) LiAlH₄ (or esterify then DIBAH $\times 2$), then H₂O $\rightarrow$ allylic 1° alcohol; (2) PBr₃ (or CBr₄/PPh₃, or MsCl then LiBr) $\rightarrow$ allylic bromide. Protecting groups untouched.

21.85 — book p. 1012 (PDF p. 19) — footnote 5 — Tetrahedron Lett. 2010, 51, 6542-6544

- Context: intermediate 3 for γ -secretase inhibitor **BMS-708163**.
- SM 1: 4-(bromomethyl)-3-fluorobenzonitrile N#Cc1ccc(CBr)c(F)c1; KOAc (CH₃CO₂K), DMF $\rightarrow$ 2, 4-(acetoxymethyl)-3-fluorobenzonitrile CC(=O)OCc1ccc(C#N)cc1F; NaOMe, MeOH $\rightarrow$ 3, 3-fluoro-4-(hydroxymethyl)benzonitrile N#Cc1ccc(CO)c(F)c1.
- Asked: (a) why NaOH(aq) cannot convert 1 $\rightarrow$ 3 directly (side products); (b) full mechanism of the two steps.
- Expected answer: (a) [M] hydroxide also hydrolyses the nitrile ($\rightarrow$ amide/carboxylate: 3-fluoro-4-(hydroxymethyl)benzamide / benzoic acid); also possible dibenzyl ether (alkoxide of 3 + 1) and S_NAr of F. (b) [H] S_N2 of acetate on benzylic bromide (one step); then transesterification: MeO⁻ adds to ester C=O $\rightarrow$ tetrahedral intermediate $\rightarrow$ expels ArCH₂O⁻ $\rightarrow$ proton transfer from MeOH gives 3 + MeOAc (catalytic methoxide regenerated).

21.86 — book p. 1012 (PDF p. 19) — footnote 6 — J. Am. Chem. Soc. 1961, 83, 4214-4216

- Reaction: m-/p-substituted methyl benzoates + NaOH, dioxane:H₂O (3:2), 35 °C $\rightarrow$ sodium benzoates + CH₃OH. Rate constants k (M⁻¹min⁻¹): p-NO₂ 102; m-NO₂ 63; m-Cl 9.1; m-Br 8.6; H 1.7; p-CH₃ 0.98; p-OCH₃ 0.42; p-NH₂ 0.06.
- Asked: structural explanation of the trend, with resonance structures.
- Expected answer [H]: rate-determining hydroxide addition builds negative charge; EWGs (p-NO₂: resonance + induction, largest; m-NO₂: induction only; m-Cl/m-Br: induction) increase C=O electrophilicity $\rightarrow$ faster. EDGs (p-CH₃ hyperconjugation/induction; p-OCH₃ and especially p-NH₂ resonance donation into the ring/carbonyl, drawn as C=O⁻ resonance forms) reduce electrophilicity $\rightarrow$ slower. (Hammett-type correlation.)

21.87 — book p. 1013 (PDF p. 20) — footnote 7 — J. Am. Chem. Soc. 1970, 92, 397-398

- Context: E. J. Corey prostaglandin synthesis.
- SM 1: 7-(methoxymethyl)bicyclo[2.2.1]hept-5-en-2-one. $\rightarrow 2 \rightarrow 3$: cis-2-[5-hydroxy-4-(methoxymethyl)cyclopent-2-enyl]acetic acid (hydroxy acid; relative stereo as drawn).
- Asked: reagents for $1 \rightarrow 2 \rightarrow 3$; structure of 2.
- Expected answer [H]: $1 \rightarrow 2$: peroxyacid (MCPBA, NaHCO_3) Baeyer-Villiger; O inserts between $\text{C}=\text{O}$ and the bridgehead (more substituted, migrates) $\rightarrow$ bicyclic δ -lactone 2 (2-oxabicyclo[3.2.1]oct-6-en-3-one bearing CH_2OMe on C8). $2 \rightarrow 3$: NaOH(aq) hydrolysis, then $\text{H}_3\text{O}^+ \rightarrow$ hydroxy acid 3.

21.88 — book p. 1013 (PDF p. 20) — footnote 8 — J. Org. Chem. 1984, 49, 4094-4095

- Context: cornerstone two-step rearrangement in stereoselective total synthesis of **quadron** (bioactive, *Aspergillus* fungus).
- SM: tricyclic (bridged) carbocycle with secondary mesylate (OMs) and CO_2H ; gem-dimethyl on bridge. Reagents: 1) NaH ; 2) H_2SO_4 , H_2O . Product: rearranged tetracyclic skeleton with γ/δ -lactone whose ring O is on a quaternary (ring-fusion) carbon.
- Asked: mechanism.
- Expected answer [L-M]: (1) NaH deprotonates CO_2H ; ionization of the mesylate is assisted by the antiperiplanar C-C bond (Wagner-Meerwein 1,2-alkyl shift, skeletal rearrangement) and loss of H^+ gives a rearranged alkene; (2) H_2SO_4 protonates the alkene (Markovnikov) $\rightarrow$ tertiary carbocation at the ring-fusion carbon; intramolecular capture by the CO_2H oxygen and deprotonation $\rightarrow$ lactone. Exact bonds should be checked against the 1984 paper (structure drawn with heavy-line perspective).

21.89 — book p. 1013 (PDF p. 20) — footnote 9 — Tetrahedron Lett. 2008, 49, 6853-6855

- Context: determination of absolute configuration of a bromohydrin; unexpected skeletal rearrangement.
- SM 1: cis-fused cyclopenta[b]furan-2-one (hexahydro, γ -lactone) bearing ring-OH (α to the CH_2 -side ring fusion carbon) and Br (on the carbon adjacent to the O-bearing ring fusion carbon), trans to the lactone O. $\text{NaH} \rightarrow 2$: isomeric fused γ -lactone (lactone O now on the former C-OH carbon) with an epoxide on the two carbons that bore the former lactone O and Br.
- Expected answer [M-H]: $\text{NaH} \rightarrow$ alkoxide; intramolecular acyl transfer (translactonization): alkoxide attacks lactone $\text{C}=\text{O} \rightarrow$ tetrahedral intermediate $\rightarrow$ expels the original ring-fusion alkoxide $\rightarrow$ new γ -lactone; liberated alkoxide is anti to adjacent C-Br $\rightarrow$ intramolecular $\text{S}_\text{N}2$ (Williamson) $\rightarrow$ epoxide 2.

21.90 — book p. 1013 (PDF p. 20) — footnote 10 — Angew. Chem. Int. Ed. 2009, 48, 9105-9108

- Context: total synthesis of **englerin A** (cytotoxic; stem bark of a Tanzanian plant); unusual base-catalysed ring contraction.
 - SM: cyclopentane-fused δ -lactone in which the alkyl O-carbon is part of an epoxide (spiro/fused epoxy-lactone: $\text{O}-\text{CH}(\text{epoxide})-\text{C}(\text{Me})(\text{epoxide})$); NaOMe (cat.) $\rightarrow$ cyclopentane-fused γ -lactone with a quaternary carbon bearing CH_3 and CHO .
 - Expected answer [H]: MeO^- adds to lactone $\text{C}=\text{O} \rightarrow$ tetrahedral intermediate $\rightarrow$ ring opens to methyl ester + alkoxide on the epoxide CH carbon; that alkoxide collapses to $\text{C}=\text{O}$ (aldehyde) breaking the epoxide C-O bond $\rightarrow$ tertiary alkoxide on C(Me); this alkoxide attacks the methyl ester (nucleophilic acyl substitution) $\rightarrow$ γ -lactone, releasing MeO^- (catalytic).
-

Chapter 22 — 알파 탄소 화학: 엔올과 엔올 음이온 (Alpha-Carbon Chemistry: Enols and Enolates)

PDF pp. 21-31 = book pp. 1057-1067.

Key reactions (반응의 복습, book pp. 1057-1058)

- α -Halogenation: ketone + Br₂, [H₃O⁺] → α -bromoketone + HBr (acid-catalysed, autocatalytic); Hell-Volhard-Zelinsky (Br₂, PBr₃; H₂O) → α -bromo acid; haloform (NaOH, Br₂; H₃O⁺) methyl ketone → acid.
- Aldol addition (NaOH, H₂O → β -hydroxy aldehyde) and aldol condensation (heat → α,β -unsaturated, E1cb); crossed aldol (HCHO + propanal); intramolecular aldol (hexane-2,5-dione → 3-methylcyclopent-2-enone).
- Claisen condensation (NaOEt; H₃O⁺ → β -keto ester); crossed Claisen (ethyl benzoate + ethyl acetate); Dieckmann (diethyl adipate → 2-carbethoxycyclopentanone).
- Alkylation via enolates: LDA, -78 °C (kinetic, less substituted side) vs NaH, 25 °C (thermodynamic, more substituted side).
- Malonic ester synthesis (NaOEt; RBr; H₃O⁺, heat → RCH₂COOH); acetoacetic ester synthesis (→ methyl ketone).
- Michael addition: R₂CuLi to enals; stabilized nucleophiles (1,3-dione + KOH + enal); Stork enamine (R₂NH/[H⁺]; MVK; H₃O⁺ → 1,5-dicarbonyl); Robinson annulation (2-methylcyclohexane-1,3-dione + MVK, NaOH, heat → Wieland-Miescher-type enone). Summary concepts: keto-enol tautomerism; enolate formation; retro-aldol; 5-/6-membered ring preference; decarboxylation of β -keto acids; 1,3- vs 1,5-difunctional relationships; tandem conjugate addition-alkylation (α,β -dialkylation).

References (참고문헌, book p. 1067 / PDF p. 31)

1. J. Am. Chem. Soc. 1954, 76, 2029-2030.
2. J. Org. Chem. 1977, 42, 1663-1664.
3. J. Org. Chem. 2012, 77, 6371-6376.
4. J. Am. Chem. Soc. 2012, 134, 9291-9295.
5. Tetrahedron Lett. 2010, 51, 6918-6920.
6. J. Am. Chem. Soc. 2011, 133, 15797-15799.
7. Tetrahedron Lett. 1969, 10, 4233-4236. (same paper as Ch. 21 ref 5)
8. Tetrahedron Lett. 2010, 51, 5291-5293.
9. J. Am. Chem. Soc. 2012, 134, 920-923.
10. Org. Lett. 2012, 14, 4738-4741.
11. Can. J. Chem. 2004, 82, 293-300.

Literature problems

22.110 — book p. 1065 (PDF p. 29) — footnote 1 — J. Am. Chem. Soc. 1954, 76, 2029-2030

- Reaction: cyclohexanone + pyrrolidine, TsOH, -H₂O → A; A + 1) CH₃I, 2) H₃O⁺ → B + pyrrolidine.
- Asked: structures of A and B (enamine as C-nucleophile; Stork alkylation).

- Expected answer [H]: A = 1-(cyclohex-1-en-1-yl)pyrrolidine C1CCN(C1)C1=CCCC1; C-methylation gives iminium; hydrolysis → B = 2-methylcyclohexanone CC1CCCC1=O.

22.111 — book p. 1065 (PDF p. 29) — footnote 2 — J. Org. Chem. 1977, 42, 1663-1664

- Reaction: cyclohexanone + optically pure trans-2,5-dimethylpyrrolidine (one Me hashed, one wedged → C2-symmetric trans isomer) C[C@@H]1CC[C@H](C)N1 (= (2R,5R); trans confirmed, absolute config vs. drawing not verified [L]), TsOH, -H₂O → enamine.
- Asked: (a) enamine structure and how many isomers can form; (b) after CH₃I then H₃O⁺, 2-methylcyclohexanone obtained in 83% ee — explain the enantiomeric excess (vs racemate in problem 21.107) and identify the major enantiomer.
- Expected answer: (a) [H] only one enamine: the C2 axis makes both N-substituent orientations (rotamers) and both faces relationships identical — single compound. (b) [M] one methyl group of the C2-symmetric pyrrolidine blocks one face of the enamine β-carbon; CH₃I approaches from the opposite face → one configuration at the new stereocentre predominates (chiral auxiliary, asymmetric induction). Specific R/S of major product [L] — depends on amine absolute configuration; not determined here (check paper).

22.112 — book p. 1065 (PDF p. 29) — footnote 3 — J. Org. Chem. 2012, 77, 6371-6376

- Context: synthesis of 1,3,6-substituted fulvenes.
- SM: 1-phenylpentane-1,4-dione CC(=O)CCC(=O)c1ccccc1. 1) NaOH/H₂O, heat; 2) PhMgCl, THF; 3) conc. H₂SO₄ → C₁₇H₁₄.
- Expected answer [H]: intramolecular aldol condensation (methyl-ketone enolate attacks benzoyl C=O, 5-ring, dehydration) → 3-phenylcyclopent-2-en-1-one O=C1CCC(c2ccccc2)=C1; PhMgCl 1,2-addition → 1,3-diphenylcyclopent-2-en-1-ol; acid dehydration → 1,3-diphenylcyclopenta-1,3-diene (C₁₇H₁₄) C1(c2ccccc2)=CC(c2ccccc2)=CC1 (fulvene precursor).

22.113 — book pp. 1065-1066 (PDF pp. 29-30) — footnote 4 — J. Am. Chem. Soc. 2012, 134, 9291-9295

- Context: synthesis of polycyclic alkaloid **haouamine B**.
- SM: N-R 2,3-dihydropyridin-4(1H)-one-type enone (1,2,3,6-tetrahydropyridin-... ketone) bearing 3-methoxyphenyl on the β-carbon and a 3,5-dimethoxybenzyl group on the adjacent sp³ C-H stereocentre (next to N). Reagent: Tf₂O, pyridine. Product: tricyclic indane-fused (hexahydro-indeno[1,2-b]pyridine) with new all-carbon quaternary centre (bearing 3-MeOC₆H₄) and a vinyl triflate (enol triflate, O-SO₂CF₃).
- Asked: mechanism; explain stereochemistry of the new chirality centre. (Tf₂O activates the Michael acceptor.)
- Expected answer [M-H]: carbonyl O attacks Tf₂O → O-triflyl oxocarbenium/allyl cation (very electrophilic β-carbon); the electron-rich dimethoxyaryl ring attacks the β-carbon intramolecularly (Friedel-Crafts-type conjugate addition, 5-membered ring) → arenium ion; pyridine removes H⁺ to restore aromaticity; product is the enol triflate. Because the tether is short (5-ring), the aryl must add on the same face as the benzyl CH₂ → cis ring fusion (new quaternary centre cis to ring-fusion H), i.e. relative stereo set by the existing stereocentre.

22.114 — book p. 1066 (PDF p. 30) — footnote 5 — *Tetrahedron Lett.* 2010, 51, 6918-6920

- SM 1: α -oxoketene dithioacetal: (E)-4-(p-tolyl)-but-3-enoyl unit attached to $C(=C(SEt)_2)-C(=O)NH(4-ClC_6H_4)$; i.e. $ArCH=CH-C(=O)-C(=C(SEt)_2)-CONHAr'$. + diethyl malonate, $K_2CO_3 \rightarrow$ 2: cyclohexenone: $C(CO_2Et)_2$ quaternary, $C(SEt)=C(CONHAr')-C(=O)-CH_2-CH(p\text{-tolyl})$.
- Expected answer [H]: carbonate deprotonates malonate; malonate enolate adds conjugately to the ketene-dithioacetal carbon, then β -elimination of EtS^- (addition-elimination, vinylic substitution); the new malonyl C-H is deprotonated; the carbanion does an intramolecular Michael (6-endo) addition onto the styryl enone β -carbon ($CH-Ar$); enolate protonation $\rightarrow$ 2.

22.115 — book p. 1066 (PDF p. 30) — footnote 6 — *J. Am. Chem. Soc.* 2011, 133, 15797-15799

- Context: total synthesis of (–)-**N-methylwelwitindolinone C isothiocyanate** (blue-green algae; drug-resistant tumours). Key intermediate X (R = protecting group) made from Y with excess $NaNH_2$.
- Y: 3-substituted cyclohexanone (quaternary C bearing Me and vinyl; adjacent C-OR) linked via $C(CH_3)_2$ to C3 of 5-bromo-N-methylindole (Br at C5). X: bridged bicyclo[4.3.1] ketone — the cyclohexanone α -carbon bonded to indole C4.
- Asked: (a) mechanism $Y \rightarrow X$ under strong base; (b) by-product = structural isomer of X lacking a ketone (hint: enolates are ambident).
- Expected answer [H]: $NaNH_2$ forms the ketone enolate and also deprotonates C4-H ortho to Br $\rightarrow$ elimination of $Br^- \rightarrow$ 4,5-indolyne (aryne); enolate α -carbon attacks the proximal aryne carbon C4 (intramolecular), aryl anion at C5 protonated $\rightarrow$ X. (b) O-attack of the enolate on the aryne $\rightarrow$ cyclic aryl enol ether (O-arylation product; $C=C-O-Ar$, no $C=O$).

22.116 — book p. 1066 (PDF p. 30) — footnote 7 — *Tetrahedron Lett.* 1969, 10, 4233-4236

- Reaction: mixture of two conformational (rotameric) enamines of cyclohexanone with ethyl prolinatate (pyrrolidine-2- CO_2Et , single enantiomer; drawn as CO_2Et hashed vs wedged in two orientations) + 1) methyl acrylate, 2) $H_3O^+ \rightarrow$ A, methyl 3-(2-oxocyclohexyl)propanoate $O=C1CCCC1CCC(=O)OC$, 21% ee.
- Asked: assuming equal amounts of both rotamers, explain why A is nearly racemic by analysing the Michael mechanism for each; choose which enantiomer (R/S) of A predominates.
- Expected answer [M]: the CO_2Et shields one face of the enamine β -carbon; in the two rotamers the $C=C$ sits on opposite sides of the N-C bond, so the shielded face is reversed $\rightarrow$ each rotamer delivers the opposite enantiomer $\rightarrow$ near-cancellation (21% ee from a small difference in reactivity/population). Predominant enantiomer [L] — not determinable without the paper; answer from facial analysis.

22.117 — book p. 1066 (PDF p. 30) — footnote 8 — *Tetrahedron Lett.* 2010, 51, 5291-5293

- Context: enantiospecific synthesis of sex pheromone of the vine mealybug **Pseudococcus viburni**.
- SM A: cyclic hemiacetal (lactol) 3-OR-4,4-dimethyltetrahydrofuran-2-ol (cf. pantolactone-derived). A + $Ph_3P=CHCO_2Et$, heat $\rightarrow$ B; $TsCl$, py $\rightarrow$ C; $Me_2CuLi \rightarrow$ D (ethyl 2-methyl-3-OR-4,4-dimethylcyclopentane... ring with CO_2Et , Me, OR, gem- Me_2); D $\rightarrow$ E $\rightarrow$ F (CH_2OAc).
- Asked: (a) B and C; (b) how C $\rightarrow$ D closes the ring; (c) reagents D $\rightarrow$ F (two steps) and E.
- Expected answer [H]: (a) lactol opens to hydroxy aldehyde; Wittig $\rightarrow$ B = (E)- $EtO_2C-CH=CH-CH(OR)-C(CH_3)_2-CH_2OH$ (R=H analog: $CCOC(=O)/C=C/C(O)C(C)(C)CO$); C = corresponding primary tosylate (CH_2OTs). (b) Me_2CuLi 1,4-adds methyl to the enoate β -carbon $\rightarrow$ ester enolate; enolate α -carbon displaces OTs

intramolecularly (SN2, 5-exo-tet) → cyclopentane D (tandem conjugate addition-alkylation). (c) 1) LiAlH₄ (xs), then H₂O → E (primary alcohol, ring-CH₂OH); 2) Ac₂O (or AcCl), pyridine → acetate F.

22.118 — book p. 1067 (PDF p. 31) — footnote 9 — J. Am. Chem. Soc. 2012, 134, 920-923

- Context: enantioselective synthesis of **fusarisetin A** (soil fungus; anti-cancer/anti-metastatic without cytotoxicity).
- SM: trans-decalin-fused bicyclic ketone; ring carbon α to the ketone bears an amide C(=O)N(CH₃)-CH(CH₂OH)-CO₂Me (N-methyl-serine methyl ester); adjacent carbon bears CH(OH)CH₃. NaOMe → pentacyclic product: pyrrolidine-2,4-dione (tetramic acid) spiro-fused, with hemiketal (C-OH) linked to O of the former CH(OH)CH₃ (tetrahydrofuran ring).
- Expected answer [H]: MeO[−] removes the acidic C-H between ketone and amide carbonyl (β-dicarbonyl) → enolate; Dieckmann: enolate attacks serine methyl ester C=O → tetrahedral intermediate → loses MeO[−] → 5-membered tetramic-acid ring (new ketone); the secondary OH (side chain) attacks this new ketone intramolecularly → hemiketal (5-ring), proton transfers.

22.119 — book p. 1067 (PDF p. 31) — footnote 10 — Org. Lett. 2012, 14, 4738-4741

- SM 1: cis-2-benzoyl-6-vinyl-3,6-dihydro-2H-pyran O=C(c1ccccc1)C1CC=CC(=C)O1 (C₁₄H₁₄O₂). 1) NaHMDS, 2) H₂O → 2: 2-hydroxy-2-phenylcycloocta-4,6-dien-1-one (8-membered ring, ketone adjacent to C(OH)Ph) O=C1CC=CC=CC1(O)c1ccccc1 (C₁₄H₁₄O₂, isomer).
- Asked: mechanism incl. stereochemistry (hint: two resonance forms of the first intermediate).
- Expected answer [M-H]: NaHMDS deprotonates C2-H (α to benzoyl) → enolate; resonance form with O[−] shows ring O-C2=C(O[−])Ph i.e. an allyl vinyl ether with the 6-vinyl group → anion-accelerated [3,3]-sigmatropic (Claisen) rearrangement: new C-C bond between vinyl terminus and C(Ph)(O[−]), C2-O becomes C=O, C6=C forms → ring-expanded cyclooctadienone alkoxide; H₂O protonates → 2. Stereo: the cis-2,6 relationship forces a boat-like TS on one face → chirality transfer to the C(OH)Ph centre (single diastereomer/enantiomer).

22.120 — book p. 1067 (PDF p. 31) — footnote 11 — Can. J. Chem. 2004, 82, 293-300

- Context: ¹⁴C-labelled (at C3) **exo-16,17-dihydro-GA5-13-acetate** (modified gibberellin 1; reduces lodging of cereals).
- SM 2: A-ring-seco gibberellin aldehyde: C10 bears O (γ-lactone O) and -CH₂CH₂-CHO (¹⁴C aldehyde); lactone C=O attached to C4-H (with CH₃); CO₂H, 13-OAc, 16-Me. 1) K₂CO₃, H₂O; 2) HCl → 3: A-ring closed, 3-OH (epimeric, wavy) at C3.
- Asked: detailed step mechanism 2 → 3.
- Expected answer [M-H]: carbonate deprotonates CO₂H, then C4-H α to the lactone carbonyl → lactone enolate; intramolecular aldol addition of C4 onto the ¹⁴C-aldehyde → new 6-membered A ring, alkoxide at C3*; protonation by water → 3-OH (both epimers); HCl reprotonates the carboxylate. (Reverse of a retro-aldol.)

Chapter 23 — 아민 (Amines)

Key reactions (반응의 복습, book pp. 1103-1104)

- Preparation: $RBr + NaCN \rightarrow RCH_2NH_2$; $RCOOH \rightarrow SOCl_2, xs\ NH_3 \rightarrow \text{amide} \rightarrow xs\ LiAlH_4 \rightarrow \text{amine}$; benzene $\rightarrow HNO_3/H_2SO_4 \rightarrow \text{nitrobenzene} \rightarrow H_2/Pt \text{ or } Fe/Zn/Sn/SnCl_2, H_3O^+; NaOH \rightarrow \text{aniline}$; azide synthesis (NaN_3 ; $H_2/Pt \text{ or } LiAlH_4$); Gabriel (phthalimide, KOH ; RBr ; H_2NNH_2); reductive amination ($NH_3 / RNH_2 / R_2NH, [H^+], NaBH_3CN$).
- Reactions: acylation ($RNH_2 + RCOCl \rightarrow \text{amide} + HCl$); Hofmann elimination ($xs\ CH_3I; Ag_2O, H_2O, \text{heat} \rightarrow \text{less-substituted alkene}$); nitrous acid: $1^\circ \text{ amine} + NaNO_2/HCl \rightarrow \text{diazonium}$; $2^\circ \rightarrow N\text{-nitrosamine}$.
- Aryl diazonium: Sandmeyer ($CuBr, CuCl, CuCN$; also CuI), Schiemann ($HBf_4 \rightarrow ArF$), $H_2O/\text{heat} \rightarrow \text{phenol}$, $H_3PO_2 \rightarrow ArH$, azo coupling with activated arenes.
- Heterocycles: pyrrole + $Br_2, 0^\circ C \rightarrow 2\text{-bromopyrrole}$; pyridine + $Br_2, 300^\circ C \rightarrow 3\text{-bromopyridine}$.
Summary concepts: amine basicity (pK_a of ammonium ions; aryl amines less basic; amides non-basic; pyridine $>$ pyrrole basicity), solvent extraction, $1/2/3^\circ$ amine IR (two/one N-H bands 3350-3500), $^1H/^13C$ shifts.

References (참고문헌, book p. 1113 / PDF p. 42)

1. Tetrahedron Lett. 2012, 53, 1119-1121.
2. Bioorg. Med. Chem. Lett. 2006, 16, 2013-2016.
3. Org. Lett. 2007, 9, 2673-2676.
4. J. Am. Chem. Soc. 1945, 67, 860-874.
5. Tetrahedron Lett. 2009, 50, 3817-3819.
6. Tetrahedron Lett. 2004, 45, 5807-5810.
7. Tetrahedron Lett. 2005, 46, 979-982.
8. J. Org. Chem. 2011, 76, 1605-1613.
9. J. Org. Chem. 1999, 64, 4966-4968.
10. J. Org. Chem. 1999, 64, 4617-4626.

Literature problems

23.89 — book p. 1111 (PDF p. 40) — footnote 1 — Tetrahedron Lett. 2012, 53, 1119-1121

- Context: first asymmetric synthesis of alkaloid (–)-(S,S)-homaline.
- SM 1: methyl (S)-3-[(3-chloropropyl)amino]-3-phenylpropanoate (2° amine). Product 2: methyl (S)-3-[(3-azidopropyl)(methyl)amino]-3-phenylpropanoate.
- Asked: reagents for the 2-step conversion.
- Expected answer [H]: (1) N-methylation by reductive amination: $HCHO (CH_2O), [H^+], NaBH_3CN$ (avoids over-alkylation to ammonium that CH_3I would cause); (2) NaN_3 (S_N2 on the primary chloride; DMF). (Order interchangeable.)

23.90 — book pp. 1111-1112 (PDF pp. 40-41) — footnote 2 — Bioorg. Med. Chem. Lett. 2006, 16, 2013-2016

- Context: nicotine-like analgesic derivatives with lower toxicity.
- SM 1: 6-chloronicotinic acid O=C(O)c1ccc(c1)nc1. Product 2: N-[(6-chloropyridin-3-yl)methyl]cyclopentanamine C1c1ccc(CNC2CCCC2)cn1.

- Expected answer [H]: (1) SOCl₂ → acid chloride; (2) cyclopentylamine (2 equiv, or + pyridine) → amide; (3) xs LiAlH₄, then H₂O → 2° amine. (Alternative: reduce acid to alcohol, oxidise to aldehyde, reductive amination with cyclopentylamine/NaBH₃CN; or 6-chloronicotinaldehyde + cyclopentanamine.)

23.91 — book p. 1112 (PDF p. 41) — footnote 3 — Org. Lett. 2007, 9, 2673-2676

- Context: **halosaline** (from *Lobelia inflata*); synthesis via azide reduction then amide reduction.
- SM: 8-membered lactone with azide: 8-propyl-6-azido-oxocan-2-one (stereo as drawn) CCC[C@H]1C[C@@H](N=[N+]=[N-])CCCC(=O)O1 (stereo indicative). H₂, Pd → 6-(2-hydroxypentyl)piperidin-2-one CCCC(=O)CC1CCCC(=O)N1; then xs LiAlH₄; H₂O → halosaline CCCC(=O)CC1CCCCN1 (1-(piperidin-2-yl)pentan-2-ol).
- Asked: show the ring-contraction (lactone → lactam) step.
- Expected answer [H]: H₂/Pd reduces N₃ → NH₂ (loses N₂); the amine attacks the lactone C=O intramolecularly (6-membered TS/ring) → tetrahedral intermediate → C-O(ring) bond breaks, releasing the secondary alkoxide → proton transfer → δ-lactam with pendant secondary OH (transamidation; stereocentres retained).

23.92 — book p. 1112 (PDF p. 41) — footnote 4 — J. Am. Chem. Soc. 1945, 67, 860-874

- Context: Woodward-Doering formal total synthesis of **quinine** (1944).
- SM: N-acetyl piperidine with cis-3-(1-aminoethyl) and 4-(CH₂CH₂CO₂Et) substituents CC(=O)N1CCC(CCC(=O)OCC)C(C(C)N)C1. 1) xs CH₃I; 2) 60% NaOH (heat) → C₁₀H₁₇NO₂.
- Expected answer [H]: exhaustive methylation of the 1° amine → -CH(CH₃)NMe₃⁺; Hofmann elimination (E₂, less-substituted alkene) → 3-vinyl; NaOH also hydrolyses the ethyl ester (→ acid) and the N-acetyl amide (→ free NH) → **homomeroquinene**: 3-(3-vinylpiperidin-4-yl)propanoic acid OC(=O)CCC1CCNCC1C=C (C₁₀H₁₇NO₂ ✓), cis-3,4.

23.93 — book p. 1112 (PDF p. 41) — footnote 5 — Tetrahedron Lett. 2009, 50, 3817-3819

- Context: piperazinone derivatives (migraine, hypertension, etc.).
- SM 1: 2-bromoacetophenone (phenacyl bromide). + L-alanine methyl ester, NaHCO₃, DMF → 2 (S_N2) CC(=O)[C@H](C)NCC(=O)c1ccccc1; + PhCH₂NH₂, NaBH₃CN, AcOH, heat → [3] → 4 (cyclic, lactamization).
- Asked: (a) structure of 3; why diastereomeric mixture; (b) structure of 4; is it a diastereomer mixture? why.
- Expected answer [H]: (a) 3 = methyl N-[2-(benzylamino)-2-phenylethyl]-L-alaninate COC(=O)[C@H](C)NCC(NCC1CCCCC1)c1ccccc1; reductive amination creates a new stereocentre (CHPh) with no facial control (remote alanine centre) → ~1:1 diastereomers. (b) 4 = 1-benzyl-3-methyl-6-phenylpiperazin-2-one C[C@H]1NCC(Cc2ccccc2)N(Cc2ccccc2)C1=O (C₁₈H₂₀N₂O); yes, still a diastereomeric (cis/trans) mixture because lactamization does not touch either stereocentre (each diastereomer of 3 cyclises separately).

23.94 — book p. 1112 (PDF p. 41) — footnote 6 — Tetrahedron Lett. 2004, 45, 5807-5810

- Context: designed anticancer alkylating agent (SAM/adenosine-based).
- SM: 5'-amino-5'-deoxy-2',3'-bis-O-(triethylsilyl)adenosine (C₂₂H₄₂N₆O₃Si₂) + ethyl (E)-4,5-dibromopent-2-enoate (EtO₂C-CH=CH-CHBr-CH₂Br), DBU → C₂₉H₅₀N₆O₅Si₂.
- Asked: structure of product (two sequential substitutions).
- Expected answer [H]: N-substituted **aziridine**: amine displaces one bromide (S_N2), then the resulting 2° amine closes intramolecularly on the vicinal bromide → 2-[(E)-2-(ethoxycarbonyl)vinyl]aziridine attached

at N to the 5'-CH₂ of adenosine. SMILES: Nc1ncnc2c1ncn2[C@@H]1O[C@H](CN3CC3/C=C/C(=O)OCC)[C@@H](O[Si](CC)(CC)C(C)[C@H]1O[Si](CC)(CC)CC — formula C₂₉H₅₀N₆O₅Si₂ ✓ (RDKit). Aziridine = alkylating group.

23.95 — book pp. 1112-1113 (PDF pp. 41-42) — footnote 7 — Tetrahedron Lett. 2005, 46, 979-982

- Context: enaminoesters (vinylogous carbamates) as N-heterocycle building blocks; one-pot.
- SM 1: tert-butyl (2-azidopropyl)carbamate (Boc-NH-CH₂-CH(CH₃)-N₃, stereocentre). H₂, Pd → [2]; with methyl acetoacetate present, 2 → imine [3] without acid; 3 tautomerises → enamine 4: methyl (Z)-3-[(1-(Boc-amino)propan-2-yl)amino]but-2-enoate COC(=O)/C=C(/C)NC(C)CNC(=O)OC(C)(C)C.
- Asked: (a) 2 and 3; (b) acid-catalysed (H₃O⁺) mechanism 3 → 4; (c) at least two reasons equilibrium favours enamine 4 over imine 3.
- Expected answer [H]: (a) 2 = tert-butyl (2-aminopropyl)carbamate CC(N)CNC(=O)OC(C)(C)C; 3 = ketimine: MeO₂C-CH₂-C(CH₃)=N-CH(CH₃)CH₂NHBoc. (b) protonate imine N → iminium; water/base removes α-C-H (CH₂ between C=N and ester) → enamine; (c) enamine C=C conjugated with ester C=O (push-pull/vinylogous carbamate resonance) and a 6-membered intramolecular N-H⋯O=C hydrogen bond (Z-isomer); also α-H is acidic (β-dicarbonyl-like).

23.96 — book p. 1113 (PDF p. 42) — footnote 8 — J. Org. Chem. 2011, 76, 1605-1613

- Context: total synthesis of **crispine A** (cytotoxic alkaloid, Mongolian thistle).
- SM 1: methyl (S)-2-amino-3-(3,4-dimethoxyphenyl)propanoate COC(=O)[C@H](N)Cc1ccc(OC)c(OC)c1; + 4-chlorobutanal, [H₃O⁺] → 2: methyl 8,9-dimethoxy-1,2,3,5,6,10b-hexahydropyrrolo[2,1-a]isoquinoline-5-carboxylate COC1CC2C(Cc1OC)C1CCCN1[C@H](C(=O)OC)C2 (C₁₆H₂₁NO₄).
- Expected answer [M-H]: imine formation (carbinolamine, -H₂O) → iminium; Pictet-Spengler: electron-rich aryl ring (para to OMe) attacks iminium carbon (6-endo) → arenium, loss of H⁺ → tetrahydroisoquinoline (2° amine); intramolecular N-alkylation (S_N2 on CH₂Cl) closes the pyrrolidine ring → 2. (Alternative order: N-alkylation first → cyclic pyrrolinium ion → Pictet-Spengler.)

23.97 — book p. 1113 (PDF p. 42) — footnote 9 — J. Org. Chem. 1999, 64, 4966-4968

- Context: ring-opened analogue of **epibatidine** (potent analgesic, Ecuadorian poison frog); fragmentation reaction.
- SM: N-ethoxycarbonyl-8-azabicyclo[3.2.1]octan-3-ol bearing 3-(6-chloropyridin-3-yl) (tertiary alcohol) CCOC(=O)N1C2CCC1CC(O)(c1ccc(Cl)nc1)C2 [M on skeleton]; CF₃CO₂H, NaBH₃CN → N-CO₂Et-2-[2-(6-chloropyridin-3-yl)allyl]pyrrolidine CCOC(=O)N1CCCC1CC(=C)c1ccc(Cl)nc1.
- Expected answer [M-H]: TFA protonates OH, loss of H₂O → tertiary (benzylic-type) carbocation at C3; Grob-type fragmentation: N lone pair pushes, C1-C2 bond cleaves → C₂=C₃ (1,1-disubstituted alkene) + N-acyliminium at C1; NaBH₃CN delivers hydride to the iminium → pyrrolidine.

23.98 — book p. 1113 (PDF p. 42) — footnote 10 — J. Org. Chem. 1999, 64, 4617-4626

- SM: 2-aminofuran tethered N(R)-CH₂CH₂-C(=CH₂)CO₂Me (2-[N-R-amino]furan with methyl 2-methylenebutanoate chain). Heat, MeOH → hexahydroindole: N-R, C_{7a}=C₇ enamine, C_{3a} quaternary CO₂Me, C₆-OMe and C₅-OH (trans, MeO hashed / OH wedged as drawn).
- Asked: mechanism and the stereochemistry of the product.
- Expected answer [M]: intramolecular Diels-Alder of the furan (diene) with the acrylate (dienophile) (IMDAF) → oxa-bridged (7-oxabicyclo[2.2.1]heptene) cycloadduct fused to a pyrrolidine; N lone pair

drives C7a-O bridge cleavage → conjugated iminium + C5-alkoxide (→ C5-OH, stereo fixed by the former bridge, cis to the CO₂Me-bearing ring fusion face); MeOH adds conjugately (1,4) to the α,β-unsaturated iminium at C6 from the face opposite the OH → enamine (C7=C7a); relative stereo: OH and OMe trans; CO₂Me configuration set by the cycloaddition.

Notes / caveats

- All 42 footnoted problems across ch. 20-23 are listed (ch. 20: footnotes 2-12, footnote 1 not in these pages; ch. 21: 1-10; ch. 22: 1-11; ch. 23: 1-10).
- Citations are transcribed exactly as printed; Ch. 20 ref 4 shows volume "43" for Tetrahedron 2014 (likely misprint for 70).
- Ch. 21 ref 5 and Ch. 22 ref 7 are the same paper (Tetrahedron Lett. 1969, 10, 4233) — both about proline-ester enamines of cyclohexanone.
- Lower-confidence items: 20.88 (Claisen route assumed), 21.88 (quadron rearrangement details), 22.111/22.116 absolute configurations, 23.98 stereo rationale.